



Survey paper

Learning strategies for particle swarm optimizer: A critical review and performance analysis

Dikshit Chauhan^a, Shivani^{b,*}, Ponnuthurai N. Suganthan^c^a Department of Electrical and Computer Engineering, National University of Singapore, 119077, Singapore^b Department of Mathematics and Computing, Dr. B.R. Ambedkar National Institute of Technology, Jalandhar 144008, Punjab, India^c College of Engineering, Qatar University, Doha, Qatar

ARTICLE INFO

Keywords:

Optimization
Evolutionary computation
Particle swarm optimizer
Learning strategies
Performance evaluation

ABSTRACT

Nature has long inspired the development of swarm intelligence (SI), a key branch of artificial intelligence that models collective behaviors observed in biological systems for solving complex optimization problems. Particle swarm optimization (PSO) is widely adopted among SI algorithms due to its simplicity and efficiency. Despite numerous learning strategies proposed to enhance PSO's performance in terms of convergence speed, robustness, and adaptability, no comprehensive and systematic analysis of these strategies exists. We review and classify various learning strategies to address this gap, assessing their impact on optimization performance. Additionally, a comparative experimental evaluation is conducted to examine how these strategies influence PSO's search dynamics. Our analysis reveals that multi-swarm strategies consistently outperform other PSO strategies in high-dimensional and multimodal problems, offering better exploration and convergence trade-offs. Finally, we discuss open challenges and future directions, emphasizing the need for self-adaptive, intelligent PSO variants capable of addressing increasingly complex real-world problems. This survey not only synthesizes the current landscape of learning-enhanced PSO but also provides actionable insights for future research and algorithmic design.

1. Introduction

Over the past few decades, optimization problems have become increasingly prevalent across various industrial and scientific domains, including mechanical engineering [1], nuclear energy [2], vehicle engineering [3], reliability analysis [4], aerospace engineering [5], and topology optimization [6]. The growing complexity of real-world problems, such as nonlinear optimal control, text clustering, DNA sequence compression, and distribution network design, has intensified the demand for efficient optimization algorithms. As a result, research on optimization techniques holds significant practical value and broad applicability [7].

Optimization problems are typically categorized into constrained and unconstrained problems. A constrained optimization problem can be expressed as: $\min \mathbf{F} = f(\mathbf{X})$, $\mathbf{X} \in \mathbb{S}$, $\mathbb{S} = \{\mathbf{X} \mid \mathbf{g}_i(\mathbf{X}) \leq 0, i = 1, \dots, m_c\}$, where $\mathbf{F} = f(\mathbf{X})$ is the objective function, $\mathbf{g}_i(\mathbf{X})$ represents the constraint functions, m is the number of constraints, and $\mathbf{X}$ is a D -dimensional vector. In contrast, unconstrained optimization problems, formulated in Euclidean n -space, take the form: $\min \mathbf{F} = f(\mathbf{X})$, $\mathbf{X} \in \mathbb{S}$, $\mathbf{X} = \{x^1, x^2, \dots, x^D\}$, $\mathbb{S} \subset \mathbb{R}^n$. Efficient optimization methods are essential for solving both types, leading to improved solutions for

complex real-world challenges. These methods can be broadly divided into gradient-based techniques and metaheuristic algorithms. While gradient-based methods rely on mathematical derivatives, metaheuristic algorithms are favored for their flexibility, derivative-free nature, and ease of implementation [8,9]. Among them, particle swarm optimizer (PSO) [10] has gained considerable attention due to its simplicity and rapid convergence.

PSO is a global stochastic optimization algorithm inspired by the collective behavior of bird flocks and biological swarms. Each particle in the swarm represents a potential solution, adjusting its position based on both personal experience and the knowledge of neighboring particles. This cooperative mechanism enables efficient search space exploration and convergence toward optimal solutions [11].

One of PSO's primary advantages is its simplicity and ease of implementation. With a straightforward mathematical structure and minimal parameter tuning, it is accessible for a wide range of optimization problems. Unlike gradient-based methods, PSO does not require derivative information, making it well-suited for non-differentiable, multimodal, and complex optimization landscapes [12,13]. Additionally,

* Corresponding author.

E-mail addresses: dikshitchauhan608@gmail.com (D. Chauhan), sainishivani2310@gmail.com (Shivani), p.n.suganthan@qu.edu.qa (P.N. Suganthan).

PSO exhibits fast convergence, particularly in lower-dimensional problems, where particles quickly identify near-optimal solutions through collaborative learning [14,15].

Despite its strengths, PSO faces several challenges, particularly in complex and high-dimensional optimization problems. One of the most significant limitations is premature convergence, where particles become trapped in local optima, leading to suboptimal solutions [16]. This issue is particularly pronounced in multimodal optimization landscapes, where the algorithm may struggle to escape local attraction basins. Moreover, as the number of dimensions increases, PSO's exploration and exploitation capabilities decline, reducing its efficiency, an effect commonly referred to as the curse of dimensionality [11,15].

Another limitation of PSO is its sensitivity to parameter settings. The performance of the algorithm heavily depends on selecting appropriate values for key parameters such as inertia weight, cognitive and social coefficients, and velocity limits [17,18]. Improper tuning of these parameters can lead to inefficient exploration, poor convergence behavior, or excessive computational costs [19]. While researchers have introduced various adaptive techniques to mitigate these issues, finding optimal parameter settings remains a challenge. Furthermore, PSO's full-dimensional learning strategy can sometimes lead to poor solutions, as it does not account for correlations between different variables, reducing its effectiveness in problems requiring coordinated learning across multiple dimensions [19].

To address these challenges, researchers have introduced various learning strategies aimed at improving the universality and robustness of PSO by dynamically balancing exploration and exploitation. One approach involves adopting multiple learning strategies, allowing particles to update their positions using different mechanisms. For example, adaptive PSO (APSO) [20] classifies the search process into four states: exploration, exploitation, convergence, and jumping out, adjusting inertia weights and acceleration coefficients accordingly. Similarly, dynamic neighbor learning PSO (DNLPSO) [21] selects an exemplar from the best positions of neighboring particles to enhance information sharing and adaptability.

Beyond neighborhood-based improvements, learning strategies have played a crucial role in advancing PSO performance. Comprehensive learning PSO (CLPSO) [22] enhances multimodal problem-solving by utilizing personal best experiences of all particles, improving diversity and global search capabilities. Extensions of CLPSO, such as heterogeneous CLPSO (HCLPSO) [23], further refine this approach by dividing the swarm into exploration and exploitation subpopulations. Other notable strategies include self-learning PSO (SLPSO) [24], which allows particles to independently adopt different learning behaviors, and genetic learning PSO (GL-PSO) [25], which integrates genetic operators (selection, mutation, and crossover) to guide particle movement more effectively.

In addition, several hybrid strategies have been developed to mitigate premature convergence and improve search efficiency. Generalized opposition-based PSO (GOPSO) [25] leverages opposition-based learning and Cauchy mutation to escape local optima, while orthogonal learning PSO (OLPSO) [26] applies orthogonal experiment design to construct learning exemplars that enhance both personal and global best experiences. Ensemble PSO (EPSO) [27] combines multiple learning strategies to optimize real-parameter problems by leveraging their complementary strengths.

Existing studies have highlighted the effectiveness of single- and multi-strategy-based PSO variants in solving optimization problems, as different strategies enhance various aspects of the algorithm. Given the vast number of learning strategies developed over the past decades, a key question arises: Which strategy or combination of strategies can significantly improve PSO's performance? While diverse learning mechanisms provide a valuable foundation, their strategic integration could lead to more robust and efficient optimization techniques. However, a critical challenge remains: how can these strategies be systematically

combined to maximize their effectiveness across different problem landscapes?

While several review papers have explored PSO enhancements and variants, as per the author's knowledge, no existing work has specifically focused on the role of learning mechanisms in PSO. Many of these mechanisms have been theoretically proposed, yet their empirical validation and comparative study remain largely underexplored. Despite the introduction of novel strategies, there is a lack of critical assessments evaluating their effectiveness, limitations, and future research directions.

To address this gap, this review provides a structured and comprehensive analysis of PSO learning mechanisms. Through a detailed examination of various strategies, ranging from adaptive and multi-swarm learning to hybrid and ensemble approaches, this study systematically evaluates their impact on PSO's exploration–exploitation balance, convergence behavior, and overall robustness. Additionally, an experimental evaluation is conducted to assess the comparative performance of different learning mechanisms, offering deeper insights into their practical applicability.

This review aims to advance the understanding of how learning strategies can be leveraged to develop next-generation PSO algorithms capable of solving increasingly complex optimization problems. By addressing existing gaps and identifying key research directions, this work contributes to the ongoing development of adaptive, intelligent, and high-performance PSO variants. A list of main contributions is as follows:

- (i) We systematically categorize existing learning strategies in PSO, including adaptive learning, comprehensive learning, and hybrid learning mechanisms, highlighting their impact on search efficiency and optimization performance.
- (ii) A structured evaluation of various PSO learning strategies is conducted, analyzing their effectiveness in improving convergence speed, robustness, and diversity across different optimization landscapes.
- (iii) The review identifies critical gaps in the current literature, such as the lack of unified evaluation frameworks and adaptive mechanisms, providing insights into promising future research directions for enhancing PSO.

The structure of this article is as follows: Section 2 introduces the fundamental theory of PSO. Section 3 reviews single-swarm-based learning strategies, while Sections 4 and 5 explore two-swarm-based and multiple-swarm-based learning strategies, respectively. Theoretical developments in PSO learning mechanisms are discussed in Section 6. Section 7 presents a comparative experimental study of different learning strategies. Open research questions and future directions are outlined in Section 8, followed by the conclusion in Section 9.

2. PSO

PSO is a nature-inspired, population-based optimization algorithm originally introduced by Kennedy and Eberhart in 1995 [10]. Modeled after the collective behavior seen in natural systems, such as bird flocking or fish schooling, PSO simulates a group of particles (potential solutions) flying through a multidimensional search space in pursuit of an optimal solution. Each particle adjusts its position based on its own experience and that of neighboring particles, making decisions through both individual learning (self-experience) and social learning (peer influence).

In PSO, each particle maintains a position vector $\mathbf{x}_i = \{x_i^1, x_i^2, \dots, x_i^D\}$ and a velocity vector $\mathbf{vel}_i = \{vel_i^1, vel_i^2, \dots, vel_i^D\}$ in a D -dimensional continuous search space. The particle updates its velocity and position using the following rule:

$$\mathbf{vel}_i^d = \omega_g \cdot \mathbf{vel}_i^d + \underbrace{c_{e1} \cdot rand_1 \cdot (\mathbf{p}_{Best_i}^d - \mathbf{x}_i^d)}_{\text{cognitive component}} + \underbrace{c_{e2} \cdot rand_2 \cdot (\mathbf{g}_{Best}^d - \mathbf{x}_i^d)}_{\text{social component}}, \quad (1)$$

where: ω_g is the inertia weight that regulates how much of the previous velocity is retained, thus controlling exploration versus exploitation, c_{e1} and c_{e2} are acceleration coefficients that weigh the cognitive (individual experience) and social (group experience) contributions, respectively, $rand_1$ and $rand_2$ are uniformly distributed random numbers between 0 and 1 that add stochasticity to the process, $\mathbf{p}_{Best_i}$ is the best solution that particle i has found so far (personal best), and $\mathbf{g}_{Best}$ is the best-known solution found by the entire swarm or a defined neighborhood of particles (global or neighborhood best).

The balance between exploration (searching new regions) and exploitation (refining known good solutions) is central to PSO's success. A higher inertia weight encourages broader exploration of the search space, while a lower one supports faster convergence to promising regions [28].

Due to its simplicity and efficiency, PSO has been widely adopted for solving a variety of real-world and theoretical optimization problems, including those in engineering design, scheduling, image processing, and machine learning [11]. Nevertheless, PSO may struggle with high-dimensional, noisy, or multimodal optimization problems, often suffering from premature convergence where the swarm gets trapped in local optima [29].

The traditional PSO algorithm updates each particle based on its own best experience and the best experience of one other particle (global or neighborhood best). However, relying solely on the best performer may ignore other valuable insights within the swarm, which could reduce diversity and slow down convergence [30–32].

To address these limitations, researchers have proposed numerous learning-based enhancements to PSO. These include adaptive learning strategies, hybrid methods incorporating other optimization algorithms, and more cooperative information-sharing schemes. Such innovations aim to improve PSO's robustness, scalability, and effectiveness across diverse optimization landscapes.

3. Single swarm-based leaning

This section presents learning strategies developed within the single-swarm (population) framework of PSO, where the entire population of particles evolves as a unified entity without being subdivided into multiple interacting swarms. In such a setup, all particles follow homogeneous learning and adaptation rules, which promote consistent information-sharing and coordinated behavior across the entire swarm. Unlike multi-swarm frameworks that incorporate multiple interacting populations or subgroups (as discussed in Sections 4 and 5), single-swarm strategies rely solely on intra-swarm dynamics to guide optimization. These methods are particularly focused on enhancing the swarm's global exploration, refining local exploitation, and improving convergence efficiency by designing better knowledge-sharing, velocity update, and exemplar selection mechanisms. Representative strategies in this category include comprehensive learning, orthogonal learning, dimensional learning, and example-based learning, each targeting different aspects of swarm intelligence. In the subsections that follow, we review these strategies in detail, analyzing their design principles, benefits, and limitations.

3.1. Comprehensive learning mechanism

In the original PSO, a particle's flying direction is guided by its individual best position ($\mathbf{p}_{Best}$) and the global best position ($\mathbf{g}_{Best}$). However, $\mathbf{g}_{Best}$ often represents a suboptimal local optimum in multimodal optimization problems, leading to premature convergence and reduced diversity in the population [33]. To address these challenges, a *comprehensive learning mechanism* for PSO (CLPSO) strategy was proposed [22].

CLPSO [22] introduces a learning mechanism where a particle does not rely solely on $\mathbf{g}_{Best}$ or its own $\mathbf{p}_{Best}$. Instead, it learns from the $\mathbf{p}_{Best}$

positions of all particles in the population. Each dimension of a particle is updated independently, learning from potentially different exemplars for different dimensions. This mechanism increases population diversity and prevents stagnation in local optima, especially in multimodal landscapes.

The velocity of the i th particle in CLPSO is updated using the equation:

$$\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_e \cdot rand_i^d \cdot \left(\mathbf{p}_{Best_{fd}}^d - \mathbf{x}_i^d \right), \quad (2)$$

where w_g is the inertia weight balancing exploration and exploitation, c_e is the acceleration coefficient determining the influence of the exemplar, $rand_i^d$ is a random number in $[0, 1]$ introducing stochasticity, and $\mathbf{p}_{Best_{fd}}^d$ represents the exemplar position for dimension d , which can be the particle's own $\mathbf{p}_{Best}$ or another particle's $\mathbf{p}_{Best}$, determined by the learning probability P_{rc} .

The decision of whether to follow its own $\mathbf{p}_{Best}$ or another particle's $\mathbf{p}_{Best}$ for each dimension is based on a learning probability $P_{rc, i}$, calculated as:

$$P_{rc, i} = \alpha + \beta \cdot \exp\left(\frac{10 \cdot (i - 1)}{N_s - 1}\right), \quad (3)$$

where $\alpha = 0.05$, $\beta = 0.45$ are predefined constants controlling the range of $P_{rc, i}$, and N_s represents the population size. For each dimension of a particle, a random number is generated and compared with $P_{rc, i}$. If the random number is smaller than $P_{rc, i}$, the particle follows another particle's $\mathbf{p}_{Best}$. Otherwise, it learns from its $\mathbf{p}_{Best}$. The exemplar $\mathbf{p}_{Best_{fd}}^d$ is determined as follows:

- (i) *Tournament selection*: Two particles are randomly chosen from the population (excluding the particle being updated).
- (ii) *Fitness comparison*: The fitness values of the two particles' $\mathbf{p}_{Best}$ are compared, and the better $\mathbf{p}_{Best}$ is chosen as the exemplar for the corresponding dimension.

This selection process ensures that particles are guided by high-quality exemplars, balancing exploration and exploitation. The step-wise procedure of the CL mechanism is given in Algorithm 1.

Algorithm 1 Pseudo code of CL strategy.

```

1: Update the velocity using Eq. (2) and update the population,
2: Evaluate the fitness of each individual,
3: if  $f_{x_i^d} < f_{\mathbf{p}_{Best_i}^d}$  then
4:    $\mathbf{p}_{Best_i}^d \leftarrow \mathbf{x}_i^d$ ,  $g_i = 0$ ,
5: else
6:    $g_i \leftarrow g_i + 1$ ,
7: end if
8: if  $g_i > R_g$  then
9:   Update  $\mathbf{p}_{Best_{fd}}^d$ 
10: end if

```

To further enhance adaptability, CLPSO incorporates a refreshing gap R_g , defining a stagnation threshold. If a particle's $\mathbf{p}_{Best}$ does not improve for R_g consecutive evaluations, its learning exemplar is reassigned, and a new $\mathbf{p}_{Best_{fd}}^d$ is generated. This mechanism avoids wasting time on poor search directions and encourages continual improvement in particle movement. The procedure for updating $\mathbf{p}_{Best}$ is illustrated in Algorithm 2.

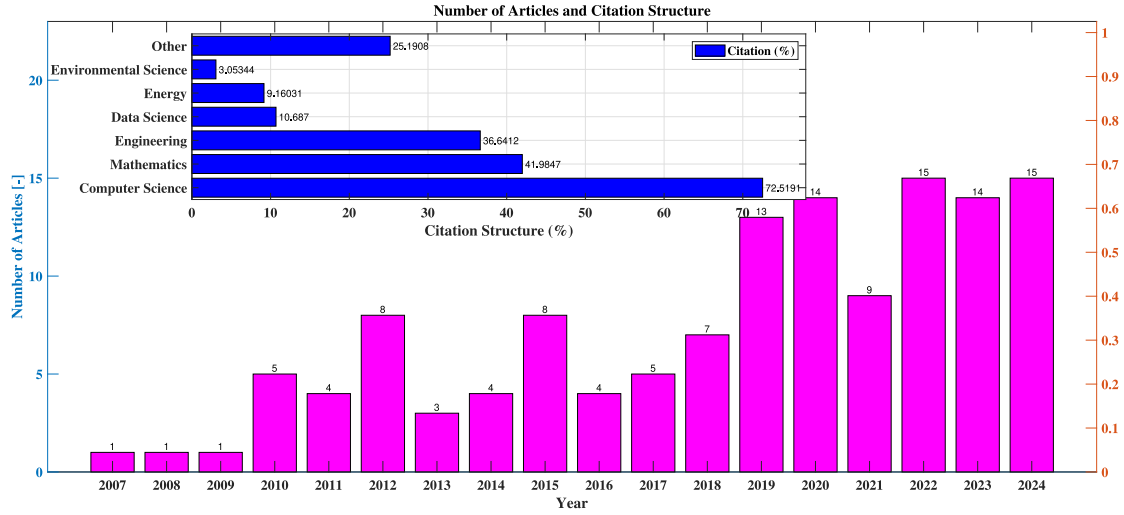


Fig. 1. Number of annually published documents on CL and application in the different fields (%) based on the Scopus platform.

Algorithm 2 Exemplar $\mathbf{p}_{Best}^d$.

```

1: for each dimension  $j$  do
2:   if rand <  $P_{c,i}$  then
3:      $a \leftarrow \text{RANDOM}(1, N_s)$ 
4:      $b \leftarrow \text{RANDOM}(1, N_s)$ 
5:     if  $f_{p_{Best_a}} \leq f_{p_{Best_b}}$  then
6:        $\mathbf{x}_i^j \leftarrow \mathbf{p}_{Best_a}^j$ 
7:     else
8:        $\mathbf{x}_i^j \leftarrow \mathbf{p}_{Best_b}^j$ 
9:     end if
10:  else
11:     $\mathbf{x}_i^j \leftarrow \mathbf{p}_{Best_i}^j$ 
12:  end if
13: end for

```

CLPSO has several differences with PSO, such as: (i) Unlike PSO, which uses the particle's $\mathbf{p}_{Best}$ and $\mathbf{g}_{Best}$ as fixed exemplars, CLPSO allows every particle to learn from the $\mathbf{p}_{Best}$ positions of any other particle. This flexibility provides a richer source of guidance, especially in complex multimodal problems. (ii) Each dimension of a particle can learn from different exemplars, providing independent and diverse updates to each coordinate. This contrasts with PSO, where the entire particle learns from the same exemplars for all dimensions. (iii) Instead of learning from both $\mathbf{p}_{Best}$ and $\mathbf{g}_{Best}$ simultaneously, each dimension in CLPSO learns from a single exemplar. This reduces the risk of conflicting updates and ensures focused learning.

By allowing each dimension to learn from different $\mathbf{p}_{Best}$ sources, CLPSO enhances population diversity and effectively mitigates premature convergence. The algorithm dynamically balances exploration (learning from others' $\mathbf{p}_{Best}$) and exploitation (refining its own $\mathbf{p}_{Best}$) through the learning probability $P_{c,i}$ and the refreshing gap R_g . This dynamic adaptability enables CLPSO to excel in multimodal optimization tasks, as it promotes the exploration of multiple optima and avoids entrapment in suboptimal solutions.

Fig. 1 depicts the annual growth of research articles on the CL mechanism and its applications across various fields, along with the citation structure (%). Between 2007 and 2018, the average number of publications remained around 4 per year, indicating limited early activity. Notable peaks occurred in 2012 and 2015, with 8 articles each year, followed by a steady phase from 2014 to 2018, with 4–7 articles annually. A significant surge began in 2019, averaging 13 articles per year from 2019 to 2024, underscoring the growing importance of CL in optimization and machine learning. Computer Science dominates with

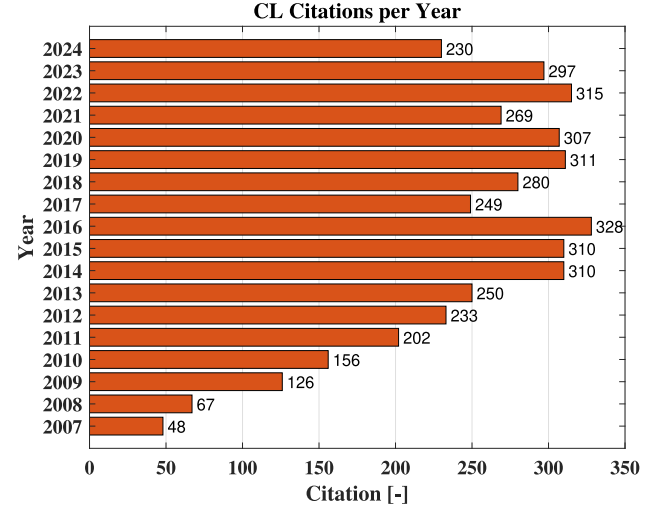


Fig. 2. CL citations per year based on the Google Scholar platform.

72.52% of citations, highlighting its primary role in machine learning and optimization. Engineering (38.64%), Mathematics (41.98%), and Data Science (10.69%) also show significant adoption, emphasizing the interdisciplinary nature of CL. Fields like Energy (9.16%), Environmental Science (3.05%), and others (25.19%) have lower citation percentages, suggesting emerging but less impactful applications. The annual citations of the original CL paper [22], as shown in Fig. 2, average 238 citations per year, showcasing its widespread recognition and global influence.

Some notable implementations of the CL mechanism are: Lynn et al. [23] integrated the CL mechanism into PSO (HCLPSO) by dividing the population into two group sizes, enhancing PSO's diversity. This approach was further extended into C-HCLPSO [34,35], incorporating chaotic strategies for photovoltaic parameter identification. Additional extensions include HCLDMSPSO [36,37], combining CL for one subpopulation and a dynamic multi-swarm strategy (discussed in Section 5.1) for another, boosting diversity. Similarly, Cao et al. [38] integrated local search and CL in CLPSO-LS to improve PSO's performance. Maitra and Chatterjee [39] combined cooperative learning and CL in PSO for multi-level image segmentation. Gulcu and Kodaz [40] proposed a parallel CLPSO (PCLPSO) using a master-slave paradigm based on multi-swarm collaboration, enhancing performance

through parallelism. Zhong et al. [41] tailored CLPSO for the Traveling Salesman Problem in D-CLPSO by introducing a simulated annealing acceptance criterion, novel flight equations, and lazy velocity computation to boost efficiency and prevent premature convergence. The CL mechanism has also been implemented in JAYA (CLJAYA) [42,43] to optimize photovoltaic parameters.

Zhang et al. [44] proposed a local optima topology (CLPSO-LOT), wherein local optima identified during iterations form a new topology, expanding the search space and accelerating convergence. Yu et al. [45] introduced two enhancements in ECLPSO: a perturbation term in velocity updates activated by personal best bounds to improve exploitation and adaptive learning probabilities based on personal best rankings to enhance convergence. Tan et al. [46] proposed an adaptive CL bacterial foraging optimization (ALCBFO) algorithm with a time-varying chemotaxis step length and CL strategy. A non-linearly decreasing modulation model balanced exploration and exploitation, while the learning mechanism enhanced diversity and prevented premature convergence. Lin et al. [47] combined an adaptive mechanism for dynamic CL probability adjustment and a cooperative archive in ACLPSO-CA. This mechanism dynamically adjusted each particle's CL probability across three levels based on performance, while the archive provided additional promising information through swarm cooperation. These studies collectively highlight the versatility and adaptability of the CL mechanism across various optimization contexts, showcasing its potential for solving complex and interdisciplinary problems.

The introduction of the learning strategy in CLPSO, where a particle's velocity is updated either using its own $\mathbf{p}_{Best}$ position or the better one from two randomly selected particles, is an innovative idea. This process is guided by the learning probability factor $P_{r_{c,j}}$, which varies exponentially with the particle's index. However, the random selection of exemplars in the learning process may not always provide optimal guidance, as the exemplars chosen might lead the particle toward incorrect directions in the search space.

The challenge arises when the momentum gained by a particle from exemplars at different stages causes significant deviation, mainly if the exemplars are located in distant and unconnected regions of the search space. This lack of interconnection between the exemplars and the particle can reduce the algorithm's effectiveness [47,48]. Consequently, refining the learning strategy is essential to provide more reliable and directed guidance in the search process. The other limitation of the CL mechanism is finding the best results on unimodal problems [23]. To address the drawbacks of CLPSO, several enhanced strategies have been proposed, as discussed above. The dynamic neighborhood learning PSO (DNLPSPSO) [21] introduces a learning strategy (Eq. (31), where parameters w_g varies 0.9 to 0.4 and $c_{e1} = c_{e2} = 1.49445$ are selected) where a particle's velocity is updated using the historical best information of particles within a dynamic neighborhood, rather than the entire population. These neighborhoods are reformed periodically, allowing participants to learn from their neighborhood or historical best practices, thus enhancing guidance and maintaining diversity.

$$\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_{e1} \cdot \mathbf{rand}_i^d \cdot (\mathbf{p}_{Best_{f,i}}^d - \mathbf{x}_i^d) + c_{e2} \cdot \mathbf{rand}_i^d \cdot (\mathbf{g}_{Best}^d - \mathbf{x}_i^d). \quad (4)$$

Additionally, $P_{r_{c,j}}$ adopts a linear variation within the same range (Eq. (5)), with the intent of enhancing each particle's learning tendency compared to the exponential variation.

$$P_{r_{c,j}} = \alpha + \beta \cdot \left(\frac{10 \cdot (i-1)}{N_s - 1} \right). \quad (5)$$

Similarly, the heterogeneous CLPSO (HCLPSO) [23,36] divides the population into two heterogeneous subpopulations: the first subpopulation generates exemplars using $\mathbf{p}_{Best}$ experiences from within its subpopulation (Eq. (2) where w_g varies from 0.99 to 0.2 and c_e varies from 3 to 1.5), preserving diversity, while the second subpopulation uses $\mathbf{p}_{Best}$ experiences from the entire population to focus on global exploration (Eq. (31) uses parameters where w_g varies from 0.99 to 0.2,

c_{e1} varies from 2.5 to 0.5, and c_{e2} varies from 0.5 to 2.5.). The $P_{r_{c,j}}$ values for each particle in the population are calculated using Eq. (3) with $\alpha = 0$, and $\beta = 0.25$. This division ensures that diversity is retained even if one subpopulation converges prematurely. Furthermore, the parallel CLPSO (PCLPSO) [40] leverages a master-slave parallel paradigm with multiple swarms working cooperatively and concurrently, improving exploration and computational efficiency. To enhance local search capabilities, a Broyden-Fletcher-Goldfarb-Shanno (BFGS) based local search operator was incorporated into CLPSO [38], combining the global exploration strengths of CLPSO with the precision of BFGS optimization. A p_{Best} -guided CLPSO (HPBPSO) [49] has also been proposed, which organizes the population similarly to HCLPSO. The first subpopulation is updated using Eq. (31) with dynamic crossover learning, while the second subpopulation is further divided into two groups and updated using Eq. (6).

$$\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_{e1} \cdot \mathbf{rand}_i^d \cdot (\mathbf{p}_{Best_{f,i}}^d - \mathbf{x}_i^d) + c_{e2} \cdot \mathbf{rand}_i^d \cdot (\mathbf{g}_{Best}^d - \mathbf{x}_i^d) + (\mathbf{rand}_i^d)^\lambda \cdot (\mathbf{g}_{Best}^d - \mathbf{p}_{Best_i}^d). \quad (6)$$

These advancements collectively enhance CLPSO's robustness and adaptability for complex optimization problems by mitigating its inherent limitations and introducing innovative learning strategies.

3.2. Orthogonal learning

In the orthogonal learning PSO (OLPSO) [26], the orthogonal learning (OL) strategy combines the personal best position $\mathbf{p}_{Best}$ and the neighborhood best position $\mathbf{n}_{Best}$ to form a guidance vector $\mathbf{o}_{Best}$. The particle updates its velocity as:

$$\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_e \cdot \mathbf{rand}_i^d \cdot (\mathbf{o}_{Best}^d - \mathbf{x}_i^d), \quad (7)$$

where $c_e = 2.0$ is the acceleration coefficient, and $\mathbf{rand}_i^d$ is a random value in $[0, 1]$. The guidance vector $\mathbf{o}_{Best}$, constructed using the orthogonal experimental design (OED) method, is given by:

$$\mathbf{o}_{Best} = \mathbf{p}_{Best} \oplus \mathbf{n}_{Best}, \quad (8)$$

where $\oplus$ denotes the OED operation. Each element of $\mathbf{o}_{Best}$ is selected from $\mathbf{p}_{Best}$ or $\mathbf{n}_{Best}$ based on the OED process. This vector serves as a dynamic learning exemplar, guiding the particle's velocity and position updates while refining its personal best $\mathbf{p}_{Best}$.

To maintain stability, $\mathbf{o}_{Best}$ is reused for multiple generations until it ceases to improve the particle's exploration. If $\mathbf{p}_{Best}$ remains unchanged for t generations, a new $\mathbf{o}_{Best}$ is reconstructed. To incorporate dynamic updates, $\mathbf{o}_{Best}$ stores only the indices of $\mathbf{p}_{Best}$ and $\mathbf{n}_{Best}$, enabling immediate integration of new information when these positions improve.

The construction of $\mathbf{o}_{Best}$ begins with generating an orthogonal array (OA), $L_M(2^d)$, where L represents a Latin square, M is the number of test cases, and d is the problem's dimensionality. Each dimension is treated as a factor with two levels: $\mathbf{p}_{Best}$ or $\mathbf{n}_{Best}$. Using the OA, M candidate solutions $\mathbf{x}_j$ ($1 \leq j \leq M$) are generated. For each factor, if the OA value is 1, the dimension is selected from $\mathbf{p}_{Best}$; otherwise, it is chosen from $\mathbf{n}_{Best}$. Candidate solutions are evaluated for fitness, and the best solution, $best_{x_j}$, is recorded. Factor effects are analyzed to determine the optimal level for each factor. A predictive solution x_p is constructed and evaluated using these levels. The fitness values $f(best_{x_j})$ and $f(x_p)$ are compared, and the better solution's level combination is used to construct $\mathbf{o}_{Best}$.

This process ensures that $\mathbf{o}_{Best}$ efficiently combines the strengths of $\mathbf{p}_{Best}$ and $\mathbf{n}_{Best}$, balancing exploration and exploitation. By leveraging the OED method, OLPSO significantly enhances the particle swarm optimization framework's learning capability.

A novel PSO variant [50], integrating Lévy flight and OL, enhances exploitation and accelerates search efficiency. Lévy flight boosts exploration, while OL strengthens exploitation and enables dynamic,

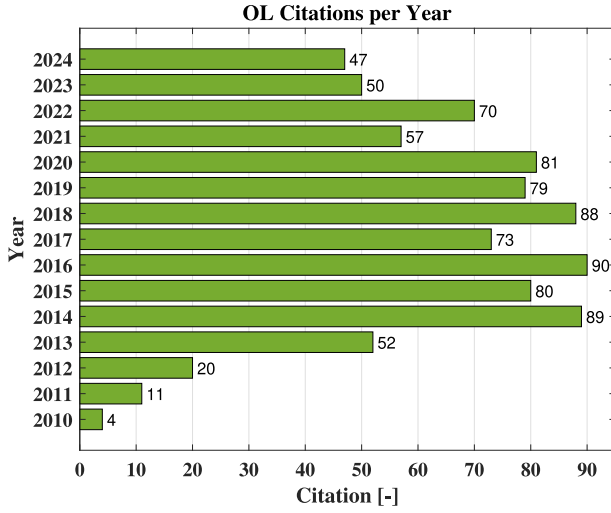


Fig. 3. OL citations per year based on the Google Scholar platform.

adaptive learning by incorporating the Lévy term based on flight probability. Parallel computation in OL further improves efficiency. Unlike PSO, this variant employs a novel strategy where particles learn from a single optimal experience, avoiding conflicts between historical and global best guides. Ho et al. [51] introduced an intelligent move mechanism (IMM) into OLPSO, utilizing OED to adjust particle velocities systematically. This divide-and-conquer approach addresses the curse of dimensionality, ensuring efficient particle movement. OLPSO has been applied to plant disease diagnosis by integrating it into a CNN with a decaying learning rate [52].

For dynamic optimization problems, OLPSO with a variable relocation strategy (VRS) [53] balances fast convergence and population diversity. VRS uses historical data for particle relocation, enabling quick adaptation to changing environments. Bai et al. [54] proposed an OLPSO integrated with DE, enhancing exploitation through deeper searches and preventing premature convergence with a stochastic star topology. OL has also improved optimization in various algorithms, including CSO [55], ABC [56,57], AEFA [58], DE [59,60], CS [61], AOA [62], and SCA [63]. These applications highlight OL's versatility in enhancing metaheuristic optimization. Fig. 3 validated its applicability in various years, which depicts the annual citation trends for OL from 2010 to 2024, based on data from Google Scholar. Citations began modestly in 2010 with only 4 citations and experienced a consistent rise, reaching a peak of 90 citations in 2016. After this peak, there was a slight decline, with citations dropping to 73 in 2018 before rebounding to 89 in 2019. The years following 2019 show a gradual decrease, with 81 citations in 2020, 70 in 2022, and 47 in 2024. This trend highlights the increasing recognition of OL in the early years, particularly between 2013 and 2016, followed by fluctuations in recent years. The decline in citations after 2020 may indicate either a shift in research focus or the stabilization of interest as the field matures.

3.3. Dimensional learning strategy

Inspired by CLPSO and OLPSO, Xu et al. [64] proposed a dimensional learning strategy (DLS) for PSO to effectively discover and preserve the promising information of the population's best solution and construct a learning exemplar for each particle. The learning exemplar is constructed by allowing each dimension of the $\mathbf{p}_{Best}$ to learn from the corresponding $\mathbf{g}_{Best}$ on a dimension-by-dimension basis to construct a learning exemplar $\mathbf{x}_{dl}$. This process ensures the excellent information of $\mathbf{g}_{Best}$ is propagated to $\mathbf{x}_{dl}$, thereby promoting the dissemination and preservation of the coding pattern of $\mathbf{g}_{Best}$. During the DLS process, DLS guarantees that $\mathbf{x}_{dl}$ is no worse than $\mathbf{p}_{Best}$.

The velocity of PSO has been changed by integrating the DLS, DLPSP, by substituting $\mathbf{p}_{Best}$ with $\mathbf{x}_{dl}$ in the PSO velocity update formula (Eq. (9)).

$$\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_{e1} \cdot \mathbf{rand}_i^d \cdot (\mathbf{x}_{dl_i}^d - \mathbf{x}_i^d) + c_{e2} \cdot \mathbf{rand}_i^d \cdot (\mathbf{g}_{Best}^d - \mathbf{x}_i^d). \quad (9)$$

Algorithm 3 Exemplar $\mathbf{x}_{dl}$.

```

1:  $\mathbf{x}_{dl} \leftarrow \mathbf{p}_{Best}$  {Initialize the learning exemplar with the personal best position}
2: for each dimension  $j$  do
3:    $\mathbf{x}_{lp}^j \leftarrow \mathbf{x}_{dl}^j$  {Copy the current learning exemplar}
4:   if  $\mathbf{x}_{lp}^j == \mathbf{g}_{Best}^j$  then
5:     continue {Skip if the current dimension already matches the global best}
6:   end if
7:    $\mathbf{x}_{lp}^j \leftarrow \mathbf{g}_{Best}^j$  {Update the current dimension with the global best value}
8:   if  $f_{\mathbf{x}_{lp}}^j < f_{\mathbf{x}_{dl}}^j$  then
9:      $\mathbf{x}_{dl}^j \leftarrow \mathbf{g}_{Best}^j$  {Accept the update if it improves fitness}
10:  end if
11: end for

```

As shown in Fig. 4, the temporary exemplar $\mathbf{x}_{dl}$ is compared to the current exemplar each time a dimension of the former is updated. If $\mathbf{x}_{dl}$ demonstrates better fitness than the current learning exemplar, the latter is replaced with $\mathbf{x}_{dl}$. Otherwise, the current learning exemplar remains unchanged and proceeds to the next dimension. This process ensures that the learning exemplar only incorporates dimensions of $\mathbf{g}_{Best}$ that enhance its fitness, guaranteeing no degradation in its quality. The detailed step-by-step procedure is outlined in Algorithm 3.

Building on the concepts of OLPSO and HCLPSO, the same paper [64] introduces a two-swarm learning PSO (TSLPSO) algorithm to balance exploration and exploitation. Like HCLPSO, the population is divided into two sub-populations: one sub-population employs the DLS, while the other adopts the CL mechanism.

Based on Google Scholar data, Fig. 5 presents the annual citation trends for the DLS from 2019 to 2024. Citations began modestly in 2019 with 8 citations, followed by a steady rise, peaking at 64 in 2022. After this peak, a slight decline was observed, with citations dropping to 36 in 2023 before rebounding to 39 in 2024. This trend highlights the growing recognition of DLS, particularly between 2020 and 2024, with some fluctuations in recent years.

To enhance optimization performance, DLS has been widely integrated into various algorithms. For instance, the DLS mechanism was incorporated into GWO [65], where three dominant wolves collaboratively construct an exemplar wolf through DLS to guide the swarm. A multi-strategy DLS approach was proposed to dynamically optimize population structure and enhance diversity during the solving process [66]. Additionally, a DLS-based squirrel search algorithm utilizing a roulette strategy was introduced [67], where dimensional learning guided squirrels' location updates on oak trees and accurately captured information from hickory trees, improving the efficiency of information flow transmission. In power systems, a DLPSP [68] was applied and tested on 13 classical benchmark problems, demonstrating its effectiveness in solving complex optimization tasks.

3.4. Adaptive learning strategy

Li et al. [69,70] proposed an adaptive learning (AL) for PSO, incorporating a variant learning strategy. Each particle learns from four sources: its own best position ($\mathbf{p}_{best}$), the closest neighbor's $\mathbf{p}_{Best_{nearest}}$, the global best ($\mathbf{g}_{best}$), and a random nearby position. This adaptive approach balances exploration and exploitation. The particle update

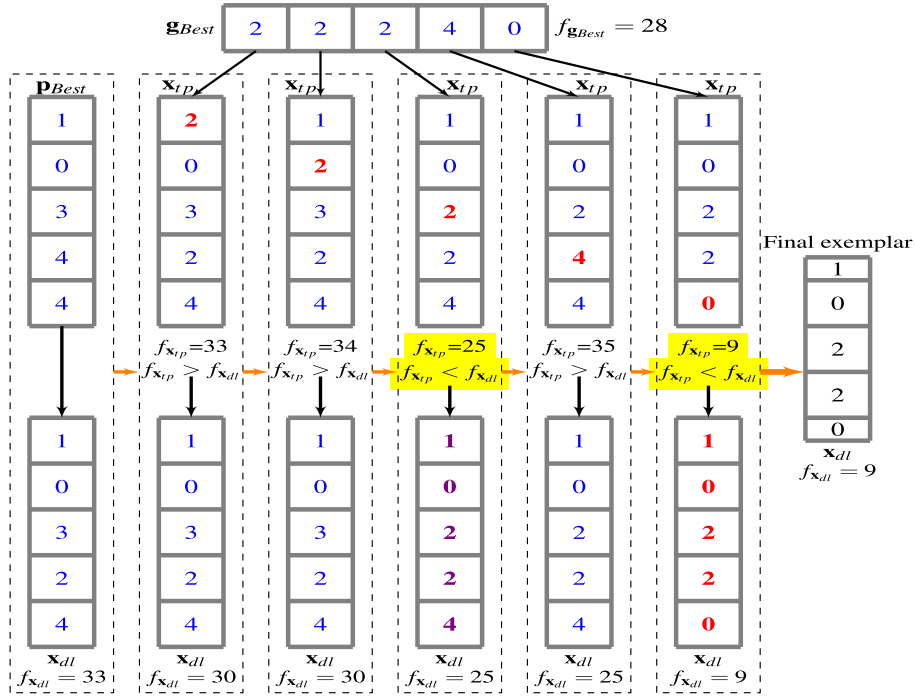
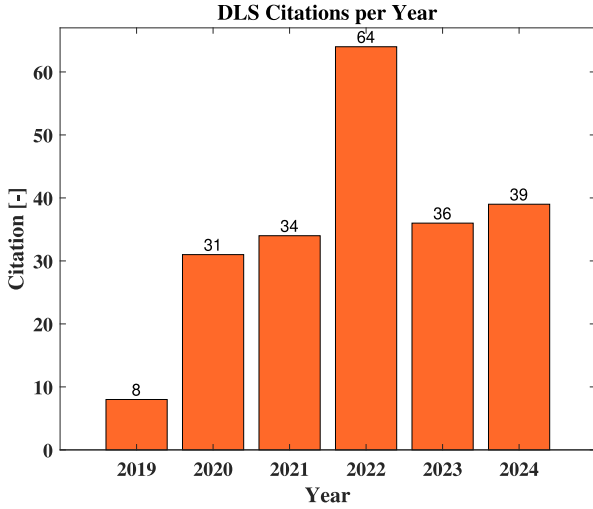
Fig. 4. An illustration of creating x_{dl} in DLS [64].

Fig. 5. DLS citations per year based on the Google Scholar platform.

rules are:

$$\begin{cases} \text{vel}_i^d = \omega_g \cdot \text{vel}_i^d + \eta \cdot r_i^d \cdot (p_{Best_i}^d - x_i^d), & (\text{self-learning}), \\ \text{vel}_i^d = \omega_g \cdot \text{vel}_i^d + \eta \cdot r_i^d \cdot (p_{Best_{nearest}}^d - x_i^d), & (\text{neighbor-learning}), \\ x_i^d = x_i^d + \text{vel}_{avg}^d \cdot N(0, 1), & (\text{random search}), \quad \text{vel}_i^d = \omega_g \cdot \text{vel}_i^d + \eta \cdot r_i^d \cdot (g_{Best}^d - x_i^d), & (\text{global-learning}), \end{cases}$$

where $p_{Best_{nearest}}$ is the best position of the closest particle, vel_{avg} is the average velocity, and $N(0, 1)$ is a standard normal random variable.

AL employs probability matching to dynamically adjust the selection ratios of the four learning strategies. Initially, each operator has an equal probability (1/4), updated based on performance. The progress value for operator i is:

$$\text{prog}_i = \sum_{j=1}^{M_i} f_{x_{ij}} - \min(f_{x_{ij}}, f_{x'_{ij}}), \quad (10)$$

where x_{ij} and $x'_{ij}(t)$ are the parent and child particles, respectively. M_i is the operator's selection times. The reward value is:

$$\text{reward}_i = \exp \left(\alpha_w \cdot \frac{\text{prog}_i}{\sum_{j=1}^{N_s} \text{prog}_j} + (1 - \alpha_w) \cdot \frac{s_i}{M_i} \right) + c_i \cdot p_i - 1, \quad (11)$$

where s_i counts successful offspring, p_i is the selection ratio, α_w is a weight factor, and c_i is a penalty term. The selection ratio (p) is updated every g generations:

$$p_i = (1 - N_s \cdot \gamma) \cdot \frac{\text{reward}_i}{\sum_{j=1}^{N_s} \text{reward}_j} + \gamma, \quad (12)$$

where γ is a minimum selection ratio set to 0.01 in experiments. This framework dynamically optimizes particle learning for improved search efficiency. The AL strategy is further used [24] to overcome the drawbacks of PSO by incorporating four learning strategies, such as CL, the difference-based velocity updating strategy (DbV), the estimation-based velocity updating strategy (EbV), and CL with p_{Best} .

Wang et al. [71] proposed a hybrid PSO with AL, combining self-learning for exploration and competitive learning for exploitation. A tolerance-based search direction adjustment mechanism (TSDM) is introduced to maintain balance. At each iteration, a particle's current position x_i is compared with its best position p_{Best_i} , updating p_{Best_i} if an improvement is found. If no particle improves across the swarm, the condition $\sum_{i=1}^{N_s} (f_{p_{Best_i}}^t - f_{p_{Best_i}}^{t-1}) = 0$ holds. To prevent premature convergence, TSDM introduces a tolerance counter T , initialized to zero and incremented by one ($T = T + 1$) when no improvements occur. As T increases, the likelihood of stagnation rises, prompting an adaptive search direction adjustment. The probability of adjustment, $\text{prob}_{adjust} = \frac{\exp(T)-1}{\exp(10)-1}$, is defined. This mechanism dynamically updates prob_{adjust} each iteration, aiding the swarm in escaping local optima. In competitive learning, particles learn from both g_{Best} and/or a *Candidate*, generated via Gaussian mutation by selecting two random particles, x_k and x_m .

3.5. Aging leader and challengers

Chen et al. [29] proposed the aging leader and challengers learning (ALC) for PSO (ALC-PSO), inspired by the natural phenomenon that

“aging weakens the leader of a colony, providing opportunities for others to challenge the leadership”. The central concept of ALC is to assign the swarm leader an age and a lifespan, enabling other particles to challenge the leadership when the leader ages. This aging mechanism is a dynamic strategy for promoting leadership transitions, ensuring that the most suitable leader guides the swarm.

The leader’s lifespan in ALC is adaptively tuned based on its leading power. Leaders with strong leadership power attract the swarm toward better positions and are granted longer lifespans. Conversely, when a leader fails to improve the swarm, it becomes aged and is replaced by challengers, which introduces diversity and helps the algorithm avoid stagnation and local optima. The influence of the leader (or challenger) on particle velocity is governed by:

$$\mathbf{vel}_i^d = \omega \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \text{rand}_1^d \cdot (\mathbf{p}_{Best_i}^d - \mathbf{x}_i^d) + c_{e_2} \cdot \text{rand}_2^d \cdot (\mathbf{L}/\mathbf{C}^d - \mathbf{x}_i^d), \quad (13)$$

where $\mathbf{L}/\mathbf{C}$ refers to either the leader ($\mathbf{Leader}_i = \{\text{leader}_i^1, \text{leader}_i^2, \dots, \text{leader}_i^d\}$) or a challenger ($\mathbf{Challenger}_i = \{\text{challenger}_i^1, \text{challenger}_i^2, \dots, \text{challenger}_i^d\}$), depending on their role in the swarm.

Chaudhary et al. [72] applied ALC-PSO to solve task scheduling problems, while Singh et al. used it to address power flow problems with FACTS devices [73–75] and optimal reactive power dispatch. Furthermore, the ALC mechanism has been integrated into differential evolution (DE) algorithms, as demonstrated in [76,77].

3.6. Elitist learning strategy

Lim and Isa [78] proposed an elitist learning strategy (ELS) for PSO, which incorporates two main components: the OED-based learning strategy (OEDLS), inspired by OLPSO [26,53,55], and the stochastic perturbation-based learning strategy (SPLS), which evolves $\mathbf{g}_{Best}$ when predefined conditions are satisfied. In OEDLS, each d th dimension is treated as a factor in a D -dimensional problem, resulting in D factors. Two levels ($Q = 2$) are defined: global best ($\mathbf{g}_{Best}$) as level 1 and improved personal best ($\mathbf{p}_{Best_i}$) as level 2. Using the $L_M(2^D)$ OA, M candidate solutions $\mathbf{x}_j$ ($1 \leq j \leq M$) are constructed. For each dimension, $\mathbf{g}_{Best}^d$ is selected if the OA level is 1; otherwise, the improved $\mathbf{p}_{Best_i}^d$ is chosen. Factor analysis (FA) is then applied to derive a predictive solution $\mathbf{x}_p$, whose fitness $f_{\mathbf{x}_p}$ is compared to $f_{\mathbf{g}_{Best}}$. If $f_{\mathbf{x}_p} < f_{\mathbf{g}_{Best}}$, $\mathbf{x}_p$ replaces $\mathbf{g}_{Best}$.

While OEDLS draws inspiration from OLPSO [26], it differs in its execution. OLPSO activates OL only when a particle’s exemplar fails to guide the search, whereas OEDLS is triggered as soon as fitness improvement is detected within the population. This immediate activation allows OEDLS to leverage information from non-global best particles more effectively, further enhancing $\mathbf{g}_{Best}$ ’s fitness. To prevent $\mathbf{g}_{Best}$ from stagnating in local optima, SPLS introduces a stochastic perturbation to $\mathbf{g}_{Best}$ if its fitness remains unchanged for m consecutive function evaluations (FEs). In SPLS, a randomly selected dimension of $\mathbf{g}_{Best}$, $\mathbf{g}_{Best}^d$, is perturbed using the following equation:

$$\mathbf{g}_{Best,pert}^d = \mathbf{g}_{Best}^d + \text{sgn}(\text{rand}_1) \cdot \text{rand}_2 \cdot (\mathbf{x}_{\max}^d - \mathbf{x}_{\min}^d), \quad (14)$$

where $\text{sgn}()$ is the sign function, $\text{rand}_1 \sim U[-1,1]$, and $\text{rand}_2 \sim N(0, R^2)$. Here, R represents the perturbation range, which decreases linearly with the number of FEs. If the fitness of the perturbed particle $\mathbf{g}_{Best,pert}$ is better than $\mathbf{g}_{Best}$, it replaces the latter, ensuring continued improvement. The same author uses the OEDLS mechanism in [79].

3.7. Example-based learning

An example-based learning (ExL) mechanism [80] for PSO (ExLPSO) is proposed to address the shortcomings of traditional PSOs by balancing convergence speed and particle diversity. The ExL mechanism is inspired by a common social phenomenon in which learning from multiple elite examples enhances individuals’ abilities more effectively

than learning from a single example or non-elite individuals. Based on this concept, a PSO utilizing multiple diverse $\mathbf{g}_{Best}$ values can outperform single- $\mathbf{g}_{Best}$ and non- $\mathbf{g}_{Best}$ PSOs in terms of both diversity and efficiency.

To implement this, a new velocity updating rule is introduced, similar to Eq. (31), but with updates to $\mathbf{g}_{Best}$ that allow learning from multiple diverse $\mathbf{g}_{Best}$ values. The velocity update is defined as:

$$\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \text{rand}_1^d \cdot (\mathbf{p}_{Best_{f_j^d}}^d - \mathbf{x}_i^d) + c_{e_2} \cdot \text{rand}_2^d \cdot (\mathbf{g}_{Best_{g_i^d}}^d - \mathbf{x}_i^d). \quad (15)$$

Unlike CLPSO, ExLPSO always selected $\mathbf{p}_{Best_{f_j^d}}^d$ with probability 1, instead of using a learning probability P_{rc} . Furthermore, in contrast to $\mathbf{g}_{Best}$ in PSO, $\mathbf{g}_{Best_{g_i^d}}^d$ in ExLPSO is not always the overall global best. Instead, it is randomly chosen from an example set $\mathbf{E}_s$ of size B_m , where g_i^d is a uniformly random integer from 1 to B_m for each d . The construction of the example set $\mathbf{E}_s$ is described in Algorithm 4.

Algorithm 4 Process of updating the example set

```

1:  $\mathbf{E}_s \leftarrow \emptyset$  {Initialize the example set}
2: if  $|\mathbf{E}_s| == 0$  then
3:   Add the current  $\mathbf{g}_{Best}$  to  $\mathbf{E}_s$  {Add the first global best to the example set}
4: else if  $0 < |\mathbf{E}_s| < B_m$  then
5:   for each  $g \in \mathbf{E}_s$  do
6:     if  $f_g > f_{\mathbf{g}_{Best}}$  then
7:       Add the current  $\mathbf{g}_{Best}$  to  $\mathbf{E}_s$  {Add  $\mathbf{g}_{Best}$  if space is available, and it's better than all examples}
8:     end if
9:   end for
10: else if  $|\mathbf{E}_s| == B_m$  then
11:   for each  $g \in \mathbf{E}_s$  do
12:     if  $f_g > f_{\mathbf{g}_{Best}}$  then
13:       Replace the oldest  $g \in \mathbf{E}_s$  with  $\mathbf{g}_{Best}$  {Replace the oldest example using FIFO order if  $\mathbf{E}_s$  is full}
14:     end if
15:   end for
16: end if
17: if  $\exists g \in \mathbf{E}_s$  such that  $f_g \leq f_{\mathbf{g}_{Best}}$  then
18:    $\mathbf{E}_s \leftarrow G$  {No changes made if a better or equal example already exists in  $\mathbf{E}_s$ }
19: end if

```

3.8. Fitness-distance ratio-based learning

Peram et al. [81] introduced a fitness-distance ratio (FDR) mechanism for PSO to address issues caused by crosstalk between particles. In FDR-PSO, each velocity dimension is updated by selecting only one particle at a time. The selected particle satisfies two criteria: (i) it must be near the particle being updated, and (ii) it should have visited a position with higher fitness.

For the d th dimension of the i th particle’s velocity, the “nearest best” particle, denoted as $\mathbf{n}_{Best}$, is chosen based on the best position of another particle ($\mathbf{p}_{Best_j}$), maximizing the FDR:

$$\mathbf{FDR}_j^d = \frac{f_{\mathbf{p}_{Best_j}^d} - f_{\mathbf{x}_i^d}}{|\mathbf{p}_{Best_j}^d - \mathbf{x}_i^d|}, \quad (16)$$

where $|\cdot|$ denotes the absolute value. The FDR-PSO velocity update is influenced by three factors: ($\mathbf{p}_{Best}$), ($\mathbf{g}_{Best}$), and ($\mathbf{n}_{Best}$). The velocity update equation is given by:

$$\mathbf{vel}_i^d = w_{g_0} \cdot \mathbf{vel}_i^d + w_{g_1} \cdot (\mathbf{p}_{Best_i}^d - \mathbf{x}_i^d) + w_{g_2} \cdot (\mathbf{g}_{Best}^d - \mathbf{x}_i^d) + w_{g_3} \cdot (\mathbf{n}_{Best}^d - \mathbf{x}_i^d). \quad (17)$$

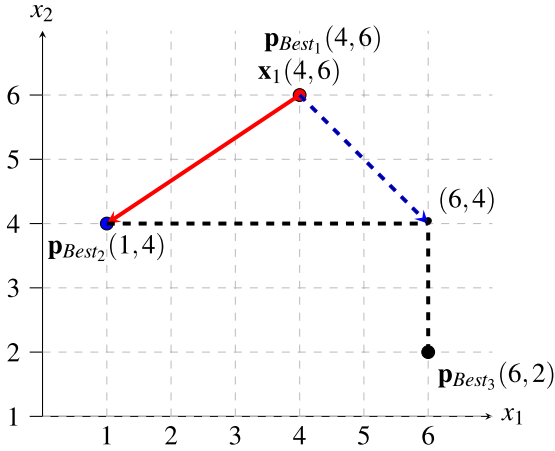


Fig. 6. An illustration of computing $\mathbf{n}_{Best}$ using FDR [83].

The coefficients $w_{g_0}, w_{g_1}, w_{g_2}$, and w_{g_3} are weights in the velocity update equation. Cleghorn and Engelbrecht [82] provided a theoretical algorithm analysis, offering insights into suitable parameter values. Xiaodong Li [83] highlighted key limitations in FDR calculation: reliance on a particle's frequently changing current position can cause unstable convergence, and using differences between other particles' personal bests and the current position often leads to suboptimal behavior.

As illustrated in Fig. 6, these limitations are evident in a 2D search space. Here, $\mathbf{n}_{Best}$ for $\mathbf{x}_1$ is calculated using personal bests $\mathbf{p}_{Best_1}$ and $\mathbf{p}_{Best_3}$. On dimension 1: $\text{FDR}_2^1 = \frac{1}{|1-4|} = \frac{1}{3}$, $\text{FDR}_3^1 = \frac{1}{|6-4|} = \frac{1}{2}$. On dimension 2: $\text{FDR}_2^2 = \frac{1}{|4-6|} = \frac{1}{2}$, $\text{FDR}_3^2 = \frac{1}{|2-6|} = \frac{1}{4}$. Maximizing the FDR values yields $\mathbf{n}_{Best}(6, 4)$, which is not an ideal attraction point. Despite $\mathbf{p}_{Best_2}$ and $\mathbf{p}_{Best_3}$ being equally fit, $\mathbf{x}_1$ converges to $\mathbf{n}_{Best}(6, 4)$ instead of either promising point.

To address this issue, Li [83] introduced the fitness-Euclidean distance ratio (FER), defined as:

$$\text{FER}_j^d = \alpha_{fer} \cdot \frac{f_{\mathbf{p}_{Best_j}^d} - f_{\mathbf{p}_{Best_i}^d}}{\|\mathbf{p}_{Best_j}^d - \mathbf{p}_{Best_i}^d\|},$$

where α_{fer} is a scaling factor, and $\|\cdot\|$ represents Euclidean distance. The particle velocity update follows Eq. (17) but excludes the global best term or FIPS [30] procedure. The FER mechanism is further used in DE [84] and ABC [85].

3.9. Scatter learning

Ren et al. [86] proposed a scatter learning (ScL) mechanism for PSO, which introduces an exemplar pool (EP) containing high-quality, diverse solutions scattered across the search space. Particles select their exemplars from EP using the roulette wheel rule. The EP is defined as:

$$\text{EP} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m \mid f_{\mathbf{x}_1} \geq f_{\mathbf{x}_2} \geq \dots \geq f_{\mathbf{x}_m}; \forall \mathbf{x}_i, \mathbf{x}_j \neq i, (\mathbf{x}_i, \mathbf{x}_j) \geq \gamma\}, \quad (18)$$

where m is the cardinality of EP, $(\cdot, \cdot)$ denotes the distance between solutions, and γ ensures a minimum distance to maintain diversity. Solutions are added to EP based on fitness and spatial separation: $\mathbf{x}_1$ is the best solution, $\mathbf{x}_2$ is the second-best satisfying $(\mathbf{x}_1, \mathbf{x}_2) \geq \gamma$, and so on. Any solution $\mathbf{x}'$ violating $(\mathbf{x}_1, \mathbf{x}') < \gamma$ is excluded. This ensures EP includes diverse and high-quality exemplars.

Particles in ScL employ one exemplar at a time to avoid oscillations. The velocity update rule is given by:

$$\mathbf{vel}_i^d = \omega_g \cdot \mathbf{vel}_i^d + c \cdot \text{rand}^d \cdot (\mathbf{x}_{e_i}^d - \mathbf{x}_i^d), \quad (19)$$

where $\mathbf{x}_{e_i}^d$ is the d th element of the current exemplar employed by particle i . Each exemplar in EP is exploited based on a selection probability,

ensuring better exemplars have a greater influence on particle behavior while maintaining diverse search patterns.

3.10. Informed-learning

Clerc and Kennedy [28] simplified the PSO velocity update equation, i.e., Eq. (1), by normalizing the personal and global best positions ($\mathbf{p}_{Best}$ and $\mathbf{g}_{Best}$), yielding:

$$\mathbf{vel}_i^d = \lambda \cdot (\mathbf{vel}_i^d + c'_e \cdot (\overline{\mathbf{p}_{Best_i}^d} - \mathbf{x}_i^d)), \quad (20)$$

where $c'_e = c_{e1} + c_{e2}$, λ is the constriction weight, and $\overline{\mathbf{p}_{Best}}$ is the normalized best position given as: $\overline{\mathbf{p}_{Best_i}^d} = \frac{c_{e1} \cdot \mathbf{p}_{Best_i}^d + c_{e2} \cdot \mathbf{g}_{Best}^d}{c_{e1} + c_{e2}}$. This method eliminates the need to balance velocity adjustments between two terms, ensuring proper convergence behavior.

Mendes et al. [30] introduced an alternative approach by incorporating a fully informed learning strategy for PSO (FIPS), where each particle utilizes information from all neighboring particles instead of a single best neighbor. The updated $\overline{\mathbf{p}_{Best}}$ is computed as:

$$\overline{\mathbf{p}_{Best_i}} = \frac{\sum_{j=1}^{n_g} \psi_k \cdot c'_{e_k} \cdot \mathbf{p}_{Best_j}}{\sum_{j=1}^{n_g} \psi_k \cdot c'_{e_k}}, \quad c'_{e_k} = U[0, c'_{e_{max}}/n_g], \quad (21)$$

where $\mathbf{p}_{Best_j}$ represents the best position of neighborhood particle j , and n_g is the number of neighboring particles. FIPS has been tested with five PSO topologies, *all*, *ring*, *four clusters*, *pyramid*, and *square* [32], demonstrating improved performance in global optimization by enhancing convergence speed and accuracy.

However, FIPS struggles in multimodal optimization since its topology-based neighborhood selection may cause particles to converge toward a single peak instead of exploring multiple optima. For instance, in a ring topology, particles interact only with adjacent indices, making it likely that neighbors belong to different niches, thereby reducing the effectiveness of peak identification [31].

To address this limitation, Qu et al. [31] proposed locally informed learning for PSO (LIPS), which, like FIPS, incorporates neighbor information but selects neighbors based on Euclidean distance rather than fixed topology. The $\overline{\mathbf{p}_{Best}}$ in LIPS is computed as:

$$\overline{\mathbf{p}_{Best_i}} = \frac{\sum_{j=1}^{n_g} \psi_k \cdot \mathbf{n}_{Best_j} / n_g}{\sum_{j=1}^{n_g} \psi_k}, \quad (22)$$

where ψ_k is a random number in the range $[0, 4.1/n_g]$. Unlike topology-based PSO, LIPS enhances exploration by adapting to the spatial distribution of particles while preserving niche integrity during exploitation. This allows LIPS to effectively locate multiple peaks with high accuracy, making it well-suited for multimodal optimization [31].

3.11. Other learning strategies

Li et al. [87] proposed an RL-based PSO incorporating a dynamic oscillation inertia weight for adaptive adjustment, a reinforcement learning-based velocity update for exploration-exploitation balance, cosine similarity-based velocity control for improved solution selection, and a local update strategy to enhance diversity and mitigate premature convergence. Qiu et al. [88] developed a Q-learning-based PSO, where Q-learning adaptively selects the optimal multi-exemplar strategy to maximize long-term rewards. An elite learning strategy improves local search and balances exploration and exploitation. Hein et al. [89] introduced a fuzzy PSO with RL, constructing fuzzy RL policies by training on simulated world models derived from past system transitions. This approach integrates self-organizing fuzzy controllers with model-based batch RL. It is suitable for domains where online learning is impractical, system dynamics are modelable, and interpretable policies are required. Multi-strategy RL has been incorporated into PSO in [90,91].

Tang et al. [92] proposed a feedback learning mechanism (FLM) for PSO, where learning probabilities are dynamically adjusted based

on a fitness feedback mechanism. The acceleration coefficients (c_{e_1} and c_{e_2}) are determined by fitness feedback and generation number, enhancing adaptability. To refine the solution further, elite stochastic disturbances are introduced to the globally best particle under adaptive probabilities, ensuring controlled exploration. The learning mechanism focuses on a single dimension of the global best particle, targeting specific refinements while maintaining computational efficiency. Gong et al. [93] proposed a genetic learning scheme for PSO (GL-PSO), integrating genetic operators to construct exemplars. The crossover operator leverages historical particle information ($\mathbf{p}_{Best}$ and $\mathbf{g}_{Best}$) to generate high-quality offspring, while mutation enhances global exploration by injecting diversity. The selection operator ensures directional evolution across generations, leading to improved search performance.

Sabat et al. [94] proposed an integrated learning (IL) approach for PSO to optimize complex multimodal functions. A particle updating strategy based on the hyperspherical coordinate system was introduced, effectively addressing evenly distributed multiple minima. This approach was combined with a CL strategy to enhance exploration. Niu et al. [95] introduced a center learning strategy, selecting the center position or the best-found position within each swarm based on learning probability. Dong et al. [96] proposed an autonomous learning adaptation (ALA) method for PSO, where each particle self-adjusts its control parameters based on past successes and failures. For successful movements, parameters increase in the forward direction while others decrease, and the opposite adjustments are applied for unsuccessful movements.

Chen et al. [97,98] developed a biogeography-based learning strategy (BLS) for PSO, where each particle updates using a combination of its own $\mathbf{p}_{Best}$ and the $\mathbf{p}_{Best}$ of others via biogeography migration. BLS generates an exemplar vector index $\mathbf{f}_i = [f_i^1, f_i^2, \dots, f_i^d]$, similar to the CL mechanism, based on migration and particle ranks. To counter stagnation, Dong et al. [99] introduced a reverse learning strategy (RLS) for PSO, where, upon $\mathbf{g}_{Best}$ remaining unchanged for a predefined number of iterations, particle velocities are updated as:

$$\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \text{rand}_1 \cdot (\mathbf{x}_i^d - \mathbf{x}_{worst_i}^d) + c_{e_2} \cdot \text{rand}_2 \cdot (\mathbf{x}_k^d - \mathbf{x}_i^d), \quad (23)$$

where $\mathbf{x}_{worst_i}$ represents the historically worst position of particle i , and $\mathbf{x}_k$ is a randomly selected particle from a low-fitness niche. Tanweer et al. [100] introduced two learning strategies: a self-regulating inertia weight for the best particle to enhance exploration and a self-perception mechanism for others to improve exploitation. Inspired by bee-foraging behavior, Chen et al. [101] proposed a bee-foraging learning PSO, incorporating three learning phases similar to those in the artificial bee colony (ABC) algorithm.

The single-swarm learning strategies discussed in this section and brief overview in Table 1 serve as foundational approaches for enhancing standard PSO. While they offer notable improvements in convergence behavior and solution quality, they also exhibit certain limitations, particularly in maintaining population diversity over time and avoiding premature convergence in complex or multimodal landscapes. These challenges have motivated the development of two-swarm and multi-swarm learning strategies, which we discuss in the next sections. In comparison to the upcoming paradigms, single-swarm methods provide simpler coordination and lower computational overhead but may lack the adaptive diversity mechanisms that arise from swarm interactions in more advanced frameworks. The insights gained from single-swarm improvements often serve as building blocks for more sophisticated learning designs involving inter-swarm cooperation and parallel evolution. Next, we transition to two-swarm strategies that introduce explicit collaborative learning between distinct swarms to further enhance robustness and adaptability.

4. Two swarm-based learning

This section presents learning strategies developed within a two-swarm (dual-population) framework for PSO. In contrast to the single-swarm strategies discussed in Section 3, where all particles operate

under a unified learning mechanism, two-swarm approaches divide the population into two distinct subpopulations, each potentially governed by different learning behaviors, parameter settings, or adaptation rules. This structural separation enhances algorithmic flexibility, allowing one swarm to prioritize global exploration while the other concentrates on local exploitation, thereby improving the swarm's capability to escape local optima and refine convergence near promising regions.

The interactions between the two swarms can vary widely. Cooperative schemes encourage periodic information exchange to maintain diversity and stimulate mutual improvement, while competitive schemes promote independent evolution, where one swarm may influence the other through dominance, replacement, or feedback mechanisms. Additionally, hybrid strategies may embed learning paradigms such as reinforcement learning, peer teaching, or adaptive exemplar transfer to further enrich the learning dynamics. Compared to single-swarm methods, the two-swarm design introduces a moderate level of structural complexity while retaining manageable coordination overhead.

In the following subsections, we systematically examine representative two-swarm learning strategies, highlighting their interaction models, adaptive behaviors, and comparative advantages in complex, multimodal, or dynamic optimization environments.

4.1. Social-learning strategy

Inspired by the social learning mechanism [103,104], Cheng and Jin [105] proposed a social learning-based PSO (SLPSO), where followers (particles with lower fitness values) learn from exhibitors (particles with higher fitness values). The position of a follower is updated as:

$$\mathbf{x}_i^d = \begin{cases} \mathbf{x}_i^d + \mathbf{vel}_i^d, & \text{if } \text{rand}_i \leq p_{r_i}^L, \\ \mathbf{x}_i^d, & \text{otherwise,} \end{cases} \quad (24)$$

where $p_{r_i}^L$ is the learning probability inspired by natural social learning. If the random value rand_i satisfies $0 \leq \text{rand}_i \leq p_{r_i}^L$, the follower learns from exhibitors, and its velocity is updated as:

$$\mathbf{vel}_i^d = \text{rand}_1^d \cdot \mathbf{vel}_i^d + \text{rand}_2^d \cdot (\mathbf{x}_k^d - \mathbf{x}_i^d) + \text{rand}_3^d \cdot \epsilon \cdot (\bar{\mathbf{x}}^d - \mathbf{x}_i^d), \quad (25)$$

where $\text{rand}_1^d, \text{rand}_2^d, \text{rand}_3^d$ are random values, ϵ is a social influence factor, $\mathbf{x}_k^d$ is the d th dimension of the exhibitor, and $\bar{\mathbf{x}}^d$ is the mean position of all particles.

To update the velocity of each follower, the population is first sorted by their fitness values. Each follower i learns from a randomly selected exhibitor k , where $i < k \leq N_s$. For a population of N_s particles sorted in ascending order of their fitness values as $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{N_s}$, the learning hierarchy is defined as follows: $\mathbf{x}_1$ (the worst particle) considers $\mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_{N_s}$ as its potential exhibitors. In contrast, $\mathbf{x}_{N_s-1}$ learns only from $\mathbf{x}_{N_s}$ (the best particle). The best particle $\mathbf{x}_{N_s}$ has no exhibitor and is, therefore, not updated. Conversely, the worst particle $\mathbf{x}_1$ cannot be an exhibitor for any other particle due to its low fitness. The SL strategy has been hybridized with exemplar learning and dynamic mutation [106]. Exemplar learning replaces the follower and social influence components of SLPSO, while a dynamic differential mutation strategy using three $\mathbf{p}_{Best}$ enhances exploration. Yu et al. [107] proposed a generation-based optimal restart strategy using radial basis functions for surrogate-assisted SLPSO. Zhao et al. [108] developed a fuzzy classification-based SLPSO to improve rule initialization and fitness function evaluation. A three-learning PSO [109] is proposed to overcome the limitations of SLPSO, such as poor searchability and low search efficiency, by intelligently incorporating three learning strategies. Additionally, the SL strategy was extended by dividing the population into two sub-groups, losers and winners, leading to the competitive swarm optimizer [110], where losers learn from winners rather than followers learn from exhibitors.

Table 1

Single population-based learning strategies for PSO.

Long name	Sort name	Year	Ref.	Comments
Comprehensive learning PSO	CLPSO	2006	[22]	CLPSO is a PSO variant where particles are not guided by $\mathbf{g}_{Best}$ but instead choose between their $\mathbf{p}_{Best}$ or a neighbor's $\mathbf{p}_{Best}$ with a probability P_r , introducing stochasticity that enhances diversity but slows convergence. The CLPSO boundary handling method is retained despite concerns about its effectiveness. Control parameters are set as $c_e = 1.49445$, ω linearly decreases from 0.9 to 0.2, and $R_g = 5$.
Orthogonal learning PSO	OLPSO	2009	[26]	The OL strategy combines the personal best position $\mathbf{p}_{Best}$ and the neighborhood best position $\mathbf{n}_{Best}$ to form a guidance vector $\mathbf{o}_{Best}$. The particle updates its velocity as: $\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_e \cdot \text{rand}_1^d \cdot (\mathbf{o}_{Best}^d - \mathbf{x}_i^d)$. The guidance vector $\mathbf{o}_{Best}$, constructed using the OED method, is given by: $\mathbf{o}_{Best} = \mathbf{p}_{Best} \oplus \mathbf{n}_{Best}$.
Dimensional learning strategy PSO	DLPSO	2019	[64]	The DLPSO effectively discovers and preserves the promising information of the population's best solution and constructs a learning exemplar for each particle. The learning exemplar is constructed by allowing each dimension of the $\mathbf{p}_{Best}$ to learn from the corresponding $\mathbf{g}_{Best}$ on a dimension-by-dimension basis to construct a learning exemplar $\mathbf{x}_{dl}$, as: $\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \text{rand}_1^d \cdot (\mathbf{x}_{dl}^d - \mathbf{x}_i^d) + c_{e_2} \cdot \text{rand}_2^d \cdot (\mathbf{g}_{Best}^d - \mathbf{x}_i^d)$. This process ensures that the learning exemplar only incorporates dimensions of $\mathbf{g}_{Best}$ that enhance its fitness, guaranteeing no degradation in its quality.
Adaptive learning PSO	ALPSO	2009	[69,70]	In ALPSO, each particle learns from four sources: its own best position ($\mathbf{p}_{best}$), the closest neighbor's $\mathbf{p}_{Best_{nearest}}$, the global best ($\mathbf{g}_{best}$), and a random nearby position. This adaptive approach balances exploration and exploitation. ALPSO employs probability matching to dynamically adjust the selection ratios of the four learning strategies.
Aging leader and challengers learning PSO	ALCPSO	2012	[29]	In this variant, a leader particle replaces $\mathbf{g}_{Best}$ for velocity updates. Initially, $\mathbf{g}_{Best}$ serves as the leader, but it ages over time and may be replaced by a competing particle to prevent stagnation in local optima. The central concept of ALC is to assign the swarm leader an age and a lifespan, enabling other particles to challenge the leadership when the leader ages. This aging mechanism is a dynamic strategy for promoting leadership transitions, ensuring that the most suitable leader guides the swarm, as: $\mathbf{vel}_i^d = \omega \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \text{rand}_1^d \cdot (\mathbf{p}_{Best}^d - \mathbf{x}_i^d) + c_{e_2} \cdot \text{rand}_2^d \cdot (\mathbf{L}/C^d - \mathbf{x}_i^d)$. Velocities are initially set to zero and restricted to 50% of the search bounds. A rebounding technique is applied as in [102]. Control parameters are $c_{e_1} = c_{e_2} = 2$, $\omega = 0.4$, initial leader lifespan = 60, $T = 2$, and $pro = 1/d$.
Elitist learning strategy	ELPSO	2014	[78]	EL combines an OED-based learning strategy, inspired by OLPSO [26,53,55], and SPLS, which adapts $\mathbf{g}_{Best}$ under predefined conditions. Unlike OLPSO, OL is triggered only when an exemplar fails, OED activates immediately upon fitness improvement, utilizing non-global best particles to enhance $\mathbf{g}_{Best}$. To prevent stagnation, SPLS applies a stochastic perturbation to $\mathbf{g}_{Best}$ if its fitness remains unchanged for m consecutive FEs. A randomly selected dimension $\mathbf{g}_{Best}^d$ is perturbed as: $\mathbf{g}_{Best,pert}^d = \mathbf{g}_{Best}^d + \text{sgn}(\text{rand}_1) \cdot \text{rand}_2 \cdot (\mathbf{x}_{max}^d - \mathbf{x}_{min}^d)$.
Example-based learning PSO	ExLPSO	2012	[80]	The ExL mechanism mimics the social phenomenon where learning from multiple elite examples enhances performance more effectively than relying on a single or non-elite individual. To achieve this, a velocity update rule is introduced, modifying $\mathbf{g}_{Best}$ to incorporate multiple diverse exemplars. The velocity update is given by: $\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \text{rand}_1^d \cdot (\mathbf{p}_{Best,i}^d - \mathbf{x}_i^d) + c_{e_2} \cdot \text{rand}_2^d \cdot (\mathbf{g}_{Best,g_i}^d - \mathbf{x}_i^d)$. This formulation enables particles to learn from multiple elite sources, promoting diverse exploration and improved optimization.

(continued on next page)

Table 1 (continued).

Fitness-distance ratio-learning based PSO	FDR-PSO	2003	[81]	In FDR-PSO, each velocity dimension is updated by selecting a single nearby particle that has visited a higher-fitness position. The velocity update is influenced by three factors: ($\mathbf{p}_{Best}$), ($\mathbf{g}_{Best}$), and ($\mathbf{n}_{Best}$). Xiaodong Li [83] identified key limitations in FDR calculation, including instability due to reliance on frequently changing positions and the potential for suboptimal behavior when using differences between other particles' personal bests and the current position.
Scatter learning PSO	ScLPSO	2013	[86]	ScL introduces an exemplar pool (EP) with high-quality, diverse solutions distributed across the search space. Particles select exemplars using a roulette wheel rule and employ one at a time to prevent oscillations. The velocity update rule is: $\mathbf{vel}_i^d = \omega_g \cdot \mathbf{vel}_i^d + c \cdot \mathbf{rand}^d \cdot (\mathbf{x}_{e_i}^d - \mathbf{x}_i^d)$.
Q-learning PSO	QLPSO	2023	[88]	Q-learning to optimize multi-exemplar selection for long-term rewards, integrating elite learning for exploration-exploitation balance.
Feedback learning PSO	FLPSO	2011	[92]	The acceleration coefficients (c_{e_1} and c_{e_2}) adapt based on fitness feedback and generation number. Elite stochastic disturbances refine the global best particle with adaptive probabilities, focusing on one dimension for targeted improvement while maintaining efficiency.
Genetic learning PSO	GLPSO	2015	[93]	GLPSO combines PSO with a composite exemplar ($\mathbf{p}_{Best}$, $\mathbf{g}_{Best}$) and genetic operators (crossover, mutation) running in parallel. Crossover exploits historical data for high-quality offspring, mutation enhances diversity, and selection ensures directional evolution. Velocity is capped at 20% of the bounds span, with randomized initial values. A rebounding technique is applied [93]. Control parameters follow [93]: $c_{e_1} = 1.49618$, $\omega_g = 0.7298$, $P_m = 0.01$, and a stopping gap of 7, after which tournament selection (20% of the population) is used.
Integrated learning PSO	ILPSO	2011	[94]	A hyperspherical coordinate-based particle update strategy was introduced to efficiently handle evenly distributed multiple minima. It was further integrated with a CL strategy to improve exploration.
Autonomous learning PSO	ALPSO	2014	[96]	Each particle, acting as an intelligent agent, adapts its parameters based on past successes and failures. Successful movements reinforce forward-directed parameters while reducing others, whereas unsuccessful movements trigger the opposite adjustments.
Biogeography-based learning PSO	BLSPSO	2017	[97,98]	Each particle updates using its own $\mathbf{p}_{Best}$ and those of others via biogeography migration. BLS constructs an exemplar vector $f_i = [f_i^1, f_i^2, \dots, f_i^d]$ based on migration and particle ranks, similar to CL.
Reverse learning PSO	RLSPSO	2018	[99]	When $\mathbf{g}_{Best}$ stagnates for a predefined number of iterations, a RLS for PSO is proposed, updating particle velocity as:
Self-regulating PSO	SRPSO	2015	[100]	$\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \mathbf{rand}_1 \cdot (\mathbf{x}_i^d - \mathbf{x}_{worst_i}^d) + c_{e_2} \cdot \mathbf{rand}_2 \cdot (\mathbf{x}_k^d - \mathbf{x}_i^d)$. Two learning strategies for PSO: a self-regulating inertia weight for the best particle to enhance exploration and a self-perception mechanism for others to improve exploitation.
Bee-foraging learning PSO	BFLPSO	2021	[101]	Inspired by bee-foraging behavior, Chen et al. [101] proposed a bee-foraging learning PSO, incorporating three learning phases similar to those in ABC.
Informed-learning PSO	FIPS	2004	[30]	Provide an alternate way to calculate $\overline{\mathbf{p}_{Best}}$ by normalized $\mathbf{p}_{Best}$ and $\mathbf{g}_{Best}$. FIPS has been tested with five PSO topologies, <i>all</i> , <i>ring</i> , <i>four clusters</i> , <i>pyramid</i> , and <i>square</i> [32], demonstrating improved performance in global optimization by enhancing convergence speed and accuracy.

4.2. Interswarm interactive learning

Inspired by social learning behavior, an interswarm interactive learning (IIL) strategy [111] is proposed for PSO. In PSO, the fitness value of $\mathbf{g}_{Best}$ is monitored, and if it remains unchanged for k consecutive iterations, i.e.,

$$f_{\mathbf{g}_{Best}}^t - f_{\mathbf{g}_{Best}}^{t-k} = 0. \quad (26)$$

The algorithm is assumed to be trapped in a local optimum, triggering the IIL strategy. Two swarms are then formed similarly to SL: one retains its original learning strategy (learned swarm), while the other (learning swarm) adapts. The softmax function determines the probability of a swarm being selected as the learned swarm.

Each particle in the learning swarm follows a learning probability P_{c_l} , defined as:

$$P_{c_l} = 0.1 + 0.5 \cdot \left(\frac{R_i}{N_s} \right)^5, \quad (27)$$

where R_i is the rank of the i th particle based on fitness, and N_s is the swarm size. The learned swarm follows the standard PSO update rule, whereas particles in the learning swarm switch between two learning modes: (1) retaining their original search strategy or (2) learning from the learned swarm. Assuming $Swarm_1$ is the learning swarm and $Swarm_2$ is the learned swarm, the velocity update follows:

$$\mathbf{vel}_{i,1}^d = \omega_g \cdot \mathbf{vel}_{i,1}^d + \begin{cases} c_{e1}' \cdot r_1 (\mathbf{p}_{Best_{i,1}}^d - \mathbf{x}_{i,1}^d) \\ + c_{e2}' \cdot r_2 (\mathbf{p}_{nBest_{i,1}}^d - \mathbf{x}_{i,1}^d) + c_{e3} \cdot r_3 (\mathbf{p}_{gBest_{i,2}}^d - \mathbf{x}_{i,1}^d), & \text{if } r_4 \leq P_{c1}, \\ c_{e1} \cdot r_5 (\mathbf{p}_{Best_{i,1}}^d - \mathbf{x}_{i,1}^d) \\ - \mathbf{x}_{i,1}^d + c_{e2} \cdot r_6 (\mathbf{p}_{nBest_{i,1}}^d - \mathbf{x}_{i,1}^d), & \text{otherwise.} \end{cases} \quad (28)$$

here, c_{e1}', c_{e2}' differ from c_1, c_2 , and c_3 controls the influence of interswarm learning, r_j , $j = 1, 2, \dots, 5$ are randomly generated numbers. This IIL strategy enhances diversity and improves the probability of escaping local optima.

4.3. Crisscross and stochastic example learning

To enhance the performance of PSO and overcome its limitations, Liang et al. [112] introduced a stochastic example learning strategy (SEL) integrated with crisscross learning. In this approach, the population is divided into two sub-populations with a ratio λ_r : an elite sub-swarm and a following sub-swarm. The elite sub-swarm guides the learning process of the following sub-swarm while simultaneously exploring information from other particles to enhance diversity.

The learning strategy for the elite sub-swarm is based on crisscross learning [113], where the personal bests of different particles are used as horizontal and vertical crossovers. This mechanism allows the elite sub-swarm to leverage its own and other particles' experiences, improving exploration and diversity. For the following sub-swarm, SEL defines the velocity updating rule as:

$$\mathbf{vel}_i^d = \omega_g \cdot \mathbf{vel}_i^d + c_1 \cdot rand^d \cdot (rand_2 \cdot \mathbf{p}_{Best_i}^d + (1 - rand_2) \cdot \mathbf{p}_{Best_k}^d - \mathbf{x}_i^d), \quad (29)$$

where k is randomly selected from an example pool $[1, i-1]$ of superior particles. The SEL strategy enables particles in the following sub-swarm to learn from superior individuals, improving diversity and enhancing their ability to exploit promising regions. This approach reduces the risk of premature convergence, fosters population diversity, and accelerates convergence.

4.4. Teaching and peer-learning

A teaching and peer-learning (TPL) strategy is introduced in PSO [114], consisting of two phases: the teaching phase and the peer-learning phase. In the teaching phase, a particle updates its velocity following the standard PSO rule (Eq. (11)). If the particle fails to improve its fitness in this phase, it enters the peer-learning phase (inspired by TLBO [115]), where it selects an exemplar particle ($\mathbf{p}_{Best_e}$) from the personal best positions of peer particles ($\mathbf{p}_{Best}$), excluding its own ($\mathbf{p}_{Best_i}$) and the global best particle ($\mathbf{g}_{Best}$). This dual-phase strategy enhances adaptability, improving the balance between exploration and exploitation. To improve the $\mathbf{g}_{Best}$ by fully exploiting the valuable information of each particle, Lim and Isa [79] used the ODELS mechanism in TPL for PSO. Many recent PSO variants lack alternative learning strategies when particles fail to improve fitness. To address this, a teaching-learning-based optimization (TLBO) framework was adapted into PSO, resulting in the teaching and peer-learning PSO (TPLPSO) algorithm.

4.5. Neighbor-based learning

Cao et al. [116] proposed a neighbor-based learning PSO, where each particle (target particle) learns from two swarm members: a

randomly selected neighbor ($\mathbf{x}_k$) and the global best particle ($\mathbf{g}_{Best}$), as:

$$\mathbf{vel}_i^d = \begin{cases} \omega_g \cdot \mathbf{vel}_i^d + F_{DE} \cdot (\mathbf{x}_k^d - \mathbf{x}_i^d) + c_e \cdot rand \cdot (\mathbf{g}_{Best}^d - \mathbf{x}_i^d), & \text{if } rand < L_{pr}, \\ \mathbf{v}_i^d, & \text{otherwise,} \end{cases} \quad (30)$$

here, F_{DE} is a scaling factor from DE and L_{pr} is the learning probability. Nasir et al. [21] proposed DNLPSO introduces a learning strategy (Eq. (31)), where parameters w_g varies 0.9 to 0.4 and $c_{e1} = c_{e2} = 1.49445$ are selected) where a particle's velocity is updated using the historical best information of particles within a dynamic neighborhood, rather than the entire population. These neighborhoods are reformed periodically, allowing participants to learn from their neighborhood or historical best practices, thus enhancing guidance and maintaining diversity.

$$\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_{e1} \cdot rand_1^d \cdot (\mathbf{p}_{Best_{f_j^d}}^d - \mathbf{x}_i^d) + c_{e2} \cdot rand_2^d \cdot (\mathbf{g}_{Best}^d - \mathbf{x}_i^d). \quad (31)$$

Additionally, P_{rc} adopts a linear variation within the same range (Eq. (5)), with the intent of enhancing each particle's learning tendency compared to the exponential variation.

4.6. Strategy adaptation learning

An ensemble learning approach is proposed in [27] for PSO, termed ensemble PSO (EPSO), which integrates multiple learning strategies into a unified framework. In EPSO, the population is divided into two groups: the first group comprises 40% of the swarm, while the second group consists of the remaining 60%. Five learning strategies are incorporated into EPSO: (i) CLPSO with $\mathbf{g}_{Best}$ [22], defined in Eq. (31) with the CLPSO probability P_{cr} , (ii) linearly decreasing ω_g PSO (LDWPSO) [10], (iii) distance-based locally informed PSO (LIPS) [31], (iv) PSO with time-varying acceleration coefficients (TVAC-PSO) [117], (v) fitness distance ratio-based PSO [81], discussed in Section 3.8.

The first group consistently employs CLPSO with $\mathbf{g}_{Best}$, while the second group dynamically selects one of the five strategies based on past performance. A self-adaptive selection mechanism is used to determine the most effective learning strategy by tracking the success and failure of each approach over a defined learning period L_p . The selection probability for each strategy is computed as:

$$p_k(g) = \frac{S_k(g)}{\sum_{k=1}^K S_k(g)}, \quad S_k(g) = \frac{\sum_{g=G-L_p}^{G-1} n_{s_k}(g)}{(\sum_{g=G-L_p}^{G-1} (n_{s_k}(g) + n_{f_k}(g)) + \epsilon)}, \quad (32)$$

here, $k = 1, 2, \dots, K$ (K is number of strategies), $G > L_p$, and $S_k(g)$ represents the success rate of the k th learning strategy. The small constant $\epsilon = 0.01$ prevents division by zero and ensures numerical stability. At each generation after L_p , the success rate of each learning strategy is updated, and strategies are selected proportionally based on their performance. This self-adaptive mechanism allows EPSO to dynamically adjust the learning strategy for each particle, enhancing its ability to balance exploration and exploitation in different optimization landscapes.

Inspired by EPSO [27], Liu and Nishi [118] proposed a strategy dynamics learning (SDL) approach for PSO, where the population is also divided into two groups: 30% and 70% of the population. Four learning strategies, such as CLPSO [22], unified PSO (UPSO) [119], LDWPSO [10], and LIPS [31], are integrated into a single framework with probabilistic selection. The first group consistently employs CL learning, while the second group adopts one of the four strategies based on a roulette wheel selection mechanism. However, the selection process may lead to rapid convergence, causing one strategy to dominate others. A restart operator is introduced to maintain strategy diversity, dividing the optimization process into r stages based on the maximum number of fitness evaluations. When entering a new stage, the probability distribution is reset as $x_h = 1/k$ for $h \in \{1, 2, \dots, k\}$.

4.7. Other learning strategies

Wang et al. [120] introduced an adaptive granularity learning (AGL) distributed PSO that incorporates machine learning techniques, such as locality-sensitive hashing and logistic regression for clustering and adaptive granularity control, respectively. The method uses a master-slave multi-subpopulation model, where subpopulations co-evolve and share evolutionary information. This dynamic adjustment of subpopulation sizes balances exploration and exploitation in large search spaces, enhancing optimization performance. Panda et al. [121] proposed a static learning strategy (SLS) to vary swarm size in PSO adaptively. The population is divided into two sub-populations, where particles interact within their neighborhoods, and survival probability ($S_{p_i} = f_i / \sum_i f_i$) determines the existence of better particles. PSO is divided into two phases: the first enhances exploration, while the second optimizes the utilization of explored information.

A cooperative-based difference learning (CDL) strategy [122] for PSO is proposed, where particles are divided into elite and common subswarms based on fitness, dynamically adapting learning strategies to enhance exploration. The elite subswarm selectively updates particles using superior information while preventing premature convergence. The velocity update rule is:

$$\mathbf{vel}_i^d = \omega_g \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \text{rand}_1 \cdot (\mathbf{p}_{Best_{rm}} - \mathbf{x}_i^d) + c_{e_2} \cdot \text{rand}_2 \cdot (\mathbf{p}_{Best_{rs}} - \mathbf{x}_i^d), \quad (33)$$

where $\mathbf{p}_{Best_{rm}}$ and $\mathbf{p}_{Best_{rs}}$ represent the best historical positions of randomly selected particles from the elite and common subswarms, respectively. The common subswarm adopts stage-based strategies to improve convergence accuracy over time. Xia et al. [123] proposed multi-exemplar-based forgetting learning (MFL) for PSO, extending the social learning process from one to two exemplars: the local best ($\mathbf{l}_{Best_i}$) and global best ($\mathbf{g}_{Best}$). Additionally, each particle is assigned a forgetting ability vector $\mathbf{f}_{g_i} = \{f_{g_i}^1, f_{g_i}^2, \dots, f_{g_i}^d\}$ to regulate information retention, mimicking human forgetting behavior. The velocity update rule is:

$$\mathbf{vel}_i = \omega_g \cdot \mathbf{vel}_i + c_{e_1} \cdot r_1 \cdot (\mathbf{p}_{Best_i} - \mathbf{x}_i) + c_{e_2} \cdot r_2 \cdot ((1 - f_{g_i}) \cdot \mathbf{l}_{Best_i} - \mathbf{x}_i) + c_{e_3} \cdot r_3 \cdot ((1 - f_{g_i}) \cdot \mathbf{g}_{Best} - \mathbf{x}_i), \quad (34)$$

where c_{e_1}, c_{e_2} , and c_{e_3} are acceleration coefficients for personal, local, and global bests, respectively, and f_{g_i} controls information loss when learning from $\mathbf{l}_{Best_i}$ and $\mathbf{g}_{Best}$. The forgetting effect is distance-dependent, meaning particles farther apart experience greater information loss. Multilevel and multi-sample learning strategies are proposed in [124–126].

Two-swarm-based learning strategies provide a flexible middle ground between the simplicity of single-swarm methods and the structural complexity of multi-swarm frameworks. By enabling differentiated learning roles and controlled interactions between two subpopulations, these strategies achieve a more effective balance between exploration and exploitation. However, their performance can be sensitive to how interaction mechanisms and learning roles are designed, and they may still face limitations in handling large-scale problems or highly dynamic landscapes. These challenges have led to the development of multi-swarm learning strategies, where more than two populations are employed to allow scalable, decentralized, and often hierarchical learning designs. Table 2 presents an overview of prominent two-population-based learning strategies developed for PSO. Each method adopts a unique approach to divide and manage the swarm into distinct sub-populations, often designed to encourage cooperative or competitive dynamics, or to incorporate hybrid learning paradigms such as exemplar-based or adaptive learning. These tables include the method name, abbreviation, publication year, citation, and a concise description of its learning mechanism and contribution to improving optimization performance. The next section explores these

advanced strategies in detail, focusing on their architecture, coordination mechanisms, and enhanced adaptability in diverse optimization scenarios.

5. Multiple swarm-based learning

This section explores multi-population PSO, where the main swarm is partitioned into several subpopulations, each potentially employing distinct learning strategies. This framework is designed to enhance population diversity, mitigate premature convergence, and improve overall search performance, particularly in high-dimensional or complex landscapes. Subpopulations may evolve independently or engage in periodic interactions to share knowledge, promote global exploration, or exploit promising regions. These interactions can be structured as cooperative exchanges, competitive dynamics, or hierarchical learning schemes. Furthermore, adaptive mechanisms are often introduced to dynamically adjust subpopulation sizes, learning parameters, or roles based on performance feedback, fostering a balance between exploration and exploitation throughout the search process.

5.1. Dynamic multi-swarm strategy

The dynamic multi-swarm strategy (DMS), first proposed for PSO [127], enhances the standard PSO by addressing its limitations in tackling complex multimodal optimization problems. This method builds on the local version of PSO and introduces a novel neighborhood topology characterized by two key features: small neighborhoods and a randomized regrouping mechanism.

DMS-PSO employs small neighborhoods, in contrast to traditional PSO approaches that often favor larger populations. By dividing the population into smaller swarms, DMS-PSO slows down the convergence velocity and enhances the diversity of solutions. Each swarm searches independently within the solution space using its members' historical information (Eq. (35), where $\mathbf{l}_{Best_i} = \{l_{Best_i}^1, l_{Best_i}^2, \dots, l_{Best_i}^d\}$), allowing the algorithm to efficiently explore and exploit the search space. This approach is particularly beneficial for complex problems, where smaller neighborhoods are better suited to navigating intricate landscapes and avoiding premature convergence.

$$\mathbf{vel}_i^d = \omega_g \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \text{rand}_i^d \cdot (\mathbf{p}_{Best_i}^d - \mathbf{x}_i^d) + c_{e_2} \cdot \text{rand}_i^d \cdot (\mathbf{l}_{Best_i}^d - \mathbf{x}_i^d). \quad (35)$$

To further mitigate the risk of stagnation or premature convergence to local optima, DMS-PSO incorporates a randomized regrouping mechanism. After every R_p generation (the regrouping period), the population is randomly reorganized into new configurations of small swarms. This process promotes information exchange across swarms, leveraging the best solutions discovered in different regions of the search space while also increasing diversity by disrupting static neighborhood structures. The value of R_p plays a pivotal role in the effectiveness of this strategy: a well-chosen R_p ensures meaningful interaction between sub-populations without hindering convergence or exploration. This regrouping achieves two critical objectives:

- (i) *Information exchange*: It allows swarms to share the valuable information accumulated during their search, thereby promoting collective intelligence across the population.
- (ii) *Increased diversity*: By restructuring neighborhoods, the algorithm disrupts stagnation and provides fresh opportunities for exploration, helping it escape local optima.

As illustrated in Fig. 9, consider a population of 12 particles divided into four sub-populations, each with three individuals. Initially, each sub-population focuses on refining its local search. After R_p generations, the population is randomly reorganized into new groups, allowing swarms to exchange valuable information and continue exploring the search space from different configurations.

Table 2
Two-population-based learning strategies for PSO.

Long name	Sort name	Year	Ref.	Comments
Social learning PSO	SLPSO	2015	[105]	Social learning occurs when followers (particles with lower fitness) learn from exhibitors (particles with higher fitness). The population is sorted by fitness, and each follower i learns from a randomly selected exhibitor k , where $i < k \leq N_s$. For a population of N_s particles sorted in ascending fitness order as $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{N_s}$, the hierarchy is: The worst particle $\mathbf{x}_1$ learns from any $\mathbf{x}_k$ where $2 \leq k \leq N_s$. The second-worst $\mathbf{x}_2$ has fewer exhibitor choices, and so on. The best particle $\mathbf{x}_{N_s}$ has no exhibitor and remains unchanged. The worst particle $\mathbf{x}_1$ cannot be an exhibitor. SL has poor searchability and low search efficiency.
Interswarm interactive learning PSO	IILPSO	2015	[111]	In IIL, two swarms are formed similarly to SL: one retains its original learning strategy (learned swarm), while the other (learning swarm) adapts. The softmax function determines the probability of a swarm being selected as the learned swarm. The learned swarm follows the standard PSO update rule, whereas particles in the learning swarm switch between two learning modes: (1) retaining their original search strategy or (2) learning from the learned swarm. The IIL strategy enhances diversity and improves the probability of escaping local optima.
Adaptive granularity learning PSO	AGLPSO	2020	[120]	AGL incorporates machine learning, including locality-sensitive hashing for clustering and logistic regression for adaptive granularity. A master-slave multi-subpopulation model enables co-evolution and information sharing, dynamically balancing exploration and exploitation for improved optimization.
Stochastic example learning PSO	SELPPO	2021	[112]	The population is split into an elite and a following sub-swarm in a ratio λ_p . The elite guides the followers while exploring diverse information using crisscross learning [113], leveraging personal bests as horizontal and vertical crossovers. The following sub-swarm updates velocity using: $\mathbf{vel}_i^d = w_e \cdot \mathbf{vel}_i^d + c_1 \cdot \mathbf{rand}_1^d \cdot \left(\mathbf{rand}_2 \cdot \mathbf{p}_{Best_i}^d + (1 - \mathbf{rand}_2) \cdot \mathbf{p}_{Best_k}^d - \mathbf{x}_i^d \right).$
Teaching and peer-learning PSO	TPLPSO	2014	[114]	This approach reduces the risk of premature convergence, fosters population diversity, and accelerates convergence. It has teaching and peer-learning phases. In the teaching phase, a particle updates velocity using the standard PSO rule (Eq. (1)). If fitness fails to improve, it enters the peer-learning phase [115], selecting an exemplar ($\mathbf{p}_{Best_e}$) from peer personal bests ($\mathbf{p}_{Best}$), excluding its own ($\mathbf{p}_{Best_i}$) and the global best ($\mathbf{g}_{Best}$). This dual-phase strategy enhances adaptability, improving the balance between exploration and exploitation.
Static learning PSO	SLSPSO	2016	[121]	The population is split into two sub-populations, where particles interact locally, and survival probability ($S_{p_i} = f_i / \sum_i f_i$) determines superior particles. PSO operates in two phases: the first emphasizes exploration, while the second refines the utilization of discovered information.
Difference learning PSO	DLPPO	2023	[122]	Particles are categorized into elite and common subswarms based on fitness, dynamically adjusting learning strategies for better exploration. The elite subswarm leverages superior information while mitigating premature convergence. The velocity update is: $\mathbf{vel}_i^d = \omega_e \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \mathbf{rand}_1 \cdot (\mathbf{p}_{Best_{en}} - \mathbf{x}_i^d) + c_{e_2} \cdot \mathbf{rand}_2 \cdot (\mathbf{p}_{Best_{en}} - \mathbf{x}_i^d).$
Neighbor-based learning PSO	NLPPO	2018	[116]	Each target particle learns from two swarm members: a randomly selected neighbor ($\mathbf{x}_k$) and the global best particle ($\mathbf{g}_{Best}$) as mentioned in Eq. (30). Nasir et al. [21] proposed dynamic neighborhood learning PSO introduces a learning strategy in Eq. (31).
Ensemble learning PSO	EPSO	2017	[27]	The population is divided into 40% and 60% groups. A unified framework integrates CLPSO with $\mathbf{g}_{Best}$ [22], TVAC-PSO [117], FDR-PSO [81], LDWPSO [10], and LIPS [31] with probabilistic selection $p_k = S_k / \sum_{k=1}^K S_k$ for $k \in \{1, 2, \dots, K\}$.
Strategy dynamic learning PSO	SDPSO	2022	[118]	The population is split into 30% and 70% groups. A unified framework integrates CLPSO [22], UPSO [119], LDWPSO [10], and LIPS [31] with probabilistic selection. At each stage transition, probabilities reset as $\mathbf{x}_h = 1/k$ for $h \in \{1, 2, \dots, k\}$.

(continued on next page)

Table 2 (continued).

Long name	Sort name	Year	Ref.	Comments
Forgetting learning PSO	MFLPSO	2020	[123]	It extends social learning to two exemplars: $\mathbf{l}_{Best_i}$ and $\mathbf{g}_{Best}$. A forgetting vector $\mathbf{f}_{g_i} = \{f_{g_i}^1, f_{g_i}^2, \dots, f_{g_i}^d\}$ regulates information retention, mimicking human forgetting. The velocity update rule is $\mathbf{vel}_i = w_g \cdot \mathbf{vel}_i + c_{e_1} \cdot r_1 \cdot (\mathbf{p}_{Best_i} - \mathbf{x}_i) + c_{e_2} \cdot r_2 \cdot ((1 - f_{g_i}) \cdot \mathbf{l}_{Best_i} - \mathbf{x}_i) + c_{e_3} \cdot r_3 \cdot ((1 - f_{g_i}) \cdot \mathbf{g}_{Best} - \mathbf{x}_i)$.

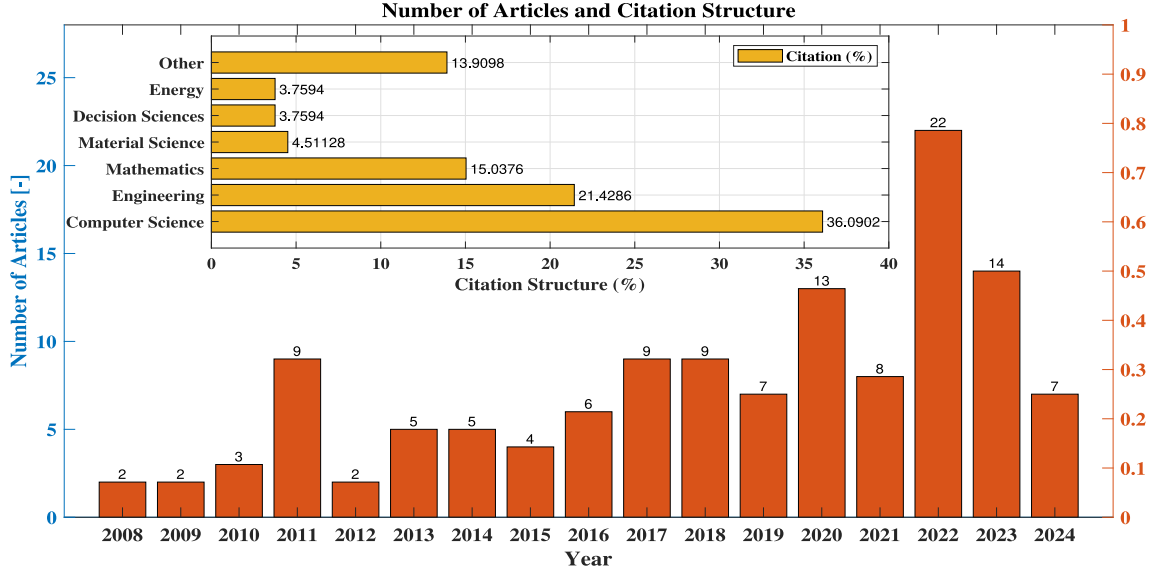


Fig. 7. Number of annually published documents on DMS and application in the different fields (%) based on the Scopus platform.

The dynamic nature of DMS-PSO, with its small-sized swarms and randomized regrouping schedule, grants it greater flexibility and adaptability compared to classical PSO variants. These innovations allow DMS-PSO to efficiently balance exploration and exploitation, making it particularly effective in handling complex, multimodal optimization problems.

Fig. 7 illustrates the annual publication trends and citation distribution of research documents on DMS and its applications across various fields, as analyzed through the Scopus platform. The bottom chart shows the annual published documents from 2008 to 2024. A gradual increase in publications is evident, starting with only two papers in 2008. Notable growth is observed after 2017, peaking in 2022 with 22 publications. This highlights a surge in research interest and developments in DMS during this period. However, a slight decline is visible post-2022, with 14 publications in 2023 and 7 in 2024 (as of now), potentially indicating a stabilization phase or a shift in research focus. The top chart provides insights into the citation structure across various fields. Computer Science leads significantly, contributing 36.09% of the total citations, showcasing its critical role in developing and implementing DMS algorithms. Engineering follows with 21.43%, reflecting the widespread application of DMS in solving engineering optimization problems. Mathematics accounts for 15.04% of citations, emphasizing its importance in theoretical advancements and algorithmic foundations. Other fields, including Material Science, Decision Sciences, and Energy, contribute to a smaller but noteworthy share of research, collectively representing the interdisciplinary applications of DMS. The annual citations of the original DMS paper [22], as per Google Scholar, as shown in Fig. 8, average 36 citations per year.

Chen et al. [128] introduced DMSDL-PSO by integrating differential mutation and the Quasi-Newton method into DMS-PSO, enhancing exploration and exploitation capabilities. The differential mutation was applied to personal historical best particles, and a modified velocity update equation addressed PSO's limitations. Liu et al. [129] extended this

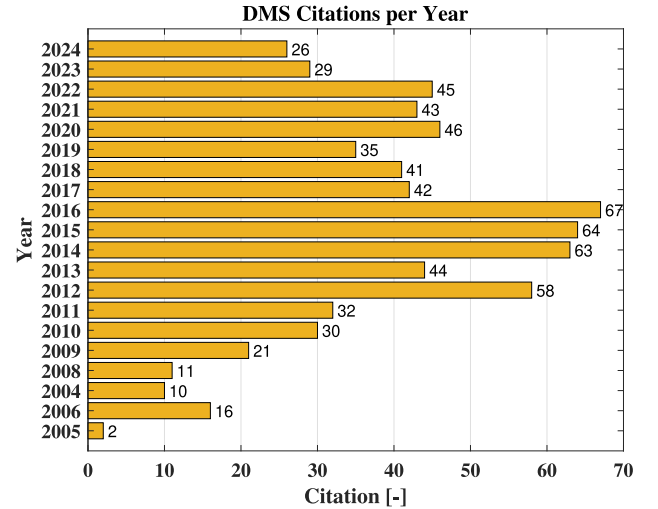


Fig. 8. DMS citations per year based on the Google Scholar platform.

idea to Harris Hawks optimization for cascade hydropower dispatch. Similarly, local search operators and cooperative learning strategies have been incorporated into DMS-PSO [130–132], further enhancing its performance. Wang et al. [36] integrated a CL mechanism into DMS-PSO by dividing the population into two groups, a strategy later adapted to the gravitational search algorithm (GSA) [133]. Yousri et al. [37] combined DMS learning with the marine predator algorithm for parameter identification in fuel cells.

Additional extensions include integrating DMS-PSO with parallel PC cluster systems [134] to enhance computational efficiency. Tao et al. [135] proposed a fitness peak clustering-based DMS-PSO with a

CL strategy, where stagnant particles merge into a global swarm for further evolution. Tang et al. [136] and Xia et al. [137] developed DMS-GPSO, where the global sub-swarm focuses on exploitation under the guidance of the optimal particle in the population. At the same time, other sub-swarms prioritize exploration guided by local best positions. Store and reset operators were also introduced to save computational resources and increase diversity. Yang et al. [138] proposed an adaptive DMS-PSO incorporating a stagnation detection mechanism (SDM) and spatial exclusion strategy. When a sub-swarm's local best solution stagnates before reaching the regrouping period, SDM generates a vitality particle to aid in discovering promising areas.

The DMS learning framework has been integrated into various optimization algorithms, including GSA [133], bird swarm algorithm [139], fireworks algorithm [140], pigeon-inspired algorithm [141], K-means clustering [142], Nelder–Mead Simplex method [143], firefly algorithm (FA) [144], harmonic search (HS) [145], and whale optimization algorithm (WOA) [146]. These integrations demonstrate the versatility and adaptability of the DMS learning strategy across diverse optimization problems.

Balancing diversity and convergence speed is a persistent challenge in PSO. While DMS significantly enhances diversity by dividing the population into smaller sub-swarms and regrouping them periodically, it sacrifices convergence speed and exploitation capability due to frequent regrouping. To address these limitations, Liang et al. [130,131] introduced the Quasi-Newton method as a local search algorithm to improve precision in promising regions. Periodically, every t generation, sub-populations are ranked by fitness, and the top 25% of groups have their $\mathbf{l}_{Best_t}$ refined using the Quasi-Newton method. Additionally, at the end of the search, the globally best solution is optimized with the Quasi-Newton method to ensure maximum accuracy. Further enhancements to DMS have been proposed to improve its performance. In [145], the HS algorithm was incorporated into each sub-swarm (DMS-PSO-HS), allowing the algorithm to effectively utilize historical information from past solutions. Another significant improvement is the hybridization of DMS-PSO with the modified multi-trajectory search (MTS) and sub-regional HS (DMS-PSO-SHS) [147], which expands the search to a larger potential space across sub-populations.

Despite advancements, the cooperative mechanism among sub-swarms in DMS-PSO remains confined to periodic regrouping, which limits its ability to fully exploit global search information. When sub-swarms are trapped in local minima, regrouping primarily exchanges local optima, hindering global optimization and exploration. Xu et al. [132] introduced a cooperative learning strategy (CLS) in DMS-PSO to address this. Before regrouping, using a tournament selection strategy, the two worst particles in each sub-swarm learn from the better particles of two randomly selected sub-swarms. This approach facilitates better information exchange among sub-swarms, enabling particles to utilize the best knowledge across the population. DMS-PSO-CLS improves the balance between global exploration and local exploitation by enhancing communication and expanding the search space.

5.2. Dynamic learning strategy

The dynamic learning strategy (DyLS) [148] introduces an innovative learning mechanism for PSO. Similar to the DMS, DyLS divides the population into multiple sub-populations. Consider a population with N_s particles, which is divided into k sub-populations denoted as $\{h_1, h_2, \dots, h_k\}$, where $N_s = \sum_{p=1}^k |h_k|$. The first sub-population consists of particles 1 to h_1 , the second includes particles $h_1 + 1$ to h_2 , and so on. This partitioning effectively splits a large PSO into several smaller, independent PSOs.

The best-performing particle is designated as $\mathbf{l}_{best}$ in each subpopulation. DyLS calculates the average of the $\mathbf{l}_{best}$ values from all sub-populations to update particle velocities using the following equation:

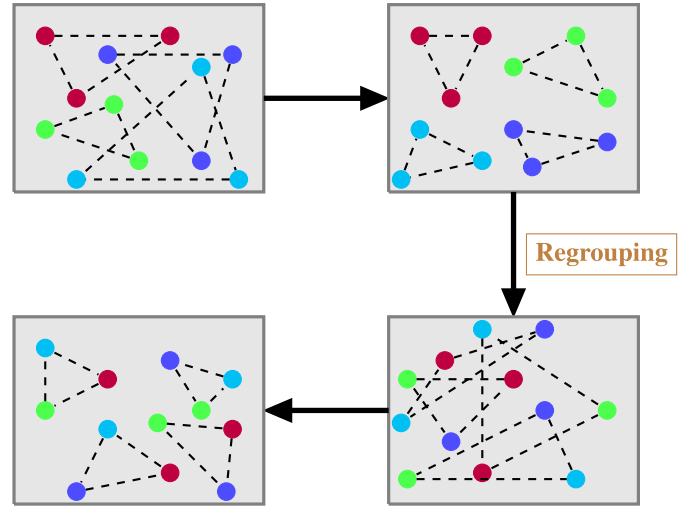


Fig. 9. The DMS learning.

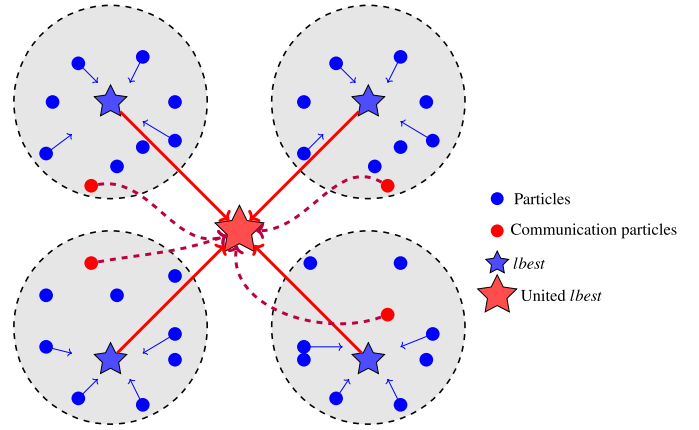


Fig. 10. DLS strategy.

$$\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \mathbf{rand}_i^d \cdot (\mathbf{p}_{Best_i}^d - \mathbf{x}_i^d) + c_{e_2} \cdot \mathbf{rand}_i^d \cdot (\overline{\mathbf{l}_{Best}^d} - \mathbf{x}_i^d), \quad (36)$$

where $\overline{\mathbf{l}_{Best}^d} = \frac{1}{k} \sum_{p=1}^k \mathbf{l}_{Best_p}^d$, and $\mathbf{l}_{Best_p}^d$ represents the best position achieved so far in the p th sub-population. The final term in Eq. (36) is the united $\mathbf{l}_{Best}$. Fig. 10 illustrates the DyLS framework in the context of a multi-population PSO. The strategy classifies particles in each sub-swarm into ordinary and communication particles. Ordinary particles search for better local bests $\mathbf{l}_{Best}$ within their sub-swarm, following the traditional framework. Communication particles, however, actively interact between sub-swarms, breaking free from $\mathbf{l}_{Best}$'s control and exploring additional useful information (Eq. (36)).

5.3. Multi-exemplar AL

Wei et al. [149] proposed a multi-exemplar AL for PSO, where particles are divided into multiple subgroups, each utilizing multi-exemplar adaptive learning (MAL) based on particle performance. Based on MALs, each subgroup particle selects distinct exemplars within a generation and across generations. After choosing candidate exemplars, a particle i updates its velocity using its learning model. In each generation, the i th particle has two exemplars: **Exemplar1_i** = $\{\text{Exemplar1}_i^1, \text{Exemplar1}_i^2, \dots, \text{Exemplar1}_i^d\}$, the best-so-far position of its swarm, and **Exemplar2_i** = $\{\text{Exemplar2}_i^1, \text{Exemplar2}_i^2, \dots,$

Exemplar_i^d , a randomly selected neighbor from its candidate learning exemplars. The velocity update rule is given by:

$$\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + \text{rand}_1 \cdot (\mathbf{Exemplar}_1^d - \mathbf{x}_i^d) + \text{rand}_2 \cdot (\mathbf{Exemplar}_2^d - \mathbf{x}_i^d). \quad (37)$$

MAL removes the acceleration coefficients used in canonical PSO to simplify the velocity update. In this framework, elite particles focus on exploitation due to high-quality exemplars, while inferior particles benefit from increased exploration by following randomly selected exemplars. Mediocre particles balance exploration and exploitation through the combined influence of their exemplars. This approach allows particles within the same subgroup to exhibit diverse learning behaviors. The other AL strategies are presented in [150–153]

5.4. Hierarchical learning

Wang et al. [154] proposed a fuzzy logic-based hierarchical learning (HL) for PSO, where particles are stratified into four layers (10%, 20%, 30%, and 40%) based on fitness. Each layer classifies particles as winners (high-energy) or losers (low-energy). Winners learn from upper-layer particles, while losers learn from winners within the same layer, promoting diversity and mitigating premature convergence.

Hierarchical energy gathering point (EM): Aggregates all particles' positions within a layer. *High-energy gathering point (EMH)*: Aggregates only high-energy particles' positions. These gathering points enable one-to-many cooperation, enhancing information exchange and search efficiency. The velocity updates for high-energy and low-energy particles are:

$$\begin{aligned} \mathbf{vel}_{MH_i}^d &= \omega_g \cdot \mathbf{vel}_{MH_i}^d + c_{e1} \cdot (\mathbf{p}_{Best_i}^d - \mathbf{x}_{MH_i}^d) + c_{e2} \cdot \epsilon \cdot (\mathbf{g}_{Best}^d - \mathbf{x}_{MH_i}^d) \\ &\quad + a \cdot (\mathbf{EM}_{M-1} - \mathbf{x}_{MH_i}^d) \end{aligned} \quad (38)$$

$$\mathbf{vel}_{ML_i}^d = \omega_g \cdot \mathbf{vel}_{ML_i}^d + c_{e1} \cdot (\mathbf{p}_{Best_i}^d - \mathbf{x}_{ML_i}^d) + r \cdot (\mathbf{EMH} - \mathbf{x}_{ML_i}^d), \quad (39)$$

where $\mathbf{x}_{MH_i}^d$ and $\mathbf{x}_{ML_i}^d$ denote the positions of high-energy and low-energy particles, respectively. The energy gathering points are defined as: $\mathbf{EM}_M = \frac{1}{N_s} \sum_{i=1}^{N_s} \mathbf{x}_{MH_i}$, $\mathbf{EMH} = \frac{1}{N_s} \sum_{i=1}^{N_s} \mathbf{x}_{MH_i}$, $\{\mathbf{x}_{MH_i} \mid f_{\mathbf{x}_{MH_i}} < \text{Average}(f_{\mathbf{x}_M})\}$. High-energy particles exploit global and hierarchical knowledge, while low-energy particles rely on local structures, ensuring a balanced exploration–exploitation tradeoff. This learning is also used by Li et al. [124], who proposed an HL-based pyramid PSO. In pyramid PSO, the population is divided into layers based on fitness values, and each layer is divided into winners and losers, where winners learn from the above-layer individuals or top-most layer individuals, and losers learn from the winners of the same layer.

5.5. Other learning strategies

Tang et al. [155] introduced a level-based inter-task learning strategy with a dynamic local topology. Particles are grouped into levels using distinct learning methods for cross-task information sharing. Diverse search preferences enhance exploration and refinement, while systematic updates to the local topology optimize inter-task neighbor selection. A reinforcement learning (RL) level approach is proposed by Wang et al. [156] for PSO, where particles are divided into l levels based on their fitness values. RL and competition strategies are used to control the number of levels, enhancing the search efficiency of PSO. Table 3 presents an overview of prominent multi-population-based learning strategies developed for PSO. This table includes the method name, abbreviation, publication year, citation, and a concise description of its learning mechanism and contribution to improving optimization performance.

In summary, multi-population PSO offers a flexible and powerful framework for addressing the limitations of single-swarm approaches, particularly in multimodal and large-scale optimization problems. By leveraging parallelism, diversity, and adaptive interaction among sub-populations, these strategies can significantly enhance the algorithm's

robustness and convergence behavior. The techniques discussed in this section highlight how multi-population designs support more scalable and resilient optimization, laying the groundwork for further hybridization with learning mechanisms, dynamic adaptation, and problem-specific customizations in future research.

6. Theoretical development

This paper reviews learning strategies for PSO and explores theoretical developments from original works. As with most PSO convergence analyses [28,157], a deterministic implementation is used for theoretical assessment, primarily establishing stability rather than guaranteeing convergence to the global optimum. These analyses focus on whether the swarm reaches a stable point without confirming if it is the best solution.

A well-balanced optimization algorithm must navigate the trade-off between exploration and convergence. Exploration enables broad search space coverage to escape local optima, while convergence ensures particles stabilize around an optimal solution. An imbalance can degrade performance, excessive exploration leads to inefficiency, while premature convergence risks stagnation in suboptimal regions. Most mathematical analyses of PSO address stability but often overlook the distinction between mature and premature convergence. Poli [158] conducted a moment-based stability analysis of PSO, assuming a stagnation phase where particles stop moving—an unrealistic assumption. This work confirmed order-1 and order-2 stability but did not determine whether the final point is globally optimal.

Building on Poli's work, Nasir et al. [21] extended the theoretical analysis to DNLPSO, an improved CLPSO variant. They showed that setting acceleration coefficients $c_{e1} = c_{e2} > 1.2$ while maximizing P_{rc} preserves exploration by adjusting the topological distance between particles, influencing their attraction toward solutions. DNLPSO further enhances search efficiency by dynamically restructuring neighborhoods, ensuring diverse exemplar selection, and preventing stagnation, unlike CLPSO, where exemplars remain static. Cheng and Jin [105] provided a convergence proof for SLPSO, modeling its dynamics as:

$$\begin{aligned} \mathbf{x}_i^d(t+1) &= \mathbf{x}_i^d(t) + \mathbf{vel}_i^d(t+1), \\ \mathbf{vel}_i^d(t+1) &= \frac{1}{2} \cdot \mathbf{vel}_i^d(t) + \frac{1}{2} \cdot (\mathbf{x}_k^d(t) - \mathbf{x}_i^d(t)) + \frac{1}{2} \cdot \epsilon \cdot (\bar{\mathbf{x}}^d(t) - \mathbf{x}_i^d(t)), \end{aligned}$$

converges to equilibrium if the social influence parameter $\epsilon > -1$. Huang et al. [80] established ELPSO convergence, linking it to PSO and CLPSO. Using Markov process theory, they demonstrated that if PSO/CLPSO converges, ELPSO does as well. Conversely, if ELPSO fails to converge, PSO/CLPSO will fail for the same optimization problem.

Inspired by the above and theoretical development mentioned in [159], we design the convergence proof for CLPSO [22]. For a one-dimensional search space, Eq. (2) of CLPSO is written as:

$$\begin{aligned} \mathbf{x}(t+1) &= \mathbf{x}(t) + \mathbf{vel}(t+1) \\ \mathbf{vel}(t+1) &= w_g \cdot \mathbf{vel}(t) + \phi \cdot (\mathbf{p}(t) - \mathbf{x}(t)), \end{aligned}$$

where $\mathbf{p}$ is nothing but $\mathbf{p}_{Best_{f_i}}$ and $\phi = c_e \cdot \text{rand}_1$. Substituting velocity into the position equation, the following equation is obtained:

$$\mathbf{x}(t+1) = (1 + \omega_g - \phi)\mathbf{x}(t) - w_g \cdot \mathbf{x}(t-1) - \phi \cdot \mathbf{p}(t) \quad (40)$$

which can be rewritten in the form of the following matrix:

$$\begin{bmatrix} \mathbf{x}(t+1) \\ \mathbf{x}(t) \\ 1 \end{bmatrix} = A \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}(t-1) \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 + \omega_g - \phi & -\omega_g & \phi \mathbf{p} \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (41)$$

The matrix A satisfies the eigenvalue equation $|A - \lambda I| = 0$, yielding eigenvalues: $\lambda_0 = 1$, $\lambda_1 = \frac{1 + \omega_g - \phi + \Gamma}{2}$, $\lambda_2 = \frac{1 + \omega_g - \phi - \Gamma}{2}$, where $\Gamma = \sqrt{(1 + \omega_g - \phi)^2 - 4\omega_g}$. Thus, Eq. (41) can be rewritten as:

$$\mathbf{x}(t) = k_1 + k_2 \lambda_1^t + k_3 \lambda_2^t, \quad (42)$$

Table 3
Multiple-population-based learning strategies for PSO.

Long name	Sort name	Year	Ref.	Comments
Dynamic multi-swarm PSO	DMS-PSO	2005	[127]	This method builds on the local version of PSO and introduces a novel neighborhood topology characterized by two key features: small neighborhoods and a randomized regrouping mechanism. DMS-PSO employs small neighborhoods, in contrast to PSO approaches that often favor larger populations. By dividing the population into smaller swarms, DMS-PSO slows down the convergence velocity and enhances the diversity of solutions. Each swarm searches independently within the solution space using its members' historical information as $\mathbf{vel}_i^d = w_g \cdot \mathbf{vel}_i^d + c_{e_1} \cdot \text{rand}_i^d \cdot (\mathbf{p}_{Best_i}^d - \mathbf{x}_i^d) + c_{e_2} \cdot \text{rand}_i^d \cdot (\mathbf{1}_{Best_i}^d - \mathbf{x}_i^d)$. The parameters $c_{e_1} = c_{e_2} = 1.49445$, $\omega_g = 0.729$, and $R_p = 5$.
Dynamic learning strategy	DyLPSO	2017	[148]	Similar to the DMS, DLS divides the population into multiple sub-populations. Consider a population with N_s particles, which is divided into k sub-populations denoted as $\{h_1, h_2, \dots, h_k\}$, where $N_s = \sum_{p=1}^k h_k $, where $\mathbf{1}_{Best}^d = \frac{1}{k} \sum_{p=1}^k \mathbf{1}_{Best_p}^d$.
Hierarchical learning PSO	HLPSO	2023	[154]	In HL, particles are stratified into four layers (10%, 20%, 30%, and 40%) based on fitness. Each layer classifies particles as winners (high-energy) or losers (low-energy). Winners learn from upper-layer particles, while losers learn from winners within the same layer, promoting diversity and mitigating premature convergence. High-energy particles exploit global and hierarchical knowledge, while low-energy particles rely on local structures, ensuring a balanced exploration-exploitation tradeoff.
Reinforcement learning PSO	RLPSO	2022	[156]	In RLPSO, particles are ranked into l levels based on fitness, with reinforcement learning (RL) and competition strategies optimizing level control for enhanced search efficiency. RL-driven velocity updates, cosine similarity for velocity control, and local updates to prevent premature convergence [87].
Inter-task learning PSO	ITLPSO	2021	[155]	A level-based inter-task learning strategy with dynamic topology groups particles into levels for cross-task learning. Diverse search preferences balance exploration and refinement, while adaptive topology updates optimize neighbor selection.

where

$$\begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \begin{bmatrix} \mathbf{p} \\ \frac{A_2 - (1+A_2)\mathbf{x}(t) + \mathbf{x}(t+1)}{\Gamma(A_1-1)} \\ \frac{-A_1 + (1+A_1)\mathbf{x}(t) - \mathbf{x}(t+1)}{\Gamma(A_2-1)} \end{bmatrix}.$$

In Eq. (42), when the condition $\max(|A_1|, |A_2|) < 1$ is satisfied [159], it follows that: $\lim_{t \rightarrow \infty} k_1 A_1^t = 0$, $\lim_{t \rightarrow \infty} k_2 A_2^t = 0$. Therefore, the particle positions eventually converge to a stable point $\mathbf{p}$, i.e.,

$$\lim_{t \rightarrow \infty} \mathbf{x}(t) = \lim_{t \rightarrow \infty} (k_1 + k_2 A_1^t + k_3 A_2^t) = \lim_{t \rightarrow \infty} (k_1 + 0 + 0) = k_1 = \mathbf{p}. \quad (43)$$

7. Experimental analysis

In this section, we compare the optimization performance of selected learning strategies, including multiple adaptive learning (MAL) [149], strategy dynamics learning (SDL) [118], dynamic neighborhood learning (DNL) [21], comprehensive learning (CL) [22], dynamic multi-swarm (DMS) [130], dimensional learning strategy (DLS) [64], social learning (SL) [105], multi-exemplar forgetting learning (MFL) [123], and adaptive strategy (AL) [153], fully-informed strategy (FIS) [30], locally-informed strategy (LIS) [31], ensemble learning (EL) [27], and unified learning (UL) [119].

These strategies were selected based on the availability of publicly accessible source codes, ensuring reproducibility. We exclude the original PSO, as their respective authors have already validated these strategies' effectiveness. All parameter settings remain unchanged except for population size and function evaluations. Details are presented in Table 4. It is noted that SDL and EL used various learning strategies to improve the impact of PSO. We evaluate the optimization performance

of the selected learning strategies on 10 unimodal, 16 multimodal, 13 hybrid, and 13 composite benchmark problems. These problems are sourced from basic test functions [105], CEC 2017 [160], and CEC 2020 [161], along with classical benchmark functions from [162,163].

The population size is set to $N_s = 75$, and the maximum number of function evaluations is 75,000, following the recommendation in [19]. Each algorithm is executed for 25 independent runs, and the mean objective function values are recorded. The optimization performance is ranked for each strategy, where rank 1 is assigned to the strategy with the lowest mean objective function value and rank 13 to the poorest-performing strategy.

For each benchmark set (basic test functions, CEC 2017, CEC 2022), the average rank of each strategy is computed by aggregating results across all problems within that set and across different dimensionalities where applicable (e.g., CEC 2013 and CEC 2017). All experiments were conducted on a Windows 11 12th Gen Intel(R) Core(TM) i5 16 GB using MATLAB 2024b.

We assessed the statistical significance of pairwise comparisons among the 13 learning strategies across different problem sets and dimensionalities using the Wilcoxon signed-rank test [164] at a 5% significance level, as recommended in [165,166]. Additionally, ranking evaluations were conducted using the Friedman test [167].

We identify learning strategies for each algorithm and problem set that are statistically significantly better, worse, or similar to the learning mechanism that performs well in the Friedman rank test. The statistical test only determines whether differences exist at $\alpha = 0.05$. However, we interpret a learning strategy as superior (inferior) if its mean rank (evaluated by the Friedman test [167]) is lower (higher) than its competitors and the test confirms statistical significance. Additionally, we evaluated the final AvgRanks values to analyze the

Table 4
Selected learning strategies for experimental analysis.

$N_s = 75, d = 5, 10, 15, 20, 30, 50, 100, MaxFE = 75,000$					
SR	Strategy Name	PSO Name	Year	Author	Ref.
1.	FIS	Fully-informed strategy-based PSO	2004	Mendes et al.	[30]
2.	DMS	Dynamic multi-swarm-based PSO	2005	Liang et al.	[130]
3.	CL	Comprehensive learning-based PSO	2006	Liang et al.	[22]
4.	DNL	Dynamic neighborhood learning-based PSO	2012	Nasir et al.	[21]
5.	LIS	Locally-informed PSO	2012	Qu et al.	[31]
6.	SL	Social learning-based PSO	2015	Cheng and Jin	[105]
7.	EL	Ensemble learning PSO	2017	Lynn et al.	[27]
8.	DLS	Dimensional learning strategy-based PSO	2019	Xu et al.	[64]
9.	UL	Unified learning PSO	2019	Parsopoulos et al.	[119]
10.	MFL	Multi-exemplar forgetting learning-based PSO	2020	Xia et al.	[123]
11.	AL	Adaptive learning-based PSO	2020	Liu et al.	[153]
12.	MAL	Multiple adaptive learning-based PSO	2020	Wei et al.	[149]
13.	SDL	Strategy dynamics learning-based PSO	2022	Liu et al.	[118]

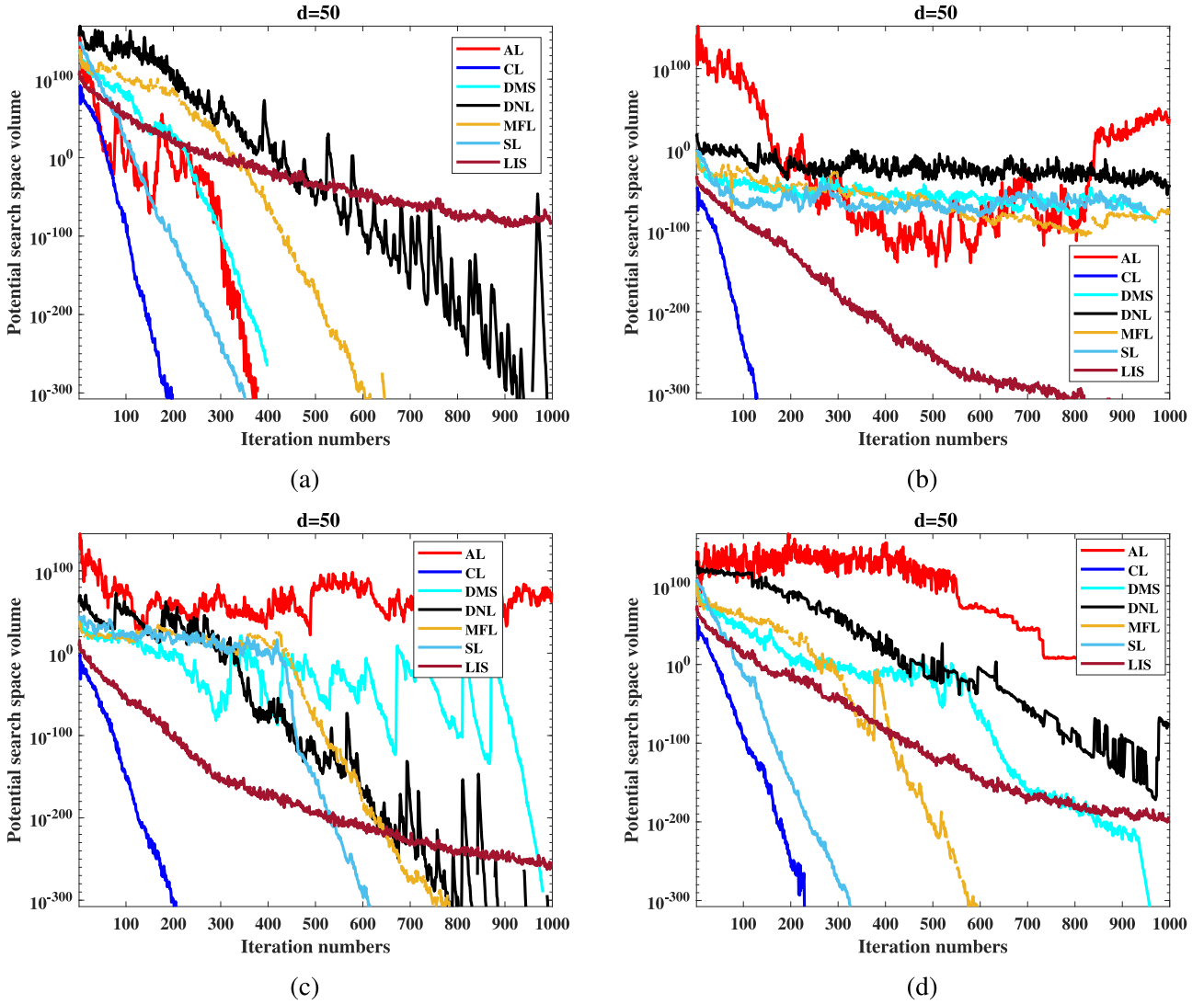


Fig. 11. Comparison of potential search space's volume for two unimodal problems ((a) Sphere and (b) Rosenbrock) and two multimodal problems ((c) Rastrigin and (d) Ackley) at 50 dimensions.

evolutionary behavior of each learning strategy. To compute Av-gRanks, we first recorded the best fitness values at each function evaluation across all runs. Then, we ranked the learning strategies for each function in every run and averaged these ranks across all functions. Additionally, the analysis of search behavior and diversity is also discussed in this article.

7.1. Search space behavior analysis

In this section, we analyze the search space volume of selected learning strategies for two unimodal and two multimodal problems in a 50-dimensional search space. We consider seven learning strategies: AL, DNL, DMS, CL, SL, MFL, and LIS. The length of the potential search

Table 5
Mathematical formulations of search space lengths.

Name	Search space length	Name	Search space length
AL:	$r_{li}^d = S_{Best_i}^d - x_i^d $	CL:	$r_{2i}^d = \max(\mathbf{p}_{Best_i^d}, x_i^d) - \min(\mathbf{p}_{Best_i^d}, x_i^d)$
LIS:	$r_{4i}^d = \mathbf{p}_{Best_i}^d - x_i^d $	SL:	$r_{3i}^d = x_k^d - x_i^d + \bar{x}^d - x_i^d $
DNL:	$r_{5i}^d = \mathbf{p}_{Best_i^d}^d - x_i^d + g_{Best}^d - x_i^d $	DMS:	$r_{6i}^d = \mathbf{p}_{Best_i}^d - x_i^d + v_{Best_i}^d - x_i^d $
MFL:	$r_{7i}^d = \mathbf{p}_{Best_i}^d - x_i^d + (1 - f_{R_i}) \cdot \mathbf{p}_{Best_i}^d - x_i^d + (1 - f_{R_i}) \cdot g_{Best}^d - x_i^d $		

space for these strategies is defined in Table 5. The potential search space volume for the i th particle under each learning strategy is given by [22]:

$$R_{i,j} = \prod_{d=1}^D r_{ji}^d, \quad j = 1, 2, 3, 4, 5, 6, 7, \quad (44)$$

where $\bar{R}_j = \text{mean}(R_{i,j})$ represents the mean volume of the potential search spaces across the entire swarm for a given learning strategy. To ensure a fair comparison, we execute each strategy 25 times on both unimodal and multimodal functions. This evaluation allows us to assess how different learning strategies influence the search space exploration and convergence behavior across different problem landscapes.

Fig. 11 presents a comparative analysis of the potential search space's volume for four benchmark optimization problems in a 50-dimensional space. The problems include two unimodal functions: Fig. 11(a) Sphere and Fig. 11(b) Rosenbrock, and two multimodal functions: Fig. 11(c) Rastrigin and Fig. 11(d) Ackley. The x -axis represents the number of iterations, while the y -axis, shown in a logarithmic scale, represents the potential search space volume. This metric helps assess how different optimization strategies navigate the search space over time.

For unimodal functions like the Sphere and Rosenbrock functions, the search space volume tends to decrease more smoothly and consistently. In Fig. 11(a), which corresponds to the Sphere function, most strategies exhibit rapid convergence, indicating that they quickly shrink the search space and focus on exploitation. This is expected, as the Sphere function has a simple structure with a single global minimum, making it easier for optimization methods to converge. In contrast, Fig. 11(b) (Rosenbrock function) shows a more gradual reduction in search space volume. This slower convergence is due to the narrow valley structure of the Rosenbrock function, which makes optimization more challenging, requiring strategies to balance exploration and exploitation carefully.

For multimodal functions, Figs. 11(c) and Fig. 11(d) show a significantly different pattern. The search space volume decreases more slowly and exhibits greater fluctuations, reflecting the presence of multiple local minima. In the Rastrigin function Fig. 11(c), the search space volume remains high for longer periods, especially for strategies that focus on exploration, such as AL (red) and DNL (black). These methods maintain diversity in the search space, preventing premature convergence to local optima. Similarly, in Fig. 11(d) (Ackley function), some strategies exhibit high variance in search space volume, suggesting that they explore a wide range of possible solutions before settling into an optimum. The fluctuating behavior observed in certain methods indicates their ability to adapt to the rugged landscape of multimodal functions.

The comparison of strategies reveals distinct behavioral patterns. Methods like CL (blue) and DMS (cyan) tend to reduce search space volume quickly, demonstrating fast convergence. While this is beneficial for unimodal problems, it can be risky for multimodal problems, where premature convergence may lead to suboptimal solutions. On the other hand, AL (red) and DNL (black) maintain higher search space volume throughout the iterations, indicating a strong exploratory capability, which is essential for solving complex multimodal functions. Additionally, some strategies, such as MFL and SL, show intermediate

behavior, balancing exploration and exploitation depending on the function's complexity.

7.2. Diversity analysis

In this section, we analyze the diversity factor introduced in our work. We employ the population diversity index to quantify the variation and spread among individuals within the population. This metric provides insight into how effectively each learning strategy enhances diversity and promotes exploration within the solution space. The mathematical formulation of this index is defined as follows [133]:

$$Div_{pop} = \frac{1}{N_s} \sum_{i=1}^{N_s} \sqrt{\left(\sum_{d=1}^D (\bar{x}_j - x_{i,j}^d)^2 \right)}, \quad \bar{x}_j = \frac{1}{N_s} \sum_{i=1}^{N_s} x_{i,j}, \quad (45)$$

where Div_{pop} represents the diversity of the entire population. The term $x_{i,j}$ denotes the position of the i th individual in the d th dimension, while $\bar{x}_j$ represents the mean position of all individuals. Diversity plays a crucial role in optimization algorithms as it directly impacts both exploration and convergence behavior.

Fig. 12 presents an analysis of diversity for two optimization problems: the Rosenbrock function (unimodal, left column) and the Rastrigin function (multimodal, right column) across different dimensions ($d = 10, 30, 50, 100$). The graphs illustrate how different learning strategies influence population diversity throughout 1000 iterations.

For the Rosenbrock function (left column), diversity generally decreases as iterations progress, indicating a gradual convergence of the search process—an expected behavior for a unimodal function. However, different learning strategies exhibit varying rates of diversity reduction. Some strategies, such as DLS, CL, and SDL, maintain higher diversity for a prolonged period, suggesting enhanced exploration capability. In contrast, strategies like SL, DMS, AL, and LIS demonstrate a rapid decline in diversity, implying faster convergence but a higher likelihood of premature stagnation.

In contrast, the Rastrigin function (right column) exhibits a significantly different diversity pattern due to its multimodal nature. Instead of a smooth decline, diversity fluctuates more prominently, particularly for strategies such as SDL, DLS, CL, UL, FIP, and AL, which exhibit sharp peaks and drops. This behavior suggests that these strategies frequently escape local minima and redistribute solutions across the search space. Unlike the Rosenbrock function, where a steady decline in diversity indicates convergence, maintaining a moderate level of diversity in Rastrigin is essential to avoid premature convergence to suboptimal solutions.

As the dimensionality increases ($d = 10, 30, 50, 100$), notable trends in diversity levels emerge. In the Rosenbrock function, higher dimensions generally lead to a slower decline in diversity, implying that larger search spaces necessitate extended exploration phases. This effect is particularly evident at $d = 100$, where strategies such as DLS, SDL, CL, AL, EL, and FIP maintain significant diversity even in the later iterations.

For the Rastrigin function, the impact of dimensionality is even more pronounced. The fluctuations in diversity become increasingly irregular as the number of dimensions grows, suggesting that higher-dimensional multimodal landscapes introduce greater challenges in maintaining a balance between exploration and exploitation. Strategies

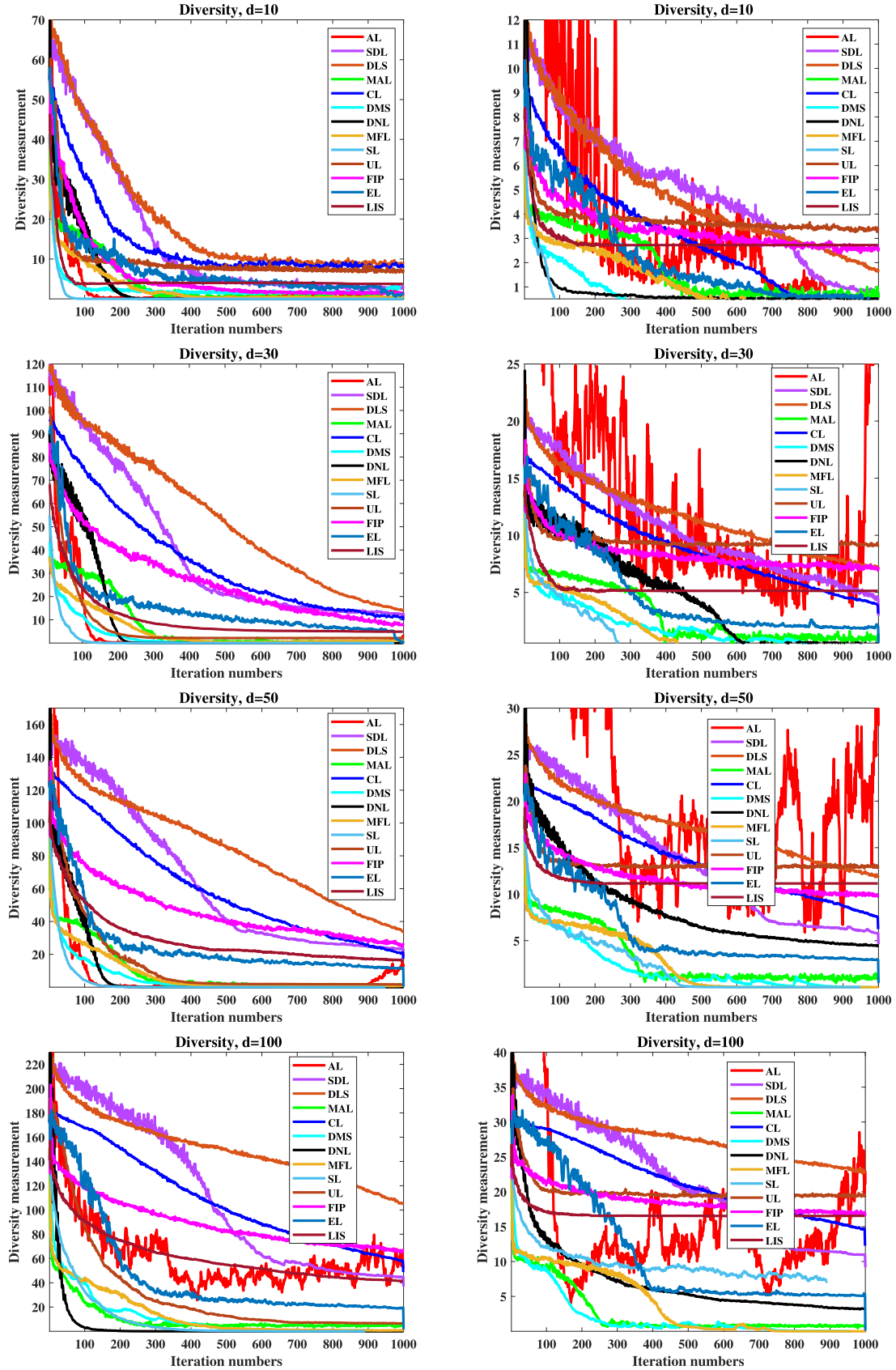


Fig. 12. Analysis of diversity in the Rosenbrock (unimodal, left) and Rastrigin (multimodal, right) functions across different dimensions.

such as AL and SDL exhibit larger oscillations, while others, such as DNL, MAL, DMS, and MFL, demonstrate a more stable diversity decline.

Overall, the results indicate that optimal diversity management is highly problem-dependent. While lower diversity is beneficial for

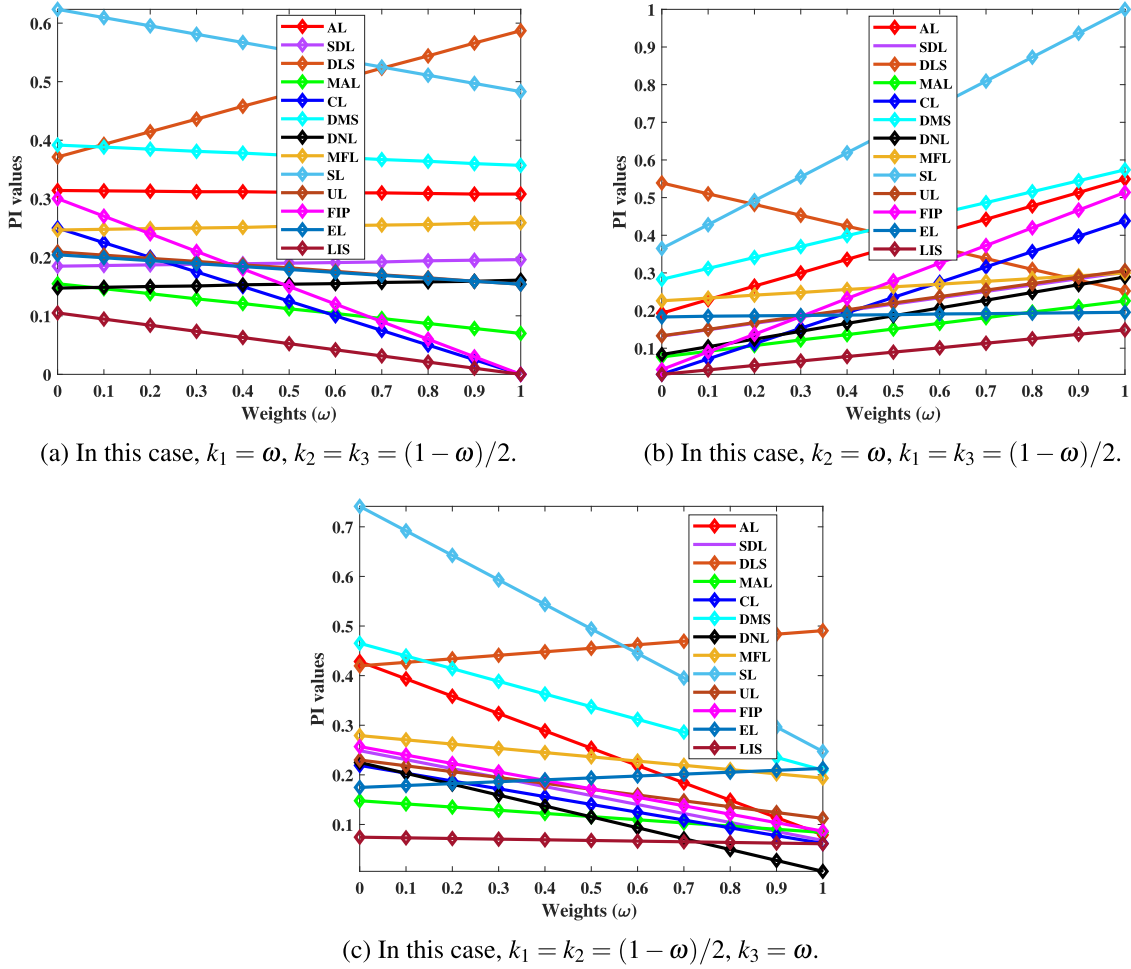


Fig. 13. Performance index of all learning strategies on basic test functions at 50 dimensions.

unimodal functions (e.g., Rosenbrock), maintaining a certain level of diversity is crucial for multimodal functions (e.g., Rastrigin) to prevent premature convergence and improve solution quality. Strategies that perform well in unimodal landscapes may not be ideal for multimodal problems, and vice versa.

7.3. Performance index analysis

In this section, we evaluate the optimization ability of each learning strategy based on the performance index (PI), which is defined as follows [168,169]:

$$PI = \frac{1}{N_p} \sum_{i=1}^{N_p} (k_1 \cdot \alpha_{1,i} + k_2 \cdot \alpha_{2,i} + k_3 \cdot \alpha_{3,i}), \quad (46)$$

$$\alpha_{1,i} = \frac{SR_i}{TR}, \quad \alpha_{2,i} = \frac{MT_i}{AT_i}, \quad \alpha_{3,i} = \frac{MF_i}{AF_i}, \quad (47)$$

where SR_i and TR represent the number of successful runs for problem i and the total number of runs, respectively. MT_i and AT_i denote the minimum average time of all strategies and the average time of the respective strategy for problem i . Similarly, MF_i and AF_i are the minimum average fitness value and the average fitness value of the strategy for the same problem. The total number of problems, N_p , is set to 13 basic test functions. In all three cases, the weight ω varies between 0 and 1 with a step of 0.1.

Fig. 13 illustrates the PI of different learning strategies across various weight values (ω) in a 50-dimensional search space. Each subplot corresponds to a different weighting scheme, highlighting how different

strategies perform under varying priority assignments for success rate (SR), execution time (MT), and fitness value (MF)

In subplot Fig. 13(a), where success rate (SR) is emphasized ($k_1 = \omega$) while execution time and fitness value are given equal weight ($k_2 = k_3 = (1 - \omega)/2$), DLS, SL, AL, and DMS consistently achieve the highest PI values, indicating their strong ability to balance success rate while maintaining reasonable computational efficiency and solution quality. In contrast, LIS, FIP, and CL performed the worst, showing a decline in PI values as ω increases. The other strategies show a moderate change in PI values during ω increases.

In subplot Fig. 13(b), where execution time (MT) is prioritized ($k_2 = \omega$), the SL, DMS, DLS, AL, FIP, and CL strategies again perform well, while strategies like UL, LIS, DNL, and MAL exhibit a rising trend, suggesting that these methods are more efficient in reducing execution time. However, EL and UL continue to show poor performance across all weight values, highlighting their inefficiency.

For subplot Fig. 13(c), where fitness value (MF) is given the highest priority ($k_3 = \omega$), SL and DLS continue to outperform other strategies, maintaining high PI values even as ω increases. MFL, UL, and EL demonstrate stable trends, while FIP, SDL, CL, MAL, and DMS show more pronounced declines. Strategies like DNL remain the least effective across all weight conditions.

In general, SL and DLS exhibit superior performance across different fitness values, maintaining consistently high PI values. SDL and MAL also demonstrate competitive performance, although their effectiveness varies depending on the test function. Conversely, FIP and LIS consistently report the lowest PI values, indicating limited search efficiency.

Table 6
Friedman ranks on the classical benchmark problems.

Dimension	AL	SDL	DLS	MAL	CL	DMS	DNL	MFL	SL	UL	FIP	EL	LIS
$d = 10$	6.6608	9.1958	5.5979	6.7378	5.5000	8.4161	5.3846	6.6608	5.5839	6.7832	10.3776	6.1434	7.9580
$d = 30$	7.4965	9.5664	5.2867	8.2972	4.7832	7.3916	3.4126	7.4965	4.3986	6.6923	11.4476	5.6923	9.0385
$d = 50$	8.1888	9.4336	4.6748	8.5420	5.0350	6.0874	2.6224	8.1888	3.6084	6.4580	11.7832	6.0490	10.3287
$d = 100$	9.0035	9.0140	5.0734	8.3147	4.7552	5.5280	2.1538	9.0035	4.1399	6.2028	11.9301	5.6014	10.2797

Table 7
Friedman ranks on the CEC 2017 test suite.

Dimension	AL	SDL	DLS	MAL	CL	DMS	DNL	MFL	SL	UL	FIP	EL	LIS
$d = 10$	8.1599	6.5361	4.0392	8.4436	7.4216	6.0251	4.5925	8.1599	7.3777	7.1489	9.5564	5.6803	7.8589
$d = 30$	7.6473	10.9248	3.3433	8.0219	5.5862	5.3950	6.1113	7.6473	4.5423	7.3730	10.6897	6.4357	7.2821
$d = 50$	8.3539	11.3117	3.2532	7.9740	5.3539	4.7013	5.5097	8.3539	3.9903	7.4286	10.9123	5.6883	8.1688
$d = 100$	8.8041	11.2320	3.3950	7.4671	4.8025	4.8119	4.7586	8.8041	4.9028	6.7994	10.8809	5.2382	9.1034

As weight values (ω) increase, certain strategies like DMS and MFL exhibit relatively stable performance, whereas UL and FIP show noticeable fluctuations. The divergence in trends across different test cases suggests that some strategies are more sensitive to weight variations, adjusting differently based on function characteristics. Overall, the results emphasize that strategy effectiveness is function-dependent, and an appropriate balance between exploration and exploitation is essential for achieving optimal search performance.

7.4. Friedman ranks analysis

The tables present Friedman rank test results on the classical benchmark problems (Table 6), the CEC 2017 test suite (Table 7), and the CEC 2020 test suite (Table 8), comparing various learning strategies across different problem dimensions. The Friedman rank test provides a ranking for each algorithm, with lower ranks indicating better performance.

In Table 6, the Friedman ranks for the classical benchmark problems show that DNL consistently achieves the lowest rank across all dimensions, indicating its superior performance compared to other learning strategies. For instance, at $d = 10$, DNL has a rank of 5.3846, which is the lowest among all methods, while at $d = 100$, its rank further improves to 2.1538, reinforcing its robustness in higher-dimensional spaces. Other strategies like SL and DLS also show competitive performance, with relatively lower ranks compared to alternatives such as AL, LIS, and FIP, which tend to have higher ranks, suggesting weaker performance.

In Tables 7 and 8, which present the Friedman ranks on the CEC 2017 and CEC 2020 test suites, the results show more variability. DLS emerges as the best-performing strategy, obtaining the lowest rank across all dimensions. For instance, at $d = 100$, DLS has a rank of 3.395, which is significantly lower than other strategies. DNL also maintains strong performance, with competitive ranks at lower dimensions, though not as dominant as in Table 6. On the other hand, strategies like AL, SDL, SL, and FIP tend to have higher ranks, suggesting that they are less effective compared to the top-performing methods.

In summary, these results indicate that DNL performs exceptionally well on classical benchmark problems, while DLS dominates in the CEC 2017 and CEC 2020 test suites. This highlights the importance of benchmarking different learning strategies across diverse test suites, as the best-performing method may vary depending on the problem set. Additionally, as the problem dimension increases, some strategies like DNL and DLS show greater robustness, reinforcing their effectiveness in high-dimensional multimodal optimization tasks.

7.5. Statistical results analysis

In this section, we statistically compare each learning strategy with the best-performing strategy in the Friedman rank test. Tables 9 to Table 11 present Wilcoxon Signed-Rank statistical test results comparing different learning strategies across three benchmark sets: classical benchmark problems, the CEC 2017 test suite, and the CEC 2020 test suite. Each table lists results for multiple learning strategies, with performance summarized in terms of wins, losses, and draws (+/≈/− across different problem dimensions (d). These results provide insights into how various learning strategies perform in different optimization settings.

In Table 9, the results of classical benchmark problems compare differential learning strategies against the DNL approach. Across different dimensions ($d = 10, 30, 50, 100$), certain strategies such as DLS, DMS, EL, MAL, and UL appear to have strong performance in comparison to DNL.

Table 10 presents statistical comparisons for the CEC 2017 test suite, where the DLS approach is tested against other learning strategies. The results suggest that SDL, DNL, FIS, LIS, MFL, and UL exhibit strong performance, particularly in lower dimensions. As problem complexity increases ($d = 30, 50, 100$), strategies like DLS and DMS maintain high win counts, indicating robustness across different problem scales.

In Table 11, results from the CEC 2020 test suite reveal similar trends. For lower dimensions ($d = 5, 10$), DLS performs better than other strategies. However, as the problem dimension increases ($d = 15, 20$), the number of draws rises, reinforcing the trend that learning strategies become harder to distinguish in higher-dimensional multimodal optimization tasks. Certain strategies, like UL and SL, appear to perform less favorably in comparison.

Overall, the results indicate that some learning strategies consistently perform well across different benchmark suites. The increasing number of draws in higher dimensions suggests that performance differences between methods become less significant as complexity grows. Additionally, the variation in rankings across different test suites highlights the importance of benchmarking across multiple problem sets to ensure a comprehensive evaluation of learning strategies in differential evolution.

7.6. Computational complexity analysis

The comparative computational complexity of various PSO learning strategies, summarized in Table 12, provides insight into how these strategies affect algorithm scalability and practical applicability. Canonical PSO serves as the baseline, with constant-time complexity

Table 8
Friedman ranks on the CEC 2020 test suite.

Dimension	AL	SDL	DLS	MAL	CL	DMS	DNL	MFL	SL	UL	FIP	EL	LIS
$d = 5$	8.0955	5.4000	4.0682	8.7455	8.1500	5.9136	4.6273	8.0955	9.7227	5.9500	7.9773	5.6227	8.6318
$d = 10$	7.2682	6.8364	4.4273	9.5727	7.1818	6.9500	4.4773	7.2682	8.1773	7.0545	9.7818	6.0182	5.9864
$d = 15$	7.2909	8.1455	3.7545	9.0000	7.0136	7.2636	4.7455	7.2909	6.0773	7.8773	10.6727	5.6955	6.1727
$d = 20$	8.1000	8.6182	3.9409	8.5000	6.4409	6.1091	4.5136	8.1000	5.5773	7.8864	10.3091	6.4955	6.4091

Table 9
Statistical test results on the classical benchmark problems, DNL *vs.*

Learning	AL	CL	DLS	DMS	EL	FIS	LIS	MAL	MFL	SDL	SL	UL
$d = 10$	4/4/5	6/4/3	2/6/5	1/7/5	1/9/3	7/5/1	1/10/2	3/8/2	1/9/3	1/8/4	1/8/4	0/9/4
$d = 30$	3/5/5	4/5/4	1/2/10	1/1/11	1/4/8	5/7/1	2/7/4	1/7/5	1/1/11	1/2/10	1/2/10	1/5/7
$d = 50$	4/5/4	3/6/4	1/1/11	1/0/12	1/3/9	6/6/1	3/8/2	1/4/8	1/1/11	1/2/10	1/1/11	1/4/8
$d = 100$	8/1/4	4/8/1	1/0/12	1/0/12	1/2/10	12/1/0	8/5/0	1/0/12	1/0/12	1/1/11	3/0/10	2/1/10

Table 10
Statistical test results on the CEC 2017 test suite, DLS *vs.*

Learning	AL	CL	SDL	DMS	DNL	EL	FIS	LIS	MAL	MFL	SL	UL
$d = 10$	20/8/1	11/18/0	14/12/3	6/14/9	18/11/0	7/20/2	23/4/2	20/7/2	10/14/5	19/8/2	5/19/5	15/9/5
$d = 30$	12/12/5	27/2/0	4/15/10	2/10/17	11/17/1	4/23/2	22/6/1	10/13/6	5/15/9	6/15/8	1/20/8	10/15/4
$d = 50$	12/15/2	29/0/0	6/8/15	3/10/16	14/13/2	5/20/4	26/2/1	16/13/0	6/12/11	6/16/7	3/15/11	15/11/3
$d = 100$	19/10/0	29/0/0	11/7/11	3/11/15	15/13/1	9/17/3	25/1/3	21/8/0	9/9/11	10/12/7	7/13/9	17/7/5

Table 11
Statistical test results on the CEC 2020 test suite, DLS *vs.*

Learning	AL	CL	SDL	DMS	DNL	EL	FIS	LIS	MAL	MFL	SL	UL
$d = 5$	6/3/1	2/7/1	8/2/0	2/5/3	7/3/0	2/7/1	9/1/0	6/4/0	4/4/2	7/3/0	3/6/1	4/6/0
$d = 10$	4/6/0	3/7/0	7/3/0	1/7/2	8/2/0	3/7/0	8/2/0	5/4/1	4/6/0	6/4/0	1/9/0	4/6/0
$d = 15$	5/4/1	7/3/0	3/6/1	2/5/3	8/2/0	1/8/1	8/2/0	4/6/0	6/3/1	2/8/0	1/8/1	4/6/0
$d = 20$	8/0/2	8/2/0	4/6/0	3/4/3	6/4/0	4/6/0	8/1/1	3/7/0	3/7/0	4/6/0	1/9/0	5/4/1

per particle per iteration ($\mathcal{O}(1)$). It is highly efficient computationally, but this simplicity can come at the cost of search diversity and may result in premature convergence, especially in complex or multimodal search spaces.

More sophisticated strategies such as CLPSO increase the complexity to $\mathcal{O}(N_s d)$, where d is the dimensionality of the problem and N_s is the population size. This arises from the need to identify the best dimensions across multiple exemplars for each particle, which enhances exploration but incurs a heavier computational burden. Similarly, dynamic multi-swarm PSO divides the swarm into interacting subpopulations and requires coordination among them. This coordination adds overhead proportional to the number of subpopulations and their interactions, resulting in a complexity of $\mathcal{O}(SN_s d + M)$, where S is subgroups and M is the migration overhead.

Self-organizing PSO strategies, which include entropy or diversity-based adaptations, typically incur a cost of $\mathcal{O}(N_s \log N_s)$ due to neighborhood analysis or information-theoretic computations. While this makes them more computationally intensive, such methods often demonstrate improved balance between exploration and exploitation through real-time structural adaptation.

Ensemble-based PSO methods select among multiple search strategies or learning rules at runtime. These introduce complexity of $\mathcal{O}(KN_s d)$, with K representing the number of candidate strategies in the ensemble. The added cost is justified by improved robustness and adaptability to diverse problem landscapes. Learning automata-based PSO strategies introduce an action-reward feedback loop, which generally adds a modest cost of $\mathcal{O}(A)$, where A is the number of automata. These methods improve dynamic decision-making in strategy selection without drastically increasing overhead.

Adaptive learning strategies, which evaluate population performance metrics to guide learning rule selection or parameter tuning, increase the cost to $\mathcal{O}(N_s)$. This cost is relatively moderate and often pays off in terms of improved convergence efficiency and responsiveness to problem changes.

Finally, surrogate-assisted PSO approaches incorporate surrogate models to approximate the objective function, reducing the number of expensive real evaluations. However, they incur significant costs due to model training and prediction, resulting in a complexity of $\mathcal{O}(T + S)$, where T and S represent training and inference costs, respectively. Despite their higher computational overhead, these approaches are particularly beneficial in applications involving expensive fitness evaluations, such as simulations or physical experiments.

In summary, while advanced PSO strategies offer improved search performance and robustness, they often come with increased computational complexity. Practitioners should consider the trade-off between exploration capability and computational efficiency based on the characteristics of their specific optimization problem. Strategies that incur higher complexity may be better suited for offline or computationally rich settings, whereas simpler, adaptive approaches may be preferable for real-time or large-scale applications.

8. Research challenges and open questions

Despite the significant advancements and widespread adoption of learning strategies across various domains, several challenges and open research questions remain. Over the past few years, these strategies have attracted growing interest, leading to notable progress in their applications. However, ongoing developments continue to reveal unresolved issues that require further exploration. Addressing these challenges demands sustained efforts from researchers to refine existing approaches and introduce innovative techniques. In the next section,

Table 12
Comparative computational complexity of learning strategies in PSO.

Learning strategy type	Example algorithms	Additional overhead	Estimated complexity (per iteration)
Basic PSO	Standard PSO	None	$\mathcal{O}(N_s d)$
Comprehensive learning	CLPSO	Learning exemplar selection from full population	$\mathcal{O}(N_s d + S_L)$ (S_L : Selection)
Two-population-based	SLPSO, IILPSO, SELPSO	Subpopulation split, inter-group learning	$\mathcal{O}(N_s d + S_R)$ (S_R : Population sorting)
Ensemble strategy	EPSO, SDPSO	Strategy pool selection, probabilistic weighting	$\mathcal{O}(KN_s d)$ (K : # strategies)
Adaptive/Granular learning	AGLPSO, TPLPSO	Clustering or logistic regression	$\mathcal{O}(N_s d + C)$ (C : clustering cost, typically $\mathcal{O}(N \log N)$)
Multi-population/Cooperative	ITLPSO, DMS-PSO	Sub-swarm interaction, migration or fusion	$\mathcal{O}(SN_s d + M)$ (S : subpopulations, M : migration overhead)
Other learning-based	MFLPSO, AGLPSO	Model training/inference (e.g., forgetting vector)	$\mathcal{O}(N_s d + M_L)$ (M_L : model cost)
Dynamic neighborhood	DNLPSO	Neighbor discovery (random or structured)	$\mathcal{O}(N_s d)$
Hybrid/Memetic PSO	Memetic PSO variants	Local search per particle	$\mathcal{O}(N_s d L)$ (L : local search complexity)

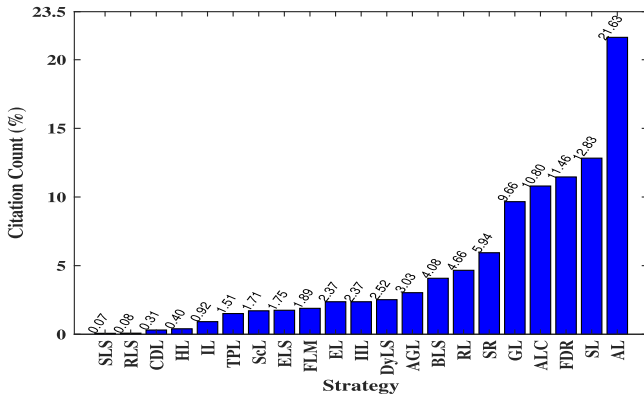


Fig. 14. Citations in the percentage of learning strategies.

we present key questions and attempt to provide possible answers based on our observations and knowledge.

8.1. Open challenges and discussion

Despite the significant advancements in PSO learning strategies (See Fig. 14), several challenges remain unresolved, limiting their effectiveness across diverse optimization problems. While numerous strategies have been proposed, their integration, adaptability, and impact on fundamental PSO behavior require deeper investigation. This section highlights key open questions and research challenges that need to be addressed to further enhance the performance and applicability of PSO.

Q.1. Which individual strategy or combination of strategies can significantly improve PSO performance across different problem domains?

As we discussed in the previous sections, PSO has been enhanced through various learning strategies, each designed to address specific challenges such as premature convergence, lack of diversity, and slow adaptability to complex landscapes. While no single approach is universally optimal across all problem domains, several strategies have proven effective in different scenarios. CL and DNL introduced a mechanism where particles learn from multiple exemplars rather than a single best neighbor, improving diversity and enhancing exploration in multimodal problems. AL dynamically adjusts key parameters based on swarm behavior, allowing the algorithm to balance exploration and

exploitation more effectively across different optimization landscapes. Hybrid learning strategies, e.g., Genetic learning, incorporate evolutionary operators like selection, crossover, and mutation to maintain diversity and prevent stagnation, making it particularly useful in large-scale optimization. Additionally, multi-swarm-based strategies, such as cooperative or hierarchical, divide the population into multiple interacting sub-swarms, allowing for better local and global search capabilities. Hybrid approaches that integrate techniques such as OBL, OL, and SL can further enhance performance by guiding particles toward promising regions of the search space while maintaining a level of randomness necessary for exploration. More recent developments, such as heterogeneous and ensemble, leverage multiple search behaviors simultaneously, making them effective in highly complex and dynamic environments. The strategy dynamics and ranking-based strategies can significantly improve PSO performance. These strategies collectively contribute to making PSO more robust, scalable, and adaptable for a wide range of applications, from engineering optimization to machine learning and real-world decision-making problems.

Q.2. How can these learning strategies be systematically integrated to maximize their effectiveness across various optimization landscapes?

To systematically integrate learning strategies and maximize their effectiveness across diverse optimization landscapes, an adaptive strategy selection framework can be developed. Such a framework would dynamically adjust the employed strategies based on the characteristics of the problem, ensuring an optimal balance between exploration and exploitation. Multi-swarm and hybrid models offer another promising approach, where different subpopulations utilize distinct learning strategies, enhancing search diversity and adaptability. Additionally, machine learning techniques, such as reinforcement learning and surrogate modeling, can be leveraged to automate strategy selection, allowing the algorithm to switch between strategies in real time based on performance feedback. Furthermore, evolving strategy ensembles, where different strategies compete and adapt based on historical performance, can provide a more flexible and efficient approach to solving complex optimization problems. By integrating these techniques, PSO can be made more robust, scalable, and capable of handling a wide range of optimization challenges.

Q.3. How do different learning mechanisms impact fundamental PSO behavior, including convergence speed, robustness, and the exploration-exploitation trade-off?

Different learning mechanisms in PSO significantly influence key aspects of its performance, including convergence speed, robustness, and

the exploration–exploitation balance. Comprehensive learning strategies, such as CLPSO, enhance diversity by allowing particles to learn from multiple exemplars, making them effective in multimodal landscapes but sometimes leading to slower convergence. Adaptive and self-learning strategies dynamically adjust search behaviors, enabling PSO to respond to changing landscapes and improving robustness in real-world applications. Two-swarm and multi-swarm models, such as HCLPSO and cooperative PSO, further refine diversity control by allowing different subpopulations to specialize in exploration or exploitation, reducing the likelihood of stagnation in local optima. Genetic learning mechanisms, such as those in GL-PSO, introduce evolutionary operators like crossover and mutation to enhance solution diversity and prevent premature convergence. However, these techniques may come at the cost of increased computational complexity. Overall, selecting an appropriate learning mechanism requires careful consideration of the optimization landscape and the trade-offs between convergence speed, robustness, and exploration–exploitation dynamics.

Q.4. What methodologies should be used to fairly and systematically assess the effectiveness of different PSO learning strategies?

To fairly and systematically assess the effectiveness of different PSO learning strategies, a multi-faceted evaluation framework is necessary. Benchmark testing on a diverse set of optimization problems, including unimodal, multimodal, high-dimensional, dynamic, and constrained problems, provides a comprehensive assessment of each strategy's strengths and limitations. Statistical analysis, using non-parametric tests such as the Wilcoxon rank-sum test and the Friedman test, ensures that performance differences are statistically significant rather than due to random variations. Ablation studies further contribute to the understanding of PSO variants by isolating the impact of individual learning components in hybrid or adaptive models. Computational efficiency metrics, including convergence speed, scalability, and memory consumption, may be essential to assess the trade-offs between solution quality and computational cost.

In addition to traditional performance evaluations, explainability methods such as SHAP (Shapley Additive Explanations), LIME (Local Interpretable Model-Agnostic Explanations), and PDP (Partial Dependence Plots) can provide insights into the decision-making process of different PSO strategies. SHAP values can help quantify the influence of different learning mechanisms on particle updates, LIME can offer local interpretability of how specific strategies affect convergence, and PDP can visualize the effect of key parameters on PSO behavior. These explainability techniques not only enhance the interpretability of PSO strategies but also aid in identifying potential biases, improving parameter tuning, and guiding future modifications. Finally, real-world applications in engineering design, machine learning, and decision-making provide practical validation, ensuring that PSO strategies are not only effective in theoretical benchmarks but also applicable to real-world optimization challenges.

Q.5. How can adaptive learning strategies dynamically adjust based on search dynamics to enhance global optimization performance?

Adaptive learning strategies dynamically adjust based on search dynamics to enhance global optimization performance through various mechanisms. One effective approach is feedback-driven parameter adaptation, where methods like Adaptive PSO (APSO) modify key parameters such as inertia weight and learning coefficients in response to real-time population behavior. By continuously analyzing convergence trends, the algorithm can dynamically shift between exploration and exploitation, improving search efficiency and preventing premature convergence.

Another promising technique is evolutionary control mechanisms, where self-adaptive and ensemble-based strategies enable PSO to transition between different learning behaviors during optimization. These strategies adjust based on the complexity of the problem and variations in the search landscape, reducing the risk of stagnation in

local optima. By integrating multiple learning mechanisms, PSO can dynamically choose the most effective approach at different stages of the optimization process.

In addition, reinforcement learning and meta-learning provide a sophisticated way to enhance adaptability. Techniques such as deep reinforcement learning (RL) and evolutionary algorithms can optimize PSO's learning strategy selection in real-time. By framing strategy selection as a learning task, these methods allow PSO to continuously refine its search behavior based on past performance, making it more effective across a wide range of problem domains.

Finally, hierarchical and multi-agent learning introduces a structured approach to swarm intelligence. By assigning different roles to particles, where some prioritize exploration while others focus on exploitation, PSO can maintain a dynamic balance throughout the search process. This structured learning mechanism is particularly beneficial in complex, high-dimensional optimization tasks, ensuring both diversity in search exploration and efficiency in convergence.

Q.6. What are the limitations of existing PSO learning mechanisms, and how can future research address them to develop self-adaptive, intelligent PSO variants?

Existing PSO learning mechanisms face several limitations that hinder their effectiveness in complex optimization scenarios. One of the primary challenges is premature convergence, which is particularly problematic in high-dimensional search spaces [11,109,170]. Many PSO variants struggle to maintain diversity in the swarm, leading to early stagnation at suboptimal solutions [64]. To address this, future research could explore adaptive multi-objective learning strategies that dynamically adjust exploration and exploitation pressures based on real-time feedback from the search process.

Another significant limitation is parameter tuning, which remains a bottleneck in PSO's performance. Many existing approaches require extensive manual fine-tuning of inertia weight, learning mechanism, and cognitive and social parameters, making them less adaptable to different problem landscapes [21]. Future advancements in self-adaptive and meta-learning-based PSO variants could automate parameter selection, allowing the algorithm to dynamically adjust its behavior based on search dynamics without extensive human intervention.

PSO also struggles with handling dynamic environments, as many real-world optimization problems involve continuously changing landscapes. Traditional PSO variants cannot adapt to shifting optima over time [18,118,135]. Learning-driven PSO models that incorporate memory-based or reinforcement learning techniques could be developed to overcome this, enabling the swarm to retain useful knowledge from past iterations and adapt more effectively to dynamic changes.

A crucial concern is the computational cost associated with advanced learning mechanisms. Many sophisticated PSO variants integrate complex learning strategies, increasing the computational burden and reducing efficiency, especially for large-scale problems. Future research could explore efficient surrogate-assisted or distributed PSO models, which leverage machine learning surrogates or parallel computing to reduce computational overhead while maintaining optimization quality. Moreover, one fundamental limitation of PSO variants lies in their lack of crossover and selection operators, which are integral to many EAs such as DE. These additional operators in DE provide greater flexibility and enhance search dynamics by enabling more diverse exploration and refined exploitation. In contrast, standard swarm algorithms rely solely on velocity and position updates, which may limit their ability to escape local optima or adapt to complex problem landscapes. As a result, PSO variants often underperform the best DE-based algorithms in terms of global search power and solution diversity, as can be observed from the CEC competitions.

Finally, a key limitation is the lack of explainability in PSO learning mechanisms. Many strategies work well empirically, but there is a limited understanding of why certain approaches perform better in

specific scenarios. Developing interpretable PSO mechanisms with explainable AI (XAI) techniques could provide deeper insights into swarm behavior, guiding the integration of PSO in real-world decision-making processes. Addressing these challenges through adaptive and intelligent learning strategies would be critical for advancing PSO's capabilities and ensuring its applicability to increasingly complex optimization problems.

8.2. Potential research directions

In the continuation of the previous section, to foster future advancements, drawing inspiration from emerging trends and incorporating more effective methodologies would be essential. Based on our literature review, we highlighted key unresolved issues that warrant deeper investigation within the swarm intelligence community. Additionally, we outline potential areas for enhancement, including refining learning strategies, improving methodologies, and expanding their applications.

Future research could prioritize the development of fully adaptive and self-learning mechanisms, enabling PSO to dynamically select and fine-tune learning strategies based on problem complexity. Integrating reinforcement learning, meta-learning, and transfer learning can facilitate real-time adaptability, enhancing PSO's performance across diverse optimization landscapes. Additionally, hybridizing PSO's learning strategies with deep learning, evolutionary algorithms (e.g., DE, GA, ACO), and ensemble-based techniques can improve search efficiency and solution quality. For instance, incorporating EA's crossover and selection operators into PSO, such as in HCLPSO variants, can improve convergence quality while maintaining PSO's natural advantage of faster convergence speed. This hybridization can potentially balance exploration and exploitation more effectively, enabling PSO to perform competitively across a wider range of optimization problems. Developing principled frameworks for such hybrid algorithms, along with theoretical studies and empirical validations, will be a promising direction to enhance the robustness and generalizability of PSO learning strategies.

Machine learning-assisted PSO can predict optimal learning strategies, while hybrid evolutionary frameworks can leverage multiple algorithms to achieve a better exploration–exploitation balance. Enhancing scalability for large-scale, high-dimensional, and multi-objective optimization problems is another critical direction. Furthermore, the development of context-aware, chaos-based, or quantum-inspired learning strategies can improve robustness in dynamic and uncertain environments, ensuring adaptability to real-world applications.

To advance the field, establishing standardized benchmarks for evaluating PSO learning strategies and conducting theoretical convergence analyses will provide deeper insights into their effectiveness. The XAI methods, such as SHAP, LIME, and PDP, can be used to analyze the insights into the decision-making process of different PSO strategies. Finally, customizing learning mechanisms for real-time optimization in domains such as robotics, energy systems, and healthcare can bridge the gap between theoretical advancements and practical implementations.

8.3. Recommendations and future directions

Building upon the comprehensive analysis of learning strategies in Particle Swarm Optimization (PSO), we propose the following recommendations to guide future research and development:

- (i) **Integration of large language models (LLMs) with EAs:** Recent studies highlight the synergistic potential of combining LLMs with EA techniques [171]. LLMs can automate the design and tuning of evolutionary algorithms, especially PSO, while EA (like PSO) can enhance LLMs through optimization of prompts, hyperparameters, and architectures. Exploring this bidirectional

integration can lead to more robust and adaptive optimization frameworks.

- (ii) **Development of meta-evolutionary frameworks:** The concept of meta-evolution [172], where evolutionary strategies evolve themselves, has shown promise in enhancing adaptability and performance across diverse optimization landscapes. Implementing such frameworks in PSO can lead to algorithms that self-optimize their parameters and strategies in real time.
- (iii) **Adoption of future-conditioned strategies:** Incorporating future-conditioned¹ approaches [173], where optimization objectives are specified in advance, can enable PSO algorithms to generate solutions that align with long-term goals. This strategy has been effective in recommendation systems and can be adapted to guide PSO in dynamic environments.
- (iv) **Enhancement of explainability and interpretability:** As PSO algorithms become more complex, integrating explainable AI techniques such as SHAP, LIME, and PDP can provide insights into their decision-making processes. This transparency is crucial for applications in sensitive domains where understanding the rationale behind solutions is essential [174].
- (v) **Standardization of benchmarking protocols:** Establishing standardized benchmarks and evaluation protocols for PSO variants will facilitate fair comparisons and accelerate progress in the field. This includes defining common datasets, performance metrics, and testing scenarios.
- (vi) **Application to real-world and dynamic problems:** Future research should focus on deploying PSO in real-world scenarios characterized by dynamic, uncertain, or constrained environments. Applications in areas such as smart grids, financial modeling, and autonomous systems can validate the practical utility of advanced PSO strategies.

These recommendations aim to foster the development of more intelligent, scalable, and versatile PSO algorithms capable of addressing the evolving challenges in both academic research and industrial optimization tasks. We summarize the key challenges and research directions in Table 13. This table provides a high-level overview of the areas requiring further investigation and outlines corresponding future strategies to address them.

9. Conclusion

This review paper has comprehensively examined and categorized a wide array of learning strategies proposed to improve the performance of particle swarm optimization (PSO). We systematically analyzed single-swarm, multi-swarm, adaptive, hybrid, and cooperative models, providing a unified taxonomy that fills a significant gap in the existing literature. Through critical comparisons and experimental insights, we have demonstrated how these learning strategies influence convergence speed, robustness, and the balance between exploration and exploitation. Despite the extensive coverage, this survey is not without limitations. While we focused on representative learning-enhanced PSO variants, some niche or application-specific strategies may not have been fully explored due to space constraints. Additionally, empirical comparisons across studies are sometimes constrained by differences in benchmark problems, parameter settings, and evaluation criteria.

From our analysis, we identify several key challenges and opportunities for future work: designing fully adaptive and intelligent PSO

¹ Future-conditioned strategies refer to optimization methods that incorporate anticipated future objectives or constraints into their search process, enabling more proactive and goal-aligned decision-making in dynamic and uncertain environments.

Table 13
Summary of challenges and future research directions in PSO learning strategies.

Challenges/Needs	Future research directions
Static or fixed learning mechanisms lack adaptability to problem complexity	Develop fully adaptive and self-learning PSO strategies using reinforcement learning, meta-learning, and transfer learning
Limited exploration–exploitation balance in existing strategies	Hybridize PSO with evolutionary operators (e.g., DE, GA crossover and selection) or ensemble learning techniques
Insufficient scalability to high-dimensional or large-scale problems	Design scalable and distributed PSO frameworks tailored for large-scale and high-dimensional spaces
Lack of robustness in dynamic or uncertain environments	Introduce context-aware, chaos-based, or quantum-inspired learning strategies for enhanced adaptability
Limited theoretical understanding of PSO dynamics and convergence	Conduct formal convergence analysis and theoretical investigations into adaptive learning behavior
Absence of standard benchmarks and comparison protocols	Establish standardized evaluation benchmarks with common datasets, metrics, and test scenarios
Opaque decision-making processes in complex strategies	Integrate explainable AI (XAI) tools (e.g., SHAP, LIME, PDP) to interpret decision-making in PSO variants
Need for PSO deployment in real-world, dynamic, and constrained environments	Customized learning mechanisms for domain-specific, real-time applications in robotics, energy systems, and healthcare
Need for higher-level automation and generalization in algorithm design	Develop meta-evolutionary frameworks where PSO can self-evolve its parameters and learning strategies
The growing importance of human-interpretable and goal-aligned optimization	Adopting future-conditioned strategies to anticipate and align with long-term or future objectives in dynamic settings
The gap in synergizing PSO with recent advances in machine learning	Exploring bidirectional integration of PSO and large language models (LLMs) for automated design, tuning, and prompt optimization

variants, improving scalability for large-scale and multi-objective problems, integrating explainability and interpretability, and standardizing benchmarking protocols for fair comparison. Emerging trends such as meta-evolution, reinforcement/meta/transfer learning integration, and future-conditioned learning strategies offer promising research avenues.

Furthermore, the practical relevance of PSO continues to grow. The advanced learning strategies reviewed here have significant potential in real-world applications, including smart grid optimization, electric vehicle scheduling, healthcare diagnostics, financial portfolio design, autonomous navigation, and scalable machine learning model tuning. Bridging the gap between theoretical innovation and real-world deployment will be critical for advancing PSO research and unlocking its full potential across dynamic and complex domains.

CRediT authorship contribution statement

Dikshit Chauhan: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Conceptualization. **Shivani:** Writing – review & editing, Writing – original draft. **Ponnuthurai N. Suganthan:** Writing – review & editing, Validation, Supervision.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Declaration of competing interest

All the authors declare that they do not have any conflict of Interest.

Acknowledgment

The authors are thankful to the National University of Singapore and Dr B R Ambedkar National Institute of Technology Jalandhar, Punjab, India for the necessary support for this work.

Data availability

No data was used for the research described in the article.

References

- [1] Bing Wang, Liming Wang, Bo Zou, Huifang Xiao, Xiaoxi Ding, Yimin Shao, A new particle swarm optimization aided evolutionary digital filter for noise cancellation in early fault diagnosis of rotating machinery, *Swarm Evol. Comput.* 83 (2023) 101407.
- [2] Yuxiang Yang, Ke Zhao, Yuelin Zhao, Feng Wu, Changyi Chen, Jun Yan, Wanxie Zhong, UA-CRD, a computational framework for uncertainty analysis of control rod drop with time-variant epistemic uncertain parameters, *Ann. Nucl. Energy* 195 (2024) 110171.
- [3] Qichao Wu, Xuwen Xia, Haojie Song, Hui Zeng, Xing Xu, Yinglong Zhang, Fei Yu, Hongrun Wu, A neighborhood comprehensive learning particle swarm optimization for the vehicle routing problem with time windows, *Swarm Evol. Comput.* 84 (2024) 101425.
- [4] Zeng Meng, Gang Li, Xuan Wang, Sadiq M Sait, Ali Rıza Yıldız, A comparative study of metaheuristic algorithms for reliability-based design optimization problems, *Arch. Comput. Methods Eng.* 28 (2021) 1853–1869.
- [5] Zeng Meng, Lin Kong, Jiaxiang Yi, Hao Peng, Optimum-pursuing method for constrained optimization and reliability-based design optimization problems using Kriging model, *Comput. Methods Appl. Mech. Engrg.* 420 (2024) 116704.
- [6] Jian Xing, Yangjun Luo, Yuan Deng, Sicheng Wu, Yundong Gai, Topology optimization design of deformable flexible thermoelectric devices for voltage enhancement, *Eng. Optim.* 55 (10) (2023) 1686–1703.
- [7] Qingke Zhang, Weiguo Liu, Xiangxu Meng, Bo Yang, Athanasios V Vasilakos, Vector coevolving particle swarm optimization algorithm, *Inform. Sci.* 394 (2017) 273–298.
- [8] Yuelin Zhao, Feng Wu, Jianhua Pang, Wanxie Zhong, New heterogeneous comprehensive learning particle swarm optimizer enhanced with low-discrepancy sequences and conjugate gradient method, *Swarm Evol. Comput.* 93 (2025) 101848.
- [9] Kenneth Sörensen, Metaheuristics—the metaphor exposed, *Int. Trans. Oper. Res.* 22 (1) (2015) 3–18.
- [10] James Kennedy, Russell Eberhart, Particle swarm optimization, in: *Proceedings of ICNN'95-International Conference on Neural Networks*, vol. 4, IEEE, 1995, pp. 1942–1948.
- [11] Essam H Houssein, Ahmed G Gad, Kashif Hussain, Ponnuthurai Nagarathnam Suganthan, Major advances in particle swarm optimization: theory, analysis, and application, *Swarm Evol. Comput.* 63 (2021) 100868.
- [12] Nianyin Zeng, Y.S. Hung, Yurong Li, Min Du, A novel switching local evolutionary PSO for quantitative analysis of lateral flow immunoassay, *Expert Syst. Appl.* 41 (4) (2014) 1708–1715.

- [13] Tareq M Shami, Ayman A El-Saleh, Mohammed Alswaitti, Qasem Al-Tashi, Mhd Amen Summakieh, Seyedali Mirjalili, Particle swarm optimization: A comprehensive survey, *IEEE Access* 10 (2022) 10031–10061.
- [14] Ahmed G. Gad, Particle swarm optimization algorithm and its applications: a systematic review, *Arch. Comput. Methods Eng.* 29 (5) (2022) 2531–2561.
- [15] Dikshit Chauhan, Shivani, Ran Cheng, Competitive swarm optimizer: a decade survey, *Swarm Evol. Comput.* 87 (2024) 101543.
- [16] Mahdihyeh Eslami, Hussain Shareef, Mohammad Khajehzadeh, Azah Mohamed, A survey of the state of the art in particle swarm optimization, *Res. J. Appl. Sci. Eng. Technol.* 4 (9) (2012) 1181–1197.
- [17] Janmenjoy Nayak, H Swapnarekha, Bighnaraj Naik, Gaurav Dhiman, S Vimal, 25 years of particle swarm optimization: Flourishing voyage of two decades, *Arch. Comput. Methods Eng.* 30 (3) (2023) 1663–1725.
- [18] Han Li, Juan Li, Peishu Wu, Yancheng You, Nianyin Zeng, A ranking-system-based switching particle swarm optimizer with dynamic learning strategies, *Neurocomputing* 494 (2022) 356–367.
- [19] Adam P Piotrowski, Jaroslaw J Napiorkowski, Agnieszka E Piotrowska, Population size in particle swarm optimization, *Swarm Evol. Comput.* 58 (2020) 100718.
- [20] Zhi-Hui Zhan, Jun Zhang, Yun Li, Henry Shu-Hung Chung, Adaptive particle swarm optimization, *IEEE Trans. Syst. Man Cybern. B* 39 (6) (2009) 1362–1381.
- [21] Md Nasir, Swagatam Das, Dipankar Maity, Soumyadip Sengupta, Udit Halder, Ponnuthurai N Suganthan, A dynamic neighborhood learning based particle swarm optimizer for global numerical optimization, *Inform. Sci.* 209 (2012) 16–36.
- [22] Jing J Liang, A Kai Qin, Ponnuthurai N Suganthan, S Baskar, Comprehensive learning particle swarm optimizer for global optimization of multimodal functions, *IEEE Trans. Evol. Comput.* 10 (3) (2006) 281–295.
- [23] Nandar Lynn, Ponnuthurai Nagaratnam Suganthan, Heterogeneous comprehensive learning particle swarm optimization with enhanced exploration and exploitation, *Swarm Evol. Comput.* 24 (2015) 11–24.
- [24] Yu Wang, Bin Li, Thomas Weise, Jianyu Wang, Bo Yuan, Qiongjie Tian, Self-adaptive learning based particle swarm optimization, *Inform. Sci.* 181 (20) (2011) 4515–4538.
- [25] Hui Wang, Zhijian Wu, Shahryar Rahnamayan, Yong Liu, Mario Ventresca, Enhancing particle swarm optimization using generalized opposition-based learning, *Inform. Sci.* 181 (20) (2011) 4699–4714.
- [26] Zhi-Hui Zhan, Jun Zhang, Ou Liu, Orthogonal learning particle swarm optimization, in: *Proceedings of the 11th Annual Conference on Genetic and Evolutionary Computation*, 2009, pp. 1763–1764.
- [27] Nandar Lynn, Ponnuthurai Nagaratnam Suganthan, Ensemble particle swarm optimizer, *Appl. Soft Comput.* 55 (2017) 533–548.
- [28] Maurice Clerc, James Kennedy, The particle swarm-explosion, stability, and convergence in a multidimensional complex space, *IEEE Trans. Evol. Comput.* 6 (1) (2002) 58–73.
- [29] Wei-Neng Chen, Jun Zhang, Ying Lin, Ni Chen, Zhi-Hui Zhan, Henry Shu-Hung Chung, Yun Li, Yu-Hui Shi, Particle swarm optimization with an aging leader and challengers, *IEEE Trans. Evol. Comput.* 17 (2) (2012) 241–258.
- [30] Rui Mendes, James Kennedy, José Neves, The fully informed particle swarm: simpler, maybe better, *IEEE Trans. Evol. Comput.* 8 (3) (2004) 204–210.
- [31] Bo-Yang Qu, Ponnuthurai Nagaratnam Suganthan, Swagatam Das, A distance-based locally informed particle swarm model for multimodal optimization, *IEEE Trans. Evol. Comput.* 17 (3) (2012) 387–402.
- [32] Nandar Lynn, Mostafa Z Ali, Ponnuthurai Nagaratnam Suganthan, Population topologies for particle swarm optimization and differential evolution, *Swarm Evol. Comput.* 39 (2018) 24–35.
- [33] Yuhui Shi, Russell C. Eberhart, Empirical study of particle swarm optimization, in: *Proceedings of the 1999 Congress on Evolutionary Computation-CEC99* (Cat. No. 99TH8406), vol. 3, IEEE, 1999, pp. 1945–1950.
- [34] Dalia Yousri, Dalia Allam, MB Eteiba, Ponnuthurai Nagaratnam Suganthan, Static and dynamic photovoltaic models' parameters identification using chaotic heterogeneous comprehensive learning particle swarm optimizer variants, *Energy Convers. Manage.* 182 (2019) 546–563.
- [35] Thu Huynh Van, Sawekchai Tangaramvong, Wei Gao, Chaotic heterogeneous comprehensive learning PSO method for size and shape optimization of structures, *Eng. Appl. Artif. Intell.* 126 (2023) 107014.
- [36] Shengliang Wang, Genyou Liu, Ming Gao, Shilong Cao, Aizhi Guo, Jiachen Wang, Heterogeneous comprehensive learning and dynamic multi-swarm particle swarm optimizer with two mutation operators, *Inform. Sci.* 540 (2020) 175–201.
- [37] Dalia Yousri, Hany M. Hasanien, Ahmed Fathy, Parameters identification of solid oxide fuel cell for static and dynamic simulation using comprehensive learning dynamic multi-swarm marine predators algorithm, *Energy Convers. Manage.* 228 (2021) 113692.
- [38] Yulian Cao, Han Zhang, Wenfeng Li, Mengchu Zhou, Yu Zhang, Wanpracha Art Chaovalitwongse, Comprehensive learning particle swarm optimization algorithm with local search for multimodal functions, *IEEE Trans. Evol. Comput.* 23 (4) (2018) 718–731.
- [39] Madhubanti Maitra, Amitava Chatterjee, A hybrid cooperative-comprehensive learning based PSO algorithm for image segmentation using multilevel thresholding, *Expert Syst. Appl.* 34 (2) (2008) 1341–1350.
- [40] Şaban Gülcü, Halife Kodaz, A novel parallel multi-swarm algorithm based on comprehensive learning particle swarm optimization, *Eng. Appl. Artif. Intell.* 45 (2015) 33–45.
- [41] Yiwen Zhong, Juan Lin, Lijin Wang, Hui Zhang, Discrete comprehensive learning particle swarm optimization algorithm with Metropolis acceptance criterion for traveling salesman problem, *Swarm Evol. Comput.* 42 (2018) 77–88.
- [42] Yiyang Zhang, Maode Ma, Zhigang Jin, Comprehensive learning jaya algorithm for parameter extraction of photovoltaic models, *Energy* 211 (2020) 118644.
- [43] Yiyang Zhang, Zhigang Jin, Comprehensive learning jaya algorithm for engineering design optimization problems, *J. Intell. Manuf.* 33 (5) (2022) 1229–1253.
- [44] Kai Zhang, Qiujun Huang, Yimin Zhang, Enhancing comprehensive learning particle swarm optimization with local optima topology, *Inform. Sci.* 471 (2019) 1–18.
- [45] Xiang Yu, Xueqing Zhang, Enhanced comprehensive learning particle swarm optimization, *Appl. Math. Comput.* 242 (2014) 265–276.
- [46] Lijing Tan, Fuyong Lin, Hong Wang, Adaptive comprehensive learning bacterial foraging optimization and its application on vehicle routing problem with time windows, *Neurocomputing* 151 (2015) 1208–1215.
- [47] Anping Lin, Wei Sun, Hongshan Yu, Guohua Wu, Hongwei Tang, Adaptive comprehensive learning particle swarm optimization with cooperative archive, *Appl. Soft Comput.* 77 (2019) 533–546.
- [48] Mohamed Abd Elaziz, Dalia Yousri, Ahmad O Aseeri, Laith Abualigah, Mohammed AA Al-qaness, Ahmed A Ewees, Fractional-order modified heterogeneous comprehensive learning particle swarm optimizer for intelligent disease detection in IoT environment, *Swarm Evol. Comput.* 84 (2024) 101430.
- [49] Xiaoding Meng, Hecheng Li, Heterogeneous pbest-guided comprehensive learning particle swarm optimization, *Appl. Soft Comput.* (2024) 111874.
- [50] Zhenyu Wang, Yichun Chen, Sheng Ding, Deli Liang, Huan He, A novel particle swarm optimization algorithm with Lévy flight and orthogonal learning, *Swarm Evol. Comput.* 75 (2022) 101207.
- [51] Shinn-Ying Ho, Hung-Sui Lin, Weei-Hurng Liauh, Shinn-Jang Ho, OPSO: Orthogonal particle swarm optimization and its application to task assignment problems, *IEEE Trans. Syst. Man, Cybernetics- Part A: Syst. Humans* 38 (2) (2008) 288–298.
- [52] Ashraf Darwish, Dalia Ezzat, Aboul Ella Hassanien, An optimized model based on convolutional neural networks and orthogonal learning particle swarm optimization algorithm for plant diseases diagnosis, *Swarm Evol. Comput.* 52 (2020) 100616.
- [53] Zi-Jia Wang, Zhi-Hui Zhan, Ke-Jing Du, Zhi-Wen Yu, Jun Zhang, Orthogonal learning particle swarm optimization with variable relocation for dynamic optimization, in: *2016 IEEE Congress on Evolutionary Computation, CEC, IEEE*, 2016, pp. 594–600.
- [54] Wenlei Bai, Fanlin Meng, Ming Sun, Haoxiang Qin, Richard Allmendinger, Kwang Y Lee, Differential evolutionary particle swarm optimization with orthogonal learning for wind integrated optimal power flow, *Appl. Soft Comput.* 160 (2024) 111662.
- [55] Guojiang Xiong, Dongyuan Shi, Orthogonal learning competitive swarm optimizer for economic dispatch problems, *Appl. Soft Comput.* 66 (2018) 134–148.
- [56] Wenlei Bai, Ibrahim Eke, Kwang Y. Lee, An improved artificial bee colony optimization algorithm based on orthogonal learning for optimal power flow problem, *Control Eng. Pract.* 61 (2017) 163–172.
- [57] Wei-feng Gao, San-yan Liu, Ling-ling Huang, A novel artificial bee colony algorithm based on modified search equation and orthogonal learning, *IEEE Trans. Cybern.* 43 (3) (2013) 1011–1024.
- [58] Dikshit Chauhan, Anupam Yadav, An archive-based self-adaptive artificial electric field algorithm with orthogonal initialization for real-parameter optimization problems, *Appl. Soft Comput.* 150 (2024) 111109.
- [59] Chunlei Li, Libao Deng, Liyan Qiao, Lili Zhang, An efficient differential evolution algorithm based on orthogonal learning and elites local search mechanisms for numerical optimization, *Knowl.-Based Syst.* 235 (2022) 107636.
- [60] Abhishek Kumar, Partha P. Biswas, Ponnuthurai N. Suganthan, Differential evolution with orthogonal array-based initialization and a novel selection strategy, *Swarm Evol. Comput.* 68 (2022) 101010.
- [61] Xiangtao Li, Jianan Wang, Minghao Yin, Enhancing the performance of cuckoo search algorithm using orthogonal learning method, *Neural Comput. Appl.* 24 (2014) 1233–1247.
- [62] Essam H Houssein, Bahaa El-din Helmy, Hegazy Rezk, Ahmed M Nassef, An enhanced archimedes optimization algorithm based on local escaping operator and orthogonal learning for PEM fuel cell parameter identification, *Eng. Appl. Artif. Intell.* 103 (2021) 104309.
- [63] Hao Chen, Ali Asghar Heidari, Xuehua Zhao, Lejun Zhang, Huiling Chen, Advanced orthogonal learning-driven multi-swarm sine cosine optimization: Framework and case studies, *Expert Syst. Appl.* 144 (2020) 113113.
- [64] Guiping Xu, Quanlong Cui, Xiaohu Shi, Hongwei Ge, Zhi-Hui Zhan, Heow Pueh Lee, Yanchun Liang, Ran Tai, Chunguo Wu, Particle swarm optimization based on dimensional learning strategy, *Swarm Evol. Comput.* 45 (2019) 33–51.

- [65] Xinyang Liu, Yifan Wang, Miao lei Zhou, Dimensional learning strategy-based grey wolf optimizer for solving the global optimization problem, *Comput. Intell. Neurosci.* 2022 (1) (2022) 3603607.
- [66] Zhendong Wang, Lei Shu, Shuxin Yang, Zhiyuan Zeng, Daojing He, Sammy Chan, Multi-population dynamic grey wolf optimizer based on dimension learning and Laplace mutation for global optimization, *Expert Syst. Appl.* 265 (2025) 125863.
- [67] Daiquan Wen, Lin Huo, A dimensional learning squirrel search algorithm based on roulette strategy, in: 2022 Asia Conference on Algorithms, Computing and Machine Learning, CACML, IEEE, 2022, pp. 32–38.
- [68] Zeyu Wang, Xizheng Zhang, Stability dispatch of microgrid based on improved dimensional learning particle swarm optimization algorithm, 2021.
- [69] Changhe Li, Shengxiang Yang, An adaptive learning particle swarm optimizer for function optimization, in: 2009 IEEE Congress on Evolutionary Computation, IEEE, 2009, pp. 381–388.
- [70] Changhe Li, Shengxiang Yang, Trung Thanh Nguyen, A self-learning particle swarm optimizer for global optimization problems, *IEEE Trans. Syst. Man Cybern. B* 42 (3) (2011) 627–646.
- [71] Feng Wang, Heng Zhang, Kangshun Li, Zhiyi Lin, Jun Yang, Xiao-Liang Shen, A hybrid particle swarm optimization algorithm using adaptive learning strategy, *Inform. Sci.* 436 (2018) 162–177.
- [72] Shikha Chaudhary, Vijay Kumar Sharma, RN Thakur, Amit Rath, Pramendra Kumar, Sachin Sharma, Modified particle swarm optimization based on aging leaders and challengers model for task scheduling in cloud computing, *Math. Probl. Eng.* 2023 (1) (2023) 3916735.
- [73] Rudra Pratap Singh, V. Mukherjee, S.P. Ghoshal, Particle swarm optimization with an aging leader and challengers algorithm for optimal power flow problem with FACTS devices, *Int. J. Electr. Power Energy Syst.* 64 (2015) 1185–1196.
- [74] Rudra Pratap Singh, V. Mukherjee, S.P. Ghoshal, Optimal reactive power dispatch by particle swarm optimization with an aging leader and challengers, *Appl. Soft Comput.* 29 (2015) 298–309.
- [75] Rudra Pratap Singh, V. Mukherjee, S.P. Ghoshal, Particle swarm optimization with an aging leader and challengers algorithm for the solution of optimal power flow problem, *Appl. Soft Comput.* 40 (2016) 161–177.
- [76] C.M. Fu, Chao Jiang, G.S. Chen, Q.M. Liu, An adaptive differential evolution algorithm with an aging leader and challengers mechanism, *Appl. Soft Comput.* 57 (2017) 60–73.
- [77] Amal Moharam, Mostafa A. El-Hosseini, Hesham A. Ali, Design of optimal PID controller using hybrid differential evolution and particle swarm optimization with an aging leader and challengers, *Appl. Soft Comput.* 38 (2016) 727–737.
- [78] Wei Hong Lim, Nor Ashidi Mat Isa, An adaptive two-layer particle swarm optimization with elitist learning strategy, *Inform. Sci.* 273 (2014) 49–72.
- [79] Wei Hong Lim, Nor Ashidi Mat Isa, Bidirectional teaching and peer-learning particle swarm optimization, *Inform. Sci.* 280 (2014) 111–134.
- [80] Han Huang, Hu Qin, Zhifeng Hao, Andrew Lim, Example-based learning particle swarm optimization for continuous optimization, *Inform. Sci.* 182 (1) (2012) 125–138.
- [81] Thanmaya Peram, Kalyan Veeramachaneni, Chilukuri K. Mohan, Fitness-distance-ratio based particle swarm optimization, in: Proceedings of the 2003 IEEE Swarm Intelligence Symposium. SIS'03 (Cat. No. 03EX706), IEEE, 2003, pp. 174–181.
- [82] Christopher W. Cleghorn, Andries P. Engelbrecht, Fitness-distance-ratio particle swarm optimization: stability analysis, in: Proceedings of the Genetic and Evolutionary Computation Conference, 2017, pp. 12–18.
- [83] Xiaodong Li, A multimodal particle swarm optimizer based on fitness euclidean-distance ratio, in: Proceedings of the 9th Annual Conference on Genetic and Evolutionary Computation, 2007, pp. 78–85.
- [84] Jing J. Liang, Bo-Yang Qu, X.B. Mao, Ben Niu, D.Y. Wang, Differential evolution based on fitness euclidean-distance ratio for multimodal optimization, *Neurocomputing* 137 (2014) 252–260.
- [85] Zhang Ming, Tian Na, Zhicheng Ji, Balanced bee colony algorithm based on fitness euclidean-distance ratio, *J. Syst. Simul.* 27 (5) (2020) 980–989.
- [86] Zhigang Ren, Aimin Zhang, Changyun Wen, Zuren Feng, A scatter learning particle swarm optimization algorithm for multimodal problems, *IEEE Trans. Cybern.* 44 (7) (2013) 1127–1140.
- [87] Wei Li, Peng Liang, Bo Sun, Yafeng Sun, Ying Huang, Reinforcement learning-based particle swarm optimization with neighborhood differential mutation strategy, *Swarm Evol. Comput.* 78 (2023) 101274.
- [88] Haiyun Qiu, Bowen Xue, Qinge Xiao, Ben Niu, Q-learning based particle swarm optimization with multi-exemplar and elite learning, in: International Conference on Intelligent Computing, Springer, 2023, pp. 310–321.
- [89] Daniel Hein, Alexander Hentschel, Thomas Runkel, Steffen Udluft, Particle swarm optimization for generating interpretable fuzzy reinforcement learning policies, *Eng. Appl. Artif. Intell.* 65 (2017) 87–98.
- [90] Xiaodong Meng, Hecheng Li, Anshan Chen, Multi-strategy self-learning particle swarm optimization algorithm based on reinforcement learning, *Math. Biosci. Eng.* 20 (5) (2023) 8498–8530.
- [91] Leyong Hou, Debin Fan, Junjie Cheng, Honglian Wu, Hu Peng, Changshou Deng, A multi-strategy particle swarm optimization framework based on deep reinforcement learning, in: 2023 15th International Conference on Advanced Computational Intelligence, ICACI, IEEE, 2023, pp. 1–8.
- [92] Yang Tang, Zidong Wang, Jian-an Fang, Feedback learning particle swarm optimization, *Appl. Soft Comput.* 11 (8) (2011) 4713–4725.
- [93] Yue-Jiao Gong, Jing-Jing Li, Yicong Zhou, Yun Li, Henry Shu-Hung Chung, Yu-Hui Shi, Jun Zhang, Genetic learning particle swarm optimization, *IEEE Trans. Cybern.* 46 (10) (2015) 2277–2290.
- [94] Samrat L. Sabat, Layak Ali, Siba K. Udgata, Integrated learning particle swarm optimizer for global optimization, *Appl. Soft Comput.* 11 (1) (2011) 574–584.
- [95] Ben Niu, Huali Huang, Lijing Tan, Jane Jing Liang, Multi-swarm particle swarm optimization with a center learning strategy, in: Advances in Swarm Intelligence: 4th International Conference, ICSI 2013, Harbin, China, June 12–15, 2013, Proceedings, Part I 4, Springer, 2013, pp. 72–78.
- [96] Wenyong Dong, Jiangsen Tian, Xu Tang, Kang Sheng, Jin Liu, Autonomous learning adaptation for particle swarm optimization, in: 2014 IEEE Congress on Evolutionary Computation, CEC, IEEE, 2014, pp. 223–228.
- [97] Xu Chen, Huaglori Tianfield, Congli Mei, Wenli Du, Guohai Liu, Biogeography-based learning particle swarm optimization, *Soft Comput.* 21 (2017) 7519–7541.
- [98] Xu Chen, Kangji Li, Bin Xu, Zhile Yang, Biogeography-based learning particle swarm optimization for combined heat and power economic dispatch problem, *Knowl.-Based Syst.* 208 (2020) 106463.
- [99] Hongbin Dong, Hua Zhang, Shuang Han, Xiaohui Li, Xiaowei Wang, Reverse-learning particle swarm optimization algorithm based on niching technology, in: 2018 IEEE/ACIS 17th International Conference on Computer and Information Science, ICIS, IEEE, 2018, pp. 405–410.
- [100] Muhammad Rizwan Tanweer, Sundaram Suresh, Narasimhan Sundararajan, Self regulating particle swarm optimization algorithm, *Inform. Sci.* 294 (2015) 182–202.
- [101] Xu Chen, Hugo Tianfield, Wenli Du, Bee-foraging learning particle swarm optimization, *Appl. Soft Comput.* 102 (2021) 107134.
- [102] Sabine Helwig, Juergen Branke, Sanaz Mostaghim, Experimental analysis of bound handling techniques in particle swarm optimization, *IEEE Trans. Evol. Comput.* 17 (2) (2012) 259–271.
- [103] Marco A Montes De Oca, Thomas Stutzle, Ken Van den Enden, Marco Dorigo, Incremental social learning in particle swarms, *IEEE Trans. Syst. Man Cybern. B* 41 (2) (2010) 368–384.
- [104] Moayed Daneshyari, Gary G. Yen, Cultural-based multiobjective particle swarm optimization, *IEEE Trans. Syst. Man Cybern. B* 41 (2) (2010) 553–567.
- [105] Ran Cheng, Yaochu Jin, A social learning particle swarm optimization algorithm for scalable optimization, *Inform. Sci.* 291 (2015) 43–60.
- [106] Xinming Zhang, Xia Wang, Qiang Kang, Jinfeng Cheng, Differential mutation and novel social learning particle swarm optimization algorithm, *Inform. Sci.* 480 (2019) 109–129.
- [107] Haibo Yu, Ying Tan, Chaoli Sun, Jianchao Zeng, A generation-based optimal restart strategy for surrogate-assisted social learning particle swarm optimization, *Knowl.-Based Syst.* 163 (2019) 14–25.
- [108] Tao Zhao, Chengsen Chen, Hongyi Cao, Evolutionary self-organizing fuzzy system using fuzzy-classification-based social learning particle swarm optimization, *Inform. Sci.* 606 (2022) 92–111.
- [109] Xinming Zhang, Qiuying Lin, Three-learning strategy particle swarm algorithm for global optimization problems, *Inform. Sci.* 593 (2022) 289–313.
- [110] Ran Cheng, Yaochu Jin, A competitive swarm optimizer for large scale optimization, *IEEE Trans. Cybern.* 45 (2) (2014) 191–204.
- [111] Quande Qin, Shi Cheng, Qingyu Zhang, Li Li, Yuhui Shi, Particle swarm optimization with interswarm interactive learning strategy, *IEEE Trans. Cybern.* 46 (10) (2015) 2238–2251.
- [112] Baoxian Liang, Yunlong Zhao, Yang Li, A hybrid particle swarm optimization with crisscross learning strategy, *Eng. Appl. Artif. Intell.* 105 (2021) 104418.
- [113] An-bo Meng, Yu-cheng Chen, Hao Yin, Si-zhe Chen, Crisscross optimization algorithm and its application, *Knowl.-Based Syst.* 67 (2014) 218–229.
- [114] Wei Hong Lim, Nor Ashidi Mat Isa, Teaching and peer-learning particle swarm optimization, *Appl. Soft Comput.* 18 (2014) 39–58.
- [115] R. Venkata Rao, Vimal J. Savsani, D.P. Vakharia, Teaching-learning-based optimization: an optimization method for continuous non-linear large scale problems, *Inform. Sci.* 183 (1) (2012) 1–15.
- [116] Leilei Cao, Lihong Xu, Erik D. Goodman, A neighbor-based learning particle swarm optimizer with short-term and long-term memory for dynamic optimization problems, *Inform. Sci.* 453 (2018) 463–485.
- [117] Asanga Ratnaweera, Saman K. Halgamuge, Harry C. Watson, Self-organizing hierarchical particle swarm optimizer with time-varying acceleration coefficients, *IEEE Trans. Evol. Comput.* 8 (3) (2004) 240–255.
- [118] Ziang Liu, Tatsushi Nishi, Strategy dynamics particle swarm optimizer, *Inform. Sci.* 582 (2022) 665–703.
- [119] Konstantinos E. Parsopoulos, Michael N. Vrahatis, UPSO: A unified particle swarm optimization scheme, in: International Conference of Computational Methods in Sciences and Engineering, ICCMSE 2004, CRC Press, 2019, pp. 868–873.
- [120] Zi-Jia Wang, Zhi-Hui Zhan, Sam Kwong, Hu Jin, Jun Zhang, Adaptive granularity learning distributed particle swarm optimization for large-scale optimization, *IEEE Trans. Cybern.* 51 (3) (2020) 1175–1188.

- [121] Aditya Panda, Srijan Ghoshal, Amit Konar, Bonny Banerjee, Atulya K Nagar, Static learning particle swarm optimization with enhanced exploration and exploitation using adaptive swarm size, in: 2016 IEEE Congress on Evolutionary Computation, CEC, IEEE, 2016, pp. 1869–1876.
- [122] Wei Li, Jianghui Jing, Yangtao Chen, Yishan Chen, A cooperative particle swarm optimization with difference learning, *Inform. Sci.* 643 (2023) 119238.
- [123] Xuewen Xia, Ling Gui, Guoliang He, Bo Wei, Yinglong Zhang, Fei Yu, Hongrun Wu, Zhi-Hui Zhan, An expanded particle swarm optimization based on multi-exemplar and forgetting ability, *Inform. Sci.* 508 (2020) 105–120.
- [124] Taiyong Li, Jiayi Shi, Wu Deng, Zhenda Hu, Pyramid particle swarm optimization with novel strategies of competition and cooperation, *Appl. Soft Comput.* 121 (2022) 108731.
- [125] Dongping Tian, Qiu Xu, Xinhui Yao, Guangnan Zhang, Yafeng Li, Chenghu Xu, Diversity-guided particle swarm optimization with multi-level learning strategy, *Swarm Evol. Comput.* 86 (2024) 101533.
- [126] Xu Yang, Hongru Li, Multi-sample learning particle swarm optimization with adaptive crossover operation, *Math. Comput. Simulation* 208 (2023) 246–282.
- [127] Jane-Jing Liang, Ponnuthurai Nagarathnam Suganthan, Dynamic multi-swarm particle swarm optimizer, in: *Proceedings 2005 IEEE Swarm Intelligence Symposium*, 2005. SIS 2005, IEEE, 2005, pp. 124–129.
- [128] Yonggang Chen, Lixiang Li, Haipeng Peng, Jinghua Xiao, Qingtao Wu, Dynamic multi-swarm differential learning particle swarm optimizer, *Swarm Evol. Comput.* 39 (2018) 209–221.
- [129] Junfeng Liu, Xinggao Liu, Yun Wu, Zhe Yang, Jian Xu, Dynamic multi-swarm differential learning harris hawks optimizer and its application to optimal dispatch problem of cascade hydropower stations, *Knowl.-Based Syst.* 242 (2022) 108281.
- [130] Jing J. Liang, Ponnuthurai N. Suganthan, Dynamic multi-swarm particle swarm optimizer with local search, in: 2005 IEEE Congress on Evolutionary Computation, vol. 1, IEEE, 2005, pp. 522–528.
- [131] Shi-Zheng Zhao, Jing J Liang, Ponnuthurai N Suganthan, Mehmet Fatih Tasgetiren, Dynamic multi-swarm particle swarm optimizer with local search for large scale global optimization, in: 2008 IEEE Congress on Evolutionary Computation (IEEE World Congress on Computational Intelligence), IEEE, 2008, pp. 3845–3852.
- [132] Xia Xu, Yinggan Tang, Junpeng Li, Changchun Hua, Xinpeng Guan, Dynamic multi-swarm particle swarm optimizer with cooperative learning strategy, *Appl. Soft Comput.* 29 (2015) 169–183.
- [133] Dikshit Chauhan, Anupam Yadav, Optimizing the parameters of hybrid active power filters through a comprehensive and dynamic multi-swarm gravitational search algorithm, *Eng. Appl. Artif. Intell.* 123 (2023) 106469.
- [134] Shu-Kai S. Fan, Ju-Ming Chang, Dynamic multi-swarm particle swarm optimizer using parallel PC cluster systems for global optimization of large-scale multimodal functions, *Eng. Optim.* 42 (5) (2010) 431–451.
- [135] Xinmin Tao, Wenjie Guo, Xiangke Li, Qing He, Rui Liu, Junrong Zou, Fitness peak clustering based dynamic multi-swarm particle swarm optimization with enhanced learning strategy, *Expert Syst. Appl.* 191 (2022) 116301.
- [136] Yichao Tang, Xiong Li, Yinglong Zhang, Xuewen Xia, Ling Gui, Dynamic multi-swarm global particle swarm optimization, in: 2019 IEEE Congress on Evolutionary Computation, CEC, IEEE, 2019, pp. 1030–1037.
- [137] Xuewen Xia, Yichao Tang, Bo Wei, Yinglong Zhang, Ling Gui, Xiong Li, Dynamic multi-swarm global particle swarm optimization, *Computing* 102 (2020) 1587–1626.
- [138] Xu Yang, Hongru Li, Youhe Huang, An adaptive dynamic multi-swarm particle swarm optimization with stagnation detection and spatial exclusion for solving continuous optimization problems, *Eng. Appl. Artif. Intell.* 123 (2023) 106215.
- [139] Jiangnan Zhang, Kewen Xia, Ziping He, Shurui Fan, Dynamic multi-swarm differential learning quantum bird swarm algorithm and its application in random forest classification model, *Comput. Intell. Neurosci.* 2020 (1) (2020) 6858541.
- [140] Ke Lei, Yonghong Wu, Dynamic multi-swarm competitive fireworks algorithm for global optimization and engineering constraint problems, *Internat. J. Uncertain. Fuzziness Knowledge-Based Systems* 31 (04) (2023) 619–648.
- [141] Yichao Tang, Bo Wei, Yinglong Zhang, Xiong Li, Xuewen Xia, Ling Gui, Dynamic multi-swarm pigeon-inspired optimisation, *Int. J. Comput. Sci. Math.* 13 (3) (2021) 267–282.
- [142] Xiaojun Yu, Dynamic multi-swarm PSO based on k-means clustering, *Natsional'nyi Hirnychiy Universitet. Nauk. Visnyk* (6) (2015) 105.
- [143] Yubin Zhong, Yi Xiang, Yuanbin Jiang, Junyan Zhong, Jiantong Shao, A hybrid dynamic multi-swarm PSO algorithm with nelder-mead simplex search method, *J. Comput. Inf. Syst.* 9 (19) (2013) 7741–7748.
- [144] Wei-Yan Chang, Prathibha Soma, Huan Chen, Hsuan Chang, Chun-Wei Tsai, A hybrid firefly with dynamic multi-swarm particle swarm optimization for WSN deployment, *J. Internet Technol.* 24 (4) (2023) 825–836.
- [145] S-Z Zhao, Ponnuthurai N Suganthan, Quan-Ke Pan, M Fatih Tasgetiren, Dynamic multi-swarm particle swarm optimizer with harmony search, *Expert Syst. Appl.* 38 (4) (2011) 3735–3742.
- [146] Fahui Miao, Yong Wu, Guanjie Yan, Xiaomeng Si, Dynamic multi-swarm whale optimization algorithm based on elite tuning for high-dimensional feature selection classification problems, *Appl. Soft Comput.* 169 (2025) 112634.
- [147] Shi-Zheng Zhao, Ponnuthurai Nagarathnam Suganthan, Swagatham Das, Dynamic multi-swarm particle swarm optimizer with sub-regional harmony search, in: *IEEE Congress on Evolutionary Computation*, IEEE, 2010, pp. 1–8.
- [148] Wenxing Ye, Weiying Feng, Suohai Fan, A novel multi-swarm particle swarm optimization with dynamic learning strategy, *Appl. Soft Comput.* 61 (2017) 832–843.
- [149] Bo Wei, Xuewen Xia, Fei Yu, Yinglong Zhang, Xing Xu, Hongrun Wu, Ling Gui, Guoliang He, Multiple adaptive strategies based particle swarm optimization algorithm, *Swarm Evol. Comput.* 57 (2020) 100731.
- [150] Lanyu Wang, Dongping Tian, Xiaorui Gou, Zhongzhi Shi, Hybrid particle swarm optimization with adaptive learning strategy, *Soft Comput.* (2024) 1–26.
- [151] Dongping Tian, Chen Liu, Zulpiya Ghenni, Particle swarm optimization based on adaptive learning strategy, *OEC2024*, in: *International Conference on Optics, Electronics, and Communication Engineering*, vol. 13395, SPIE, 2024, pp. 464–473.
- [152] Yuheng Wu, Xingyu Zhu, Wenxu Zhao, Xiaona Xia, A novel particle swarm optimization algorithm for meta-heuristic analysis mechanism based on population learning strategies and adaptive selection of leadership particles, in: 2024 IEEE 11th International Conference on Data Science and Advanced Analytics, DSAA, IEEE, 2024, pp. 1–9.
- [153] Hao Liu, Xu-Wei Zhang, Liang-Ping Tu, A modified particle swarm optimization using adaptive strategy, *Expert Syst. Appl.* 152 (2020) 113353.
- [154] Yong Wang, Zhihao Wang, Gai-Ge Wang, Hierarchical learning particle swarm optimization using fuzzy logic, *Expert Syst. Appl.* 232 (2023) 120759.
- [155] Zedong Tang, Maoguo Gong, Yu Xie, Hao Li, Alex Kai Qin, Multi-task particle swarm optimization with dynamic neighbor and level-based inter-task learning, *IEEE Trans. Emerg. Top. Comput. Intell.* 6 (2) (2021) 300–314.
- [156] Feng Wang, Xujie Wang, Shilei Sun, A reinforcement learning level-based particle swarm optimization algorithm for large-scale optimization, *Inform. Sci.* 602 (2022) 298–312.
- [157] Juan Luis Fernandez-Martinez, Esperanza Garcia-Gonzalo, Stochastic stability analysis of the linear continuous and discrete PSO models, *IEEE Trans. Evol. Comput.* 15 (3) (2010) 405–423.
- [158] Riccardo Poli, Dynamics and stability of the sampling distribution of particle swarm optimisers via moment analysis, *J. Artif. Evol. Appl.* (2008).
- [159] Frans Van den Bergh, Andries P. Engelbrecht, A study of particle swarm optimization particle trajectories, *Inform. Sci.* 176 (8) (2006) 937–971.
- [160] N. Awad, M. Ali, J. Liang, B. Qu, P. Suganthan, Problem definitions and evaluation criteria for the cec 2017 special session and competition on single objective bound constrained real-parameter numerical optimization, 2016.
- [161] CT Yue, Kenneth V Price, Ponnuthurai N Suganthan, JJ Liang, Mostafa Z Ali, BY Qu, Noor H Awad, Partha P Biswas, Problem definitions and evaluation criteria for the CEC 2020 special session and competition on single objective bound constrained numerical optimization, *Tech. Rep.* 201911, *Comput. Intell. Lab.*, Zhengzhou Univ., Zhengzhou, China, 2019.
- [162] Janez Brest, Sao Greiner, Borko Boskovic, Marjan Mernik, Viljem Zumer, Self-adapting control parameters in differential evolution: A comparative study on numerical benchmark problems, *IEEE Trans. Evol. Comput.* 10 (6) (2006) 646–657.
- [163] Jingqiao Zhang, Arthur C. Sanderson, JADE: adaptive differential evolution with optional external archive, *IEEE Trans. Evol. Comput.* 13 (5) (2009) 945–958.
- [164] Joaquín Derrac, Salvador García, Daniel Molina, Francisco Herrera, A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms, *Swarm Evol. Comput.* 1 (1) (2011) 3–18.
- [165] Salvador García, Francisco Herrera, An extension on “statistical comparisons of classifiers over multiple data sets” for all pairwise comparisons., *J. Mach. Learn. Res.* 9 (12) (2008).
- [166] Aydin Ulaş, Olcay Taner Yıldız, Ethem Alpaydın, Cost-conscious comparison of supervised learning algorithms over multiple data sets, *Pattern Recognit.* 45 (4) (2012) 1772–1781.
- [167] Jerome Friedman, Trevor Hastie, Robert Tibshirani, *The Elements of Statistical Learning*, vol. 1, (10) Springer series in statistics New York, 2001.
- [168] Kusum Deep, Manoj Thakur, A new crossover operator for real coded genetic algorithms, *Appl. Math. Comput.* 188 (1) (2007) 895–911.
- [169] Dikshit Chauhan, Anupam Yadav, A comprehensive survey on artificial electric field algorithm: theories and applications, *Arch. Comput. Methods Eng.* 31 (5) (2024) 2663–2715.
- [170] Xuewen Xia, Ling Gui, Zhi-Hui Zhan, A multi-swarm particle swarm optimization algorithm based on dynamical topology and purposeful detecting, *Appl. Soft Comput.* 67 (2018) 126–140.
- [171] Dikshit Chauhan, Bapi Dutta, Indu Bala, Niki van Stein, Thomas Bäck, Anupam Yadav, Evolutionary computation and large language models: A survey of methods, synergies, and applications, 2025, *arXiv preprint arXiv:2505.15741*.
- [172] Minyang Chen, Chenchen Feng, Ran Cheng, MetaDE: Evolving differential evolution by differential evolution, *IEEE Trans. Evol. Comput.* (2025).

- [173] Chongming Gao, Kexin Huang, Ziang Fei, Jiaju Chen, Jiawei Chen, Jianshan Sun, Shuchang Liu, Qingpeng Cai, Peng Jiang, Future-conditioned recommendations with multi-objective controllable decision transformer, 2025, arXiv preprint [arXiv:2501.07212](https://arxiv.org/abs/2501.07212).
- [174] Dikshit Chauhan, Nitin Gupta, Anupam Yadav, An explainable reconfiguration-based optimization algorithm for industrial and reliability-redundancy allocation problems, 2025, arXiv preprint [arXiv:2504.01331](https://arxiv.org/abs/2504.01331).